DATA SCIENCE WITH EXCEL

Exploratory Data Analysis for Banking Analysts

From raw tables to reliable insight — using the tools already available in Excel.

STRUCTURED DATA | EDA | STATISTICS | DATA QUALITY

Banking context · Excel-first examples · Practical formulas and charts

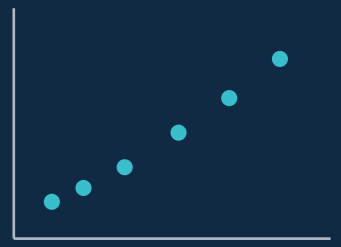
X

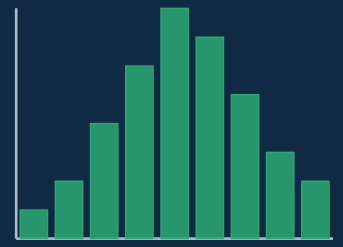
X Retail Banking EDA

fx =CORREL(Table1[Income],Table1[Balance])

ID	Age	Income	Balance
C1001	42	65k	12.5k
C1002	35	54k	8.2k
C1003	61	92k	45.5k
C1004	29		3.2k
C1005	47	71k	165k

Correlation





Training deck

Course Schedule

Dual timetable for Italy / Hungary and Romania

Same learning opportunities. Different time zones. Shared goals.

Italy / Hungary

SAME COURSE
SHARED PROGRESS

09:00 – 10:30	Session
10:30 – 10:40	Morning break (10 min)
10:40 – 12:00	Session
12:00 – 13:00	Lunch break
13:00 – 15:00	Session
15:00 – 15:15	Afternoon break (15 min)
15:15 – 17:00	Session

Daily timetable: 09:00 – 12:00 / 13:00 – 17:00

Romania

SAME CONTENT
LOCAL CONVENIENCE

10:00 – 11:30	Session
11:30 – 11:40	Morning break (10 min)
11:40 – 13:00	Session
13:00 – 14:00	Lunch break
14:00 – 16:00	Session
16:00 – 16:15	Afternoon break (15 min)
16:15 – 18:00	Session

Daily timetable: 10:00 – 13:00 / 14:00 – 18:00

Why this structure?

The **morning block** is slightly shorter and the afternoon block longer. This is intentional: it helps avoid pushing the lunch break too late in Romania. In exchange, the mid-morning break is only 10 minutes, while the afternoon break is 15 minutes.



The first draft of this set of slides was generated using AI tools under the direction of Antonio Piemontese and in accordance with his general and specific instructions and guidelines. The material was subsequently reviewed, revised, and further refined through additional interactions with AI tools, based on Antonio Piemontese's feedback and specific requests for changes.
Throughout the process, Antonio Piemontese retained responsibility for the overall direction, content, review, and final approval of the material.

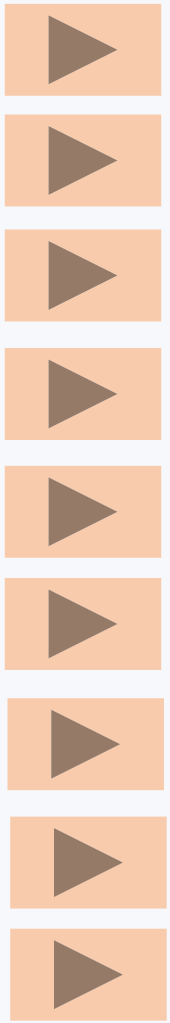
This declaration is provided in accordance with the transparency requirements of the European Union's AI Act.

the first two day of the course

Course map: what we will build together

A conceptual framework for data analysis, anchored in Excel and banking examples.

0	**Preliminaries**	Technical Requirements, Language, Anatomy of Excel
1	**Foundations**	Data Science, analytics, banking context
2	**Data & Types**	Structured tables and data types
3	**EDA**	Profiling, summaries, pivots, visual exploration
4	**Probability & Statistics**	Random variables, distributions, uncertainty
5	**Relationships**	Correlation, correlation matrices and patterns
6	**Visualisation**	Plots that reveal structure and problems, dashboards
7	**Data issues**	Outliers, missing values, quality
8	**Prediction Models**	Regression, Classification (by the Solver)



Already done last time

Goal: make the analysis explainable, auditable and reproducible — not just "get a chart".

Chapter title



00 - Preliminaries

Technical requirements: Excel version matters

Minimum setup to run every example in this training deck.

1. **Microsoft Excel 2021 or later — minimum for full compatibility.**
2. **Recommended: Microsoft Excel 2024 or Microsoft 365.**
3. Enable the Analysis ToolPak (Data → Data Analysis).
4. Windows 10 or Windows 11 recommended for the classroom setup.
5. PowerPoint 2016+ or a compatible presentation viewer.
6. Dynamic-array functions are used in examples: SORT and UNIQUE.
7. Excel Tables (Ctrl+T) are used throughout the EDA workflow.
8. No Python, R, Power BI or other Data Science software required.
9. No VBA or macros required.
10. No Internet connection required for the exercises.

KEY REQUIREMENT **Full compatibility requires Excel 2021 or later • Recommended: Excel 2024 / Microsoft 365**

How to change the language in Excel

Display language controls menus and tabs; authoring language affects editing and localized function names.

1 Open Options

File → Options → Language (with possible download for the new language)

2 Choose the language

Office display language

Changes Ribbon tabs, menus, commands and dialog boxes.

Office authoring languages

Controls proofing/editing language and can affect localized function names.

Select the desired language → Set as Preferred / Default → restart Excel if requested.

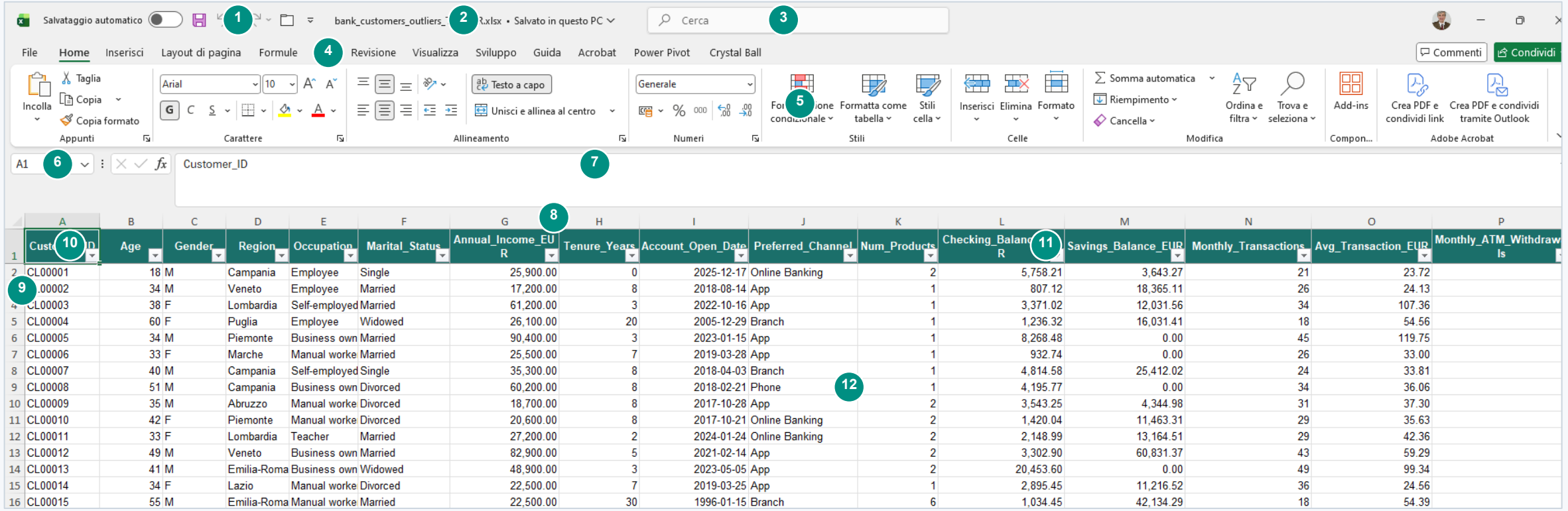
What changes?

Ribbon	Home ↔ Home Data ↔ Dati
Menus	File ↔ File Options ↔ Opzioni
Functions	SUM ↔ SOMMA AVERAGE ↔ MEDIA
Separators	Comma / semicolon and decimal conventions may depend on regional settings.

If the language is not installed, Excel may offer Add a Language / install the Office language pack.

Anatomy of the Excel interface (2024)

The main components you will use when exploring and analysing tabular data



	Customer_ID	Age	Gender	Region	Occupation	Marital_Status	Annual_Income_EUR	Tenure_Years	Account_Open_Date	Preferred_Channel	Num_Products	Checking_Balance_EUR	Savings_Balance_EUR	Monthly_Transactions	Avg_Transaction_EUR	Monthly_ATM_Withdrawals
2	CL00001	18	M	Campania	Employee	Single	25,900.00	0	2025-12-17	Online Banking	2	5,758.21	3,643.27	21	23.72	
3	CL00002	34	M	Veneto	Employee	Married	17,200.00	8	2018-08-14	App	1	807.12	18,365.11	26	24.13	
4	CL00003	38	F	Lombardia	Self-employed	Married	61,200.00	3	2022-10-16	App	1	3,371.02	12,031.56	34	107.36	
5	CL00004	60	F	Puglia	Employee	Widowed	26,100.00	20	2005-12-29	Branch	1	1,236.32	16,031.41	18	54.56	
6	CL00005	34	M	Piemonte	Business own	Married	90,400.00	3	2023-01-15	App	1	8,268.48	0.00	45	119.75	
7	CL00006	33	F	Marche	Manual worke	Married	25,500.00	7	2019-03-28	App	1	932.74	0.00	26	33.00	
8	CL00007	40	M	Campania	Self-employed	Single	35,300.00	8	2018-04-03	Branch	1	4,814.58	25,412.02	24	33.81	
9	CL00008	51	M	Campania	Business own	Divorced	60,200.00	8	2018-02-21	Phone	1	4,195.77	0.00	34	36.06	
10	CL00009	35	M	Abruzzo	Manual worke	Divorced	18,700.00	8	2017-10-28	App	2	3,543.25	4,344.98	31	37.30	
11	CL00010	42	F	Piemonte	Manual worke	Divorced	20,600.00	8	2017-10-21	Online Banking	2	1,420.04	11,463.31	29	35.63	
12	CL00011	33	F	Lombardia	Teacher	Married	27,200.00	2	2024-01-24	Online Banking	2	2,148.99	13,164.51	29	42.36	
13	CL00012	49	M	Veneto	Business own	Married	82,900.00	5	2021-02-14	App	2	3,302.90	60,831.37	43	59.29	
14	CL00013	41	M	Emilia-Roma	Business own	Widowed	48,900.00	3	2023-05-05	App	2	20,453.60	0.00	49	99.34	
15	CL00014	34	F	Lazio	Manual worke	Divorced	22,500.00	7	2019-03-25	App	1	2,895.45	11,216.52	36	24.56	
16	CL00015	55	M	Emilia-Roma	Manual worke	Married	22,500.00	30	1996-01-15	Branch	6	1,034.45	42,134.29	18	54.39	

1 Quick Access Toolbar
Frequently used commands such as Save, Undo and Redo.

2 Title bar
Shows the workbook name and the Excel application window.

3 Search box
Find commands, help topics and Excel features quickly.

4 Ribbon tabs
Home, Insert, Formulas, Data, Review, View, etc.

5 Ribbon
Context-sensitive command area organised into groups.

6 Name Box
Shows the active cell reference; can also jump to a cell/range.

7 Formula Bar
View, enter and edit values or formulas for the active cell.

8 Column headings
Letters identify columns; click to select an entire column.

9 Row headings
Numbers identify rows; click to select an entire row.

10 Active cell
The currently selected cell; its reference appears in the Name Box.

11 Table header & filters
Field names plus sorting/filter controls in an Excel Table.

12 Worksheet grid
The cell area where tabular data, formulas and results live.

Also important (not visible in this screenshot): sheet tabs, horizontal/vertical scroll bars, the status bar and zoom controls.

Excel Formulas in Italian and English

The main matchings

English Excel	Italian Excel	Where we use it
IF	SE	Checks: every flag
AND / OR	E / O	Checks: mortgage vs payment, age, date, score
COUNT	CONTA.NUMERI	Statistics: N numeric values
COUNTA	CONTA.VALORI	Key: total injected anomalies
COUNTBLANK	CONTA.VUOTE	Statistics: blank cells; sentinels and blanks
COUNTIF	CONTA.SE	Statistics: values beyond the fences; Checks: duplicate ID, summary
COUNTIFS	CONTA.PIÙ.SE	Checks: duplicate record; cluster (48 × 199)
AVERAGE	MEDIA	Statistics: mean; z-score
MEDIAN	MEDIANA	Statistics: median
STDEV	DEV.ST	Statistics: standard deviation; z-score
MIN / MAX	MIN / MAX	Statistics; range checks
QUARTILE	QUARTILE	Statistics: Q1, Q3 → IQR
SUMPRODUCT	MATR.SOMMA.PRODOTTO	Checks: N flags and summary
ABS	ASS	Checks: date vs tenure
INT	INT	Checks: date vs tenure
VLOOKUP	CERCA.VERT	Checks: look up a customer by ID in another sheet (e.g. Key)
HLOOKUP	CERCA.ORIZZ	Checks: look up a value by column header in a horizontal table
DATE	DATA	Checks: extraction date DATE(2026,9,13)
YEAR	ANNO	Checks: plausible date
TODAY	OGGI	Key: filtering future dates
DATEDIF	DATA.DIFF	Key: date/tenure consistency
LEN	LUNGHEZZA	Key: trailing spaces in categories
TRIM	ANNULLA.SPAZI	Key: trailing spaces in categories
UPPER	MAIUSC	Key: standardising categories

Three modern variants participants may type instead of the "classic" functions used in the file: STDEV.S → DEV.ST.C, QUARTILE.INC → INC.QUARTILE, STANDARDIZE → NORMALIZZA (computes the z-score directly).

The same formula in both languages, separators included — English Excel uses `,` between arguments and `.` for decimals, Italian Excel uses `;` and `,`: `=IF(OR(B2<18,B2>100),"Implausible age","")` ↔ `=SE(O(B2<18;B2>100);"Implausible age";"")` and `=Q1-1.5*IQR` ↔ `=Q1-1,5*IQR`.

Excel Formulas in English and Polish

The main matchings

English Excel	Polish Excel	Where we use it
IF	JEŻELI	Checks: every flag
AND / OR	ORAZ / LUB	Checks: mortgage vs payment, age, date, score
COUNT	ILE.LICZB	Statistics: N numeric values
COUNTA	ILE.NIEPUSTYCH	Key: total injected anomalies
COUNTBLANK	LICZ.PUSTE	Statistics: blank cells, sentinels and blanks
COUNTIF	LICZ.JEŻELI	Statistics: values beyond the fences; duplicate ID
COUNTIFS	LICZ.WARUNKI	Checks: duplicate record; cluster (48 × 199)
AVERAGE	ŚREDNIA	Statistics: mean; z-score
MEDIAN	MEDIANA	Statistics: median
STDEV	ODCH.STANDARDOWE	Statistics: standard deviation; z-score
MIN / MAX	MIN / MAX	Statistics: range checks
QUARTILE	KWARTYL	Statistics: Q1, Q3 → IQR
SUMPRODUCT	SUMA.ILOCZYNÓW	Checks: N flags and summary
ABS	MODUŁ.LICZBY	Checks: date vs tenure
INT	ZAOKR.DO.CAŁK	Checks: date vs tenure

English Excel	Polish Excel	Where we use it
VLOOKUP	WYSZUKAJ.PIONOWO	Checks: look up a customer by ID in another sheet
HLOOKUP	WYSZUKAJ.POZIOMO	Checks: find a value in a horizontal lookup table
DATE	DATA	Checks: extraction date, DATE(2026,9,13)
YEAR	ROK	Checks: plausible date
TODAY	DZIŚ	Key: filtering future dates
DATEDIF	DATA.RÓŻNICA	Key: date/tenure consistency
LEN	DŁ	Key: trailing spaces in categories
TRIM	USUŃ.ZBĘDNE.ODSTĘPY	Key: trailing spaces in categories
UPPER	LITERY.WIELKIE	Key: standardising categories

Modern equivalents: STDEV.S → ODCH.STANDARD.PRÓBKI; QUARTILE.INC → KWARTYL.PRZEDZ.ZAMK; STANDARDIZE → NORMALIZUJ.

Regional formula syntax: English Excel uses comma (,) between arguments and period (.) for decimals. Polish Excel typically uses semicolon (;) between arguments and comma (,) for decimals.

English Excel

```
=IF(OR(B2<18,B2>100),"Implausible age","")
```

```
=Q1-1.5*IQR
```

Polish Excel

```
=JEŻELI(LUB(B2<18;B2>100);"Implausible age";"")
```

```
=Q1-1,5*IQR
```

Excel Formulas in English and Romanian

The main matchings

English Excel	Romanian Excel	Where we use it
IF	IF	Checks: every flag
AND / OR	AND / OR	Checks: age, date, score
COUNT	COUNT	Statistics: numeric values
COUNTA	COUNTA	Key: nonempty cells
COUNTBLANK	COUNTBLANK	Statistics: blanks
COUNTIF	COUNTIF	Statistics: values beyond thresholds
COUNTIFS	COUNTIFS	Checks: duplicate records
AVERAGE	AVERAGE	Statistics: mean
MEDIAN	MEDIAN	Statistics: median
STDEV	STDEV	Statistics: standard deviation
MIN / MAX	MIN / MAX	Statistics: range checks
QUARTILE	QUARTILE	Statistics: Q1, Q3 and IQR
SUMPRODUCT	SUMPRODUCT	Checks: flag totals
ABS	ABS	Checks: date vs tenure
INT	INT	Checks: date vs tenure

English Excel	Romanian Excel	Where we use it
VLOOKUP	VLOOKUP	Customer lookup by ID in another sheet
HLOOKUP	HLOOKUP	Horizontal lookup by header
DATE	DATE	Construct a date (DATE(2026,9,13))
YEAR	YEAR	Extract the year from a date
TODAY	TODAY	Current date
DATEDIF	DATEDIF	Date and tenure consistency
LEN	LEN	Text length
TRIM	TRIM	Remove extra spaces in text
UPPER	UPPER	Uppercase categories

Important: Romanian Excel generally keeps English function names.
The locale mainly changes separators and number formats.

Modern equivalents: STDEV.S, QUARTILE.INC, STANDARDIZE — names unchanged.

Regional formula syntax:
English setup: argument separator comma (,) and decimal point (.).
Romanian regional setup: typically semicolon (;) and decimal comma (,). Settings may vary.

English Excel

```
=IF(OR(B2<18,B2>100);"Implausible age","")
=Q1-1,5*IQR
```

Romanian Excel

```
=IF(OR(B2<18;B2>100);"Implausible age";"")
=Q1-1,5*IQR
```

Two main behaviours when editing a formula

Especially regarding the arrow keys

When you are editing a formula in Excel, there are **two main behaviors**, especially for the arrow keys:

- **Edit mode**: the ← → ↑ ↓ keys **move the cursor inside the formula**, without changing the cell references.
- **Point mode**: the arrow keys **select/move the reference to a cell** in the formula.

The **F2** key allows you to switch to the formula text editing behavior.

For example, with:

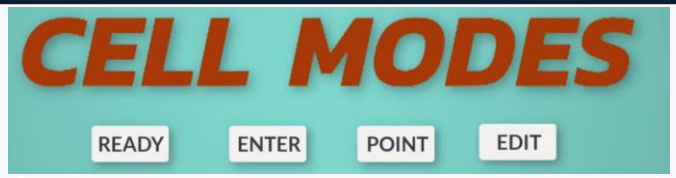
```
=A1+$B$2
```

if the cursor is on `A1`:

- in **Edit mode**, pressing → moves the cursor between the characters;
- in **Point mode**, pressing → can change the selected reference from `A1` to `B1`.

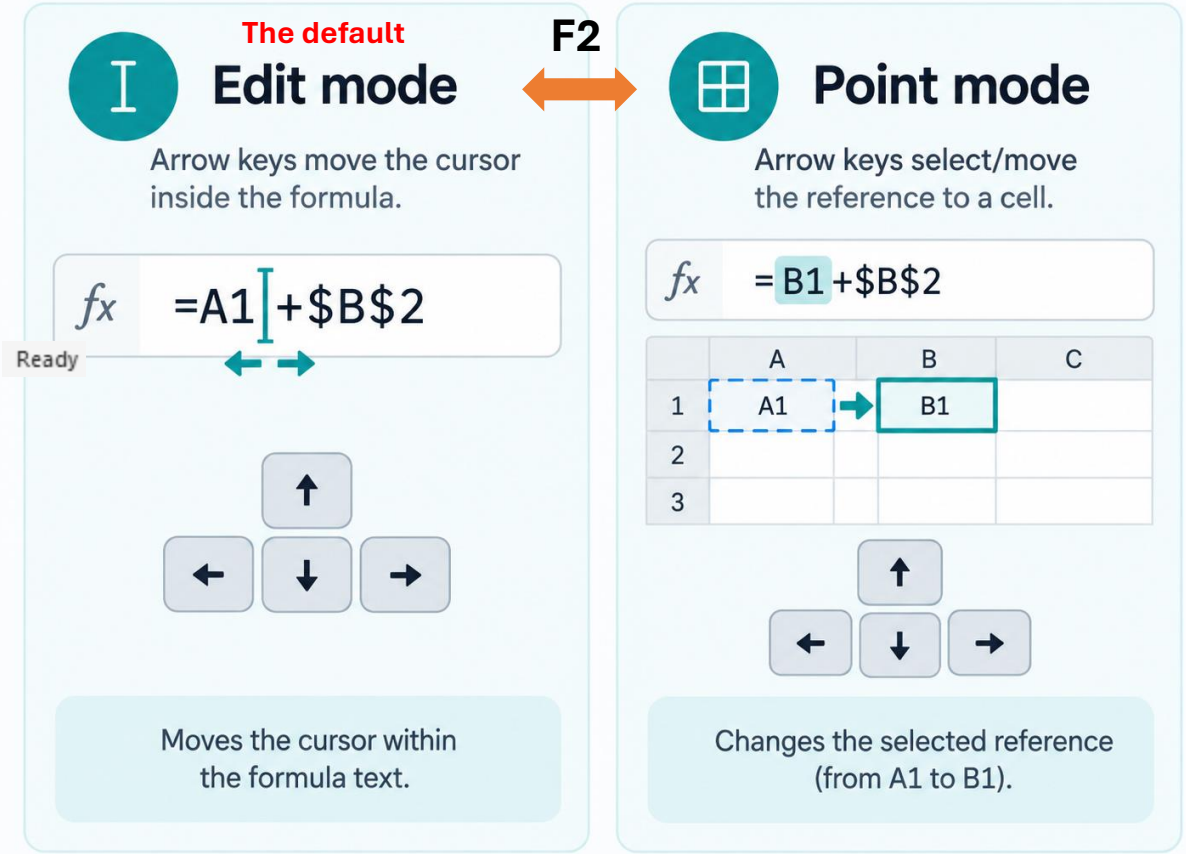
POINT MODE: FORMULA SELECTION

EDIT MODE: IN-CELL EDITING



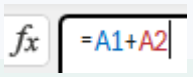
(Pronto, Invio, Puntatore, Modifica)

Knowing Excel cell modes (Ready, Enter, Point, and Edit) helps you avoid frustrating navigation errors and fix broken formula references quickly



click on right mouse button in the left bottom corner (and then on the top line) to see cell modes

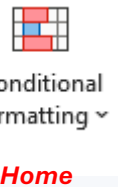
Ready



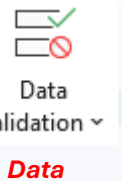
Editing formulas



Formulas



Home



Data

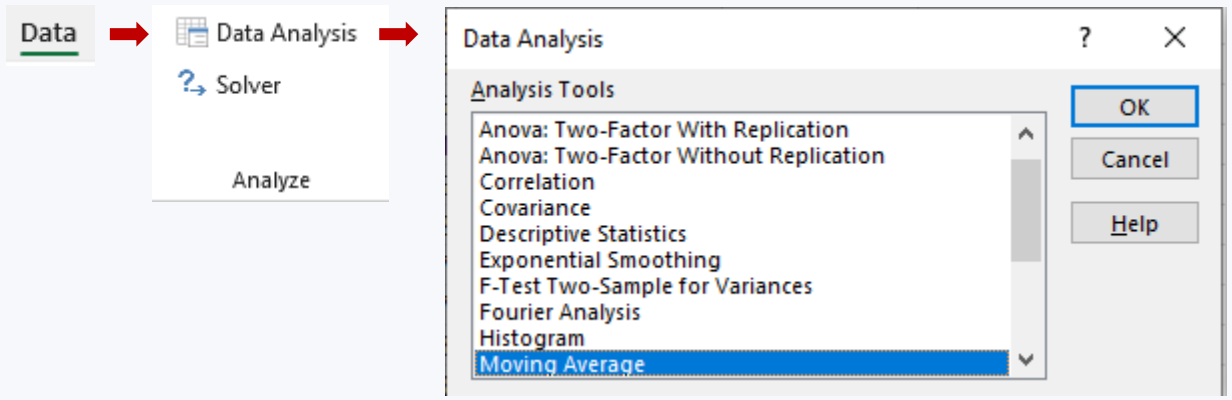


Insert

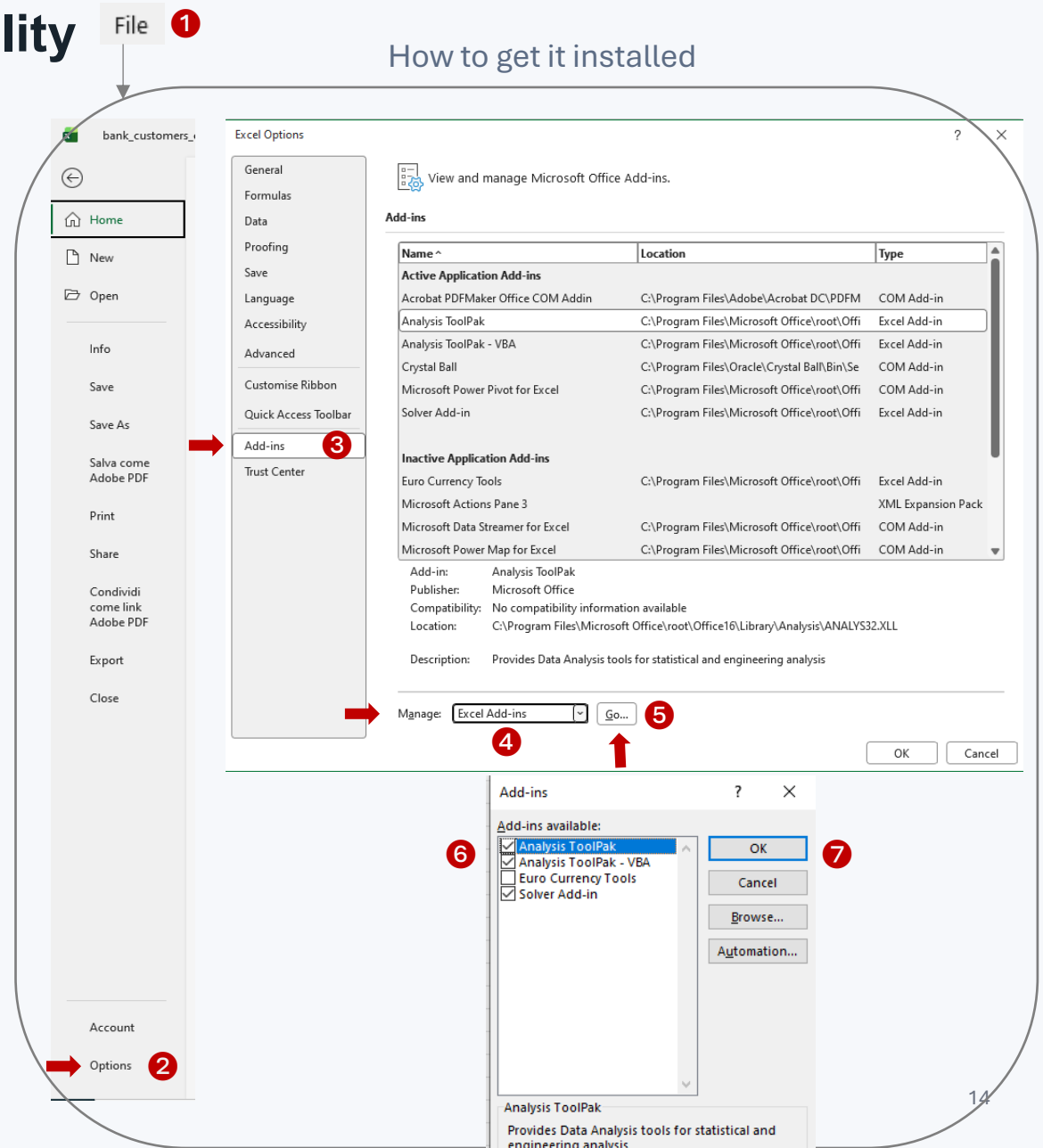
link to a good explanation video

Analysis tool –a key piece of functionality

What it is (our target)



How to get it installed



Cell References

In Excel, **A1 references** are the standard way to identify cells using:

- a **column letter**
- followed by a **row number**

Examples:

```
A1
B5
M22
AA100
```

So:

- `A1` = column A, row 1
- `M22` = column M, row 22

This is called **A1 reference style**.

It is also used for ranges:

```
A1:A10
B2:D20
M22:M31
```

Excel also distinguishes:

- **relative reference** → `A1`
- **absolute reference** → `$A$1`
- **mixed reference** → `$A1` or `A$1`

Example:

```excel
=A1*$B$1
```

If copied down, `A1` changes, while `$B$1` stays fixed.

The alternative is the **R1C1 reference style**, for example:

```
R1C1
R22C13
```

where `R` means row and `C` means column.

For most users and for the training deck, **A1 notation is the normal/default Excel notation.**

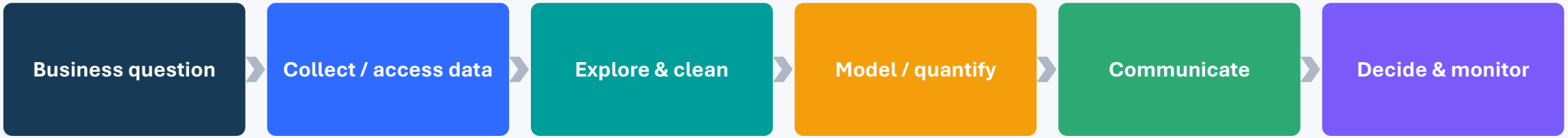
Chapter title



01 - Foundations

Data Science in one slide

Data Science combines business questions, data, statistical reasoning and computational tools to support decisions.



What matters most

A model or dashboard is useful only if it answers a relevant question with trustworthy data.

Where Excel fits

Excel is excellent for inspection, quick EDA, small-to-medium datasets, validation and communication.

Where Excel stops

Very large datasets, heavy automation and advanced modelling often need SQL, Python, R or cloud platforms.

```
Question → Data → Evidence → Decision
```

The typical Data Science flow

Why Excel still matters for bank analysts

Excel is not “less scientific”; it is a practical analysis surface when used with disciplined methods.

Fast inspection

Open a table and immediately inspect values, types, gaps and extremes.

Business proximity

Analysts, risk teams and finance users already exchange workbooks.

Transparent logic

Formulas, filters and PivotTables make many transformations visible.

Excellent EDA

Descriptive statistics, histograms, scatter plots, box plots and correlations.

Audit support

Checks, reconciliation and exception lists can be made explicit.

Bridge tool

Useful before/after SQL, Python or modelling pipelines. Excel can be easily put into a data pipeline.

The key limitation to Data Science is not the tool: it is uncontrolled data handling, hidden assumptions and poor reproducibility.

Data Science, statistics, BI and Machine Learning

They overlap, but they answer different kinds of questions.

Business Intelligence / Data Warehouse

What happened?

Reports, KPIs, dashboards, aggregation

Everything referred to the PAST

Statistics

What can we infer?

Variation, uncertainty, estimation, testing

Knowledge of the SAMPLE (*descriptive* statistics) and knowledge from SAMPLE to POPULATION (*inferential* statistics)

Machine Learning

What can we predict?

Prediction, classification, clustering

Predictive Modeling is about the FUTURE or NEW data (quite limited in Excel)

Data Science

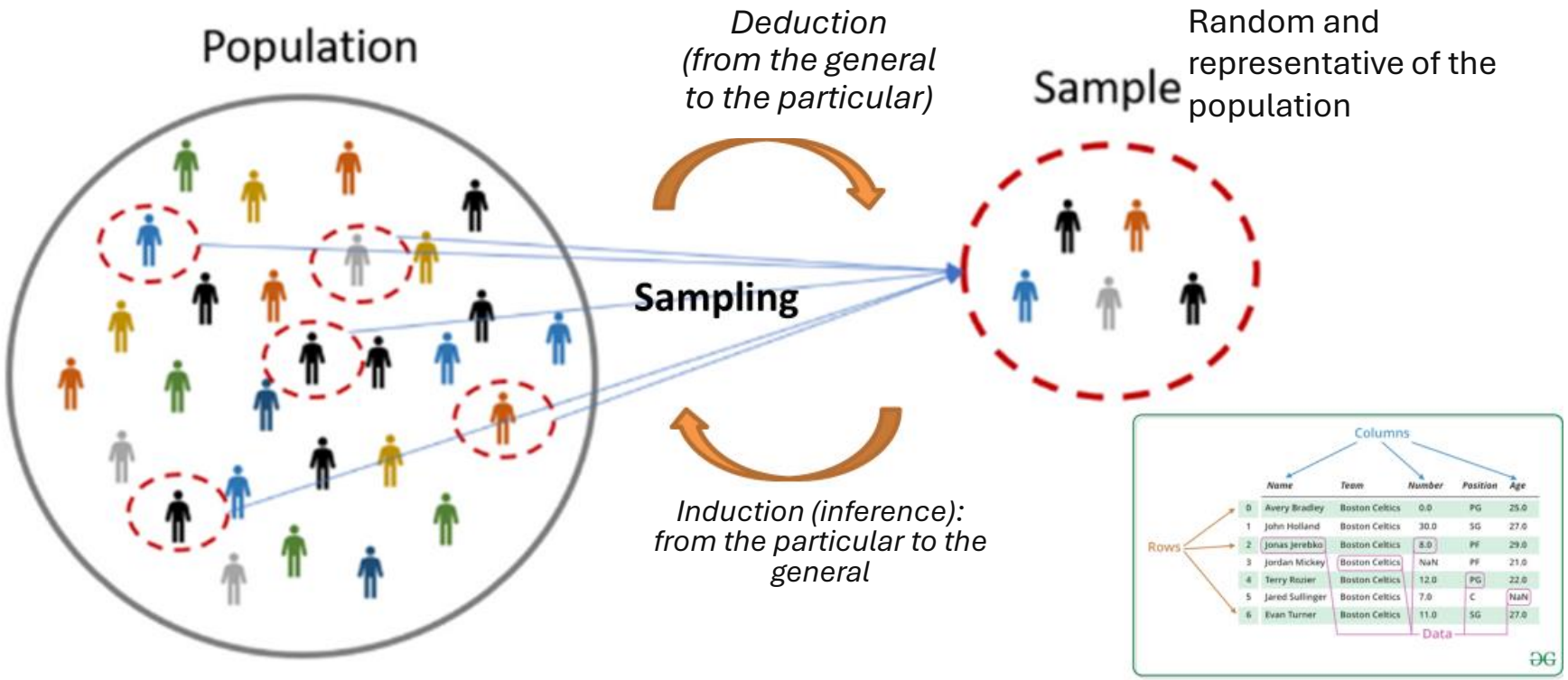
How do we turn data into decisions?

End-to-end problem framing, data, analysis, modelling, communication

From raw data to knowledge to actions

Excel can support all four — but EDA and descriptive statistics are where it is especially strong.

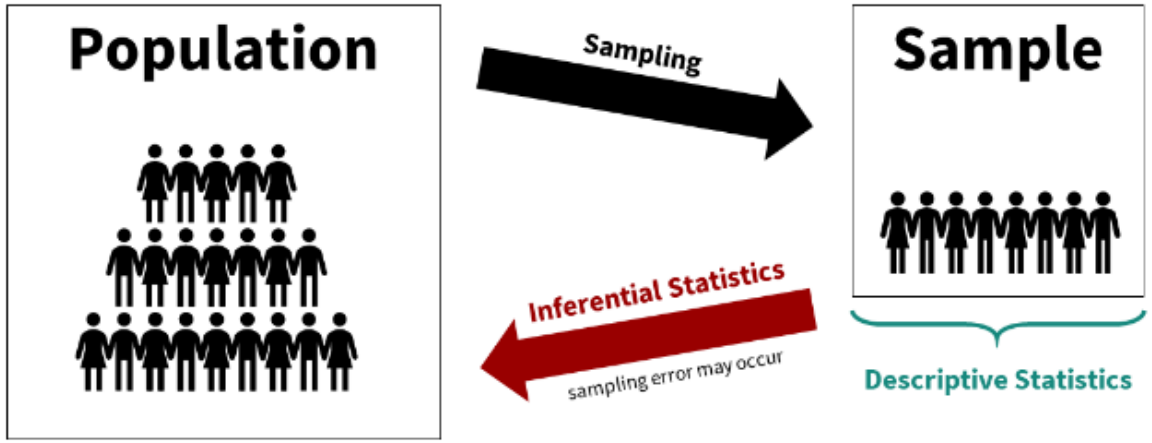
Sample vs. Population



The dataframe

Excel 2024 limit per single worksheet is 1,048,576 rows.
Computing EDA and building predictive models over the full population might be too demanding for our hardware.

Sampling selects observations from the population; statistical inference uses the sample to learn about the population.

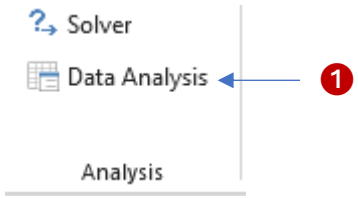


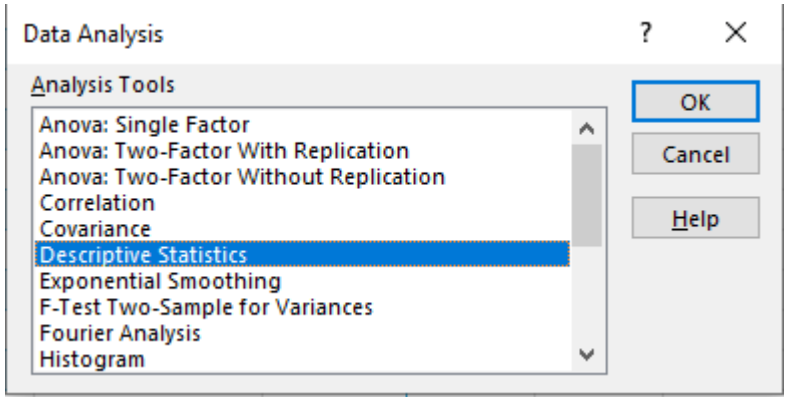
Sometimes we **cannot access** the whole population (eg, for security reasons), or processing the whole population requires **suitable hardware & software** (we maybe don't have).

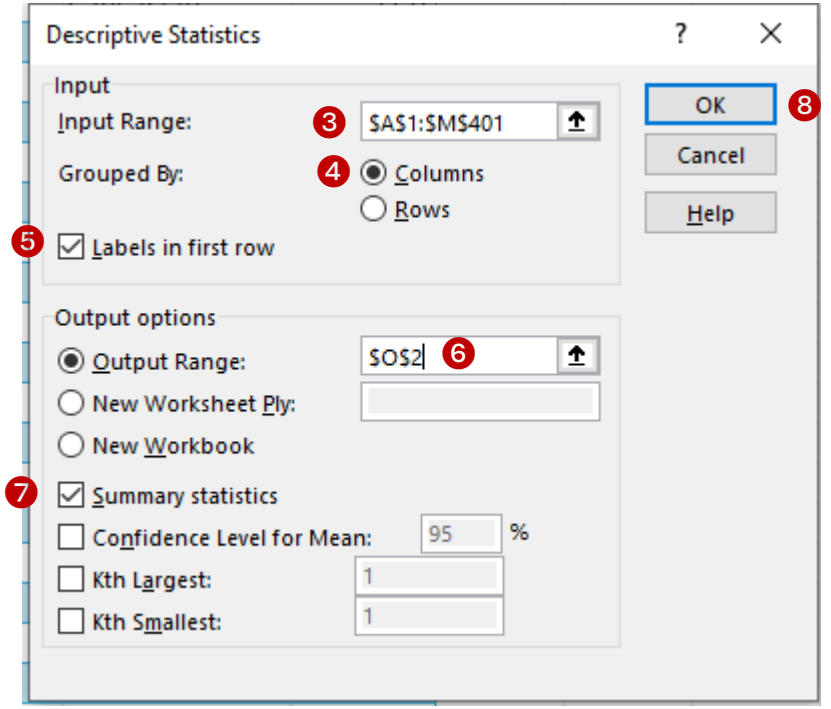
In banking, a **population** is the complete set of individuals, accounts, transactions, loans, or events that a statistical analysis aims to study. A **sample** is a subset of that population, selected for analysis.

Examples of populations and samples in banking

Banking area	Population	Sample
Mortgage lending	All 50,000 mortgage applications received by a bank in 2025	2,000 randomly selected applications
Credit risk	All 120,000 active personal loans on December 31, 2025	5,000 selected loans
Credit card transactions	All 20 million card transactions processed in 2025	100,000 sampled transactions
Fraud detection	All 10 million online payments executed during a quarter	50,000 selected payments
Customer satisfaction	All 800,000 retail banking customers	4,000 customers invited to participate in a survey
Loan defaults	All 40,000 loans originated in 2023, tracked for 24 months	3,000 loans selected for default analysis
Banking operations	All 500,000 wire transfers executed during a month	10,000 sampled transfers
ATM performance	All ATM withdrawals across the bank's network in 2025	30,000 sampled withdrawals
Anti-money laundering (AML)	All 2 million customer transactions within the investigation period	20,000 transactions selected for review
Bank employees	All 15,000 employees of the banking group	800 employees selected for a workplace survey







Income	
Mean	40672.31
Standard Error	1776.19
Median	29715.00
Mode	23793.00
Standard Deviation	35523.70
Sample Variance	1261933598.32
Kurtosis	3.31
Skewness	1.76
Range	186622.00
Minimum	12.00
Maximum	186634.00
Sum	16268924.00
Count	400.00

How to Read Excel Descriptive Statistics

Example: Income (n = 400)

Central tendency
Typical or central values in the data.

Mean (average)	40,672.31	The average income across all 400 observations.
Median (middle value)	29,715.00	The middle income. Less affected by extreme values than the mean.
Mode (most frequent)	23,793.00	The most common income in the dataset.

Totals
Overall amount and number of observations.

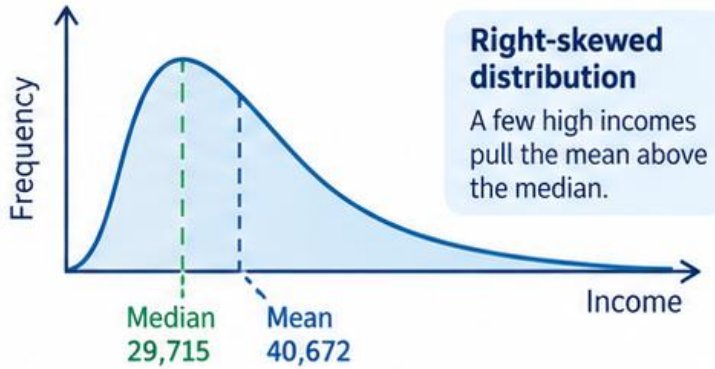
Sum	16,268,924.00	The total of all income values in the dataset.
Count	400	The number of income values (observations).

Spread / variability
How much the values vary.

Standard Deviation	35,523.70	Typical distance of incomes from the mean. Larger values indicate more variability.
Sample Variance	1,261,933,598.32	The squared standard deviation. (Units are income².)
Standard Error	1,776.19	Quantifies the uncertainty of the sample mean. Smaller values indicate a more precise estimate of the mean.
Range	186,622.00	The difference between the maximum and minimum. Range = 186,634 − 12 = 186,622.
Minimum	12.00	The smallest income in the dataset.
Maximum	186,634.00	The largest income in the dataset.

Shape of distribution
Information about the shape of the data.

Skewness	1.76	Positive value indicates a right-skewed distribution with a long upper tail (higher incomes).
Kurtosis	3.31	Greater than 3 suggests heavier tails (more extreme values) than a normal distribution.



Right-skewed distribution
A few high incomes pull the mean above the median.

What this dataset suggests

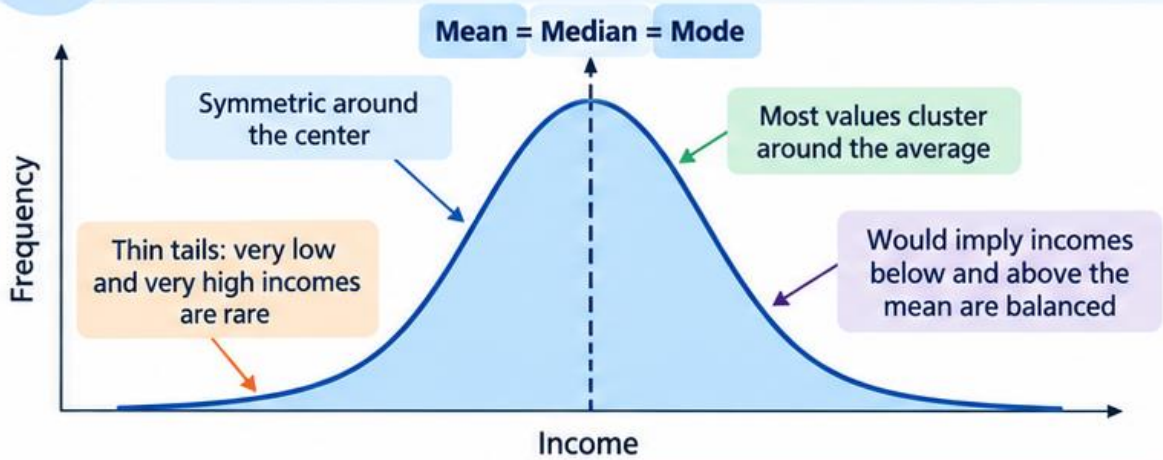
(the insights)

- The mean (40,672.31) is well above the median (29,715.00), suggesting that high incomes likely pull the average upward. (id est, incomes are right-skewed)
- Incomes show strong variability, with a large standard deviation (35,523.70) and a very wide range (186,622.00).
- The minimum (12.00) and maximum (186,634.00) are far apart, indicating a wide spread of incomes.
- The distribution is positively skewed (skewness = 1.76) with heavier tails (kurtosis = 3.31), meaning there are more extreme high-income values than in a normal distribution.

Income Distributions: Hypothetical Gaussian vs Real-Life Right-Skewed

Why salaries usually do not follow a normal distribution

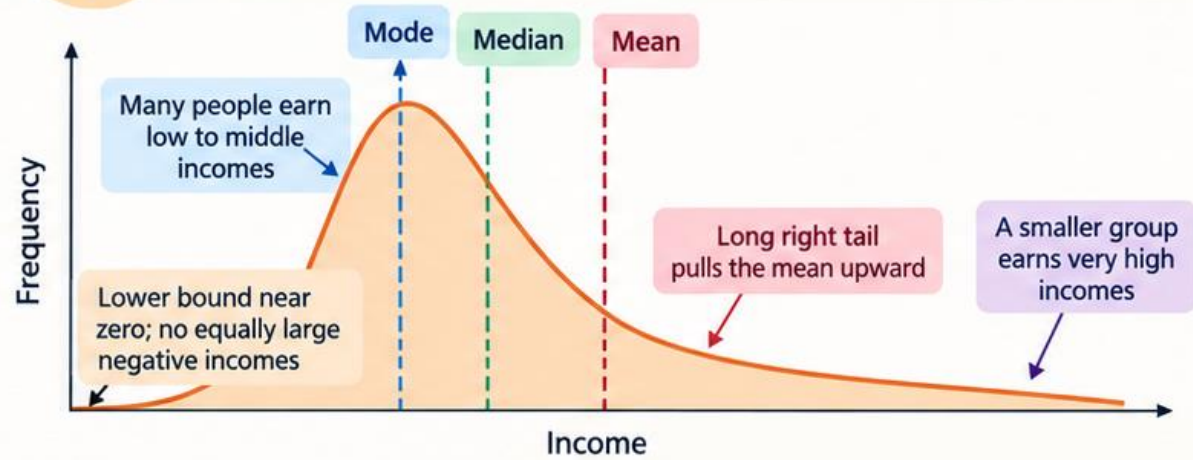
Hypothetical Gaussian (Normal) distribution



Useful as a **theoretical benchmark**, but unrealistic for incomes.

If incomes were Gaussian, the "average income" would describe a typical worker fairly well.

Real-life income distribution



Better describes real salary data.

The mean can overstate what a typical worker earns; the median is often more representative.

Main differences

	Hypothetical Gaussian (Normal)	Real-life income distribution
Shape	Symmetric bell	Asymmetric, long right tail
Position of summary measures	Mean = Median = Mode	Mean > Median > Mode
Interpretation of average	Average represents the center well	Average pulled up by top earners
Realism for incomes	Rarely observed	Common in real data
Policy implication	Mean and median similar	Need median and percentiles to describe typical earnings

Key takeaway: For income data, the median usually describes the typical person better than the mean, because a few high incomes distort the average.

Banking examples: business questions before tools

Start with the decision or risk question, then choose the analysis.

Domain	Business question	Excel-first EDA
Retail customers	Which segments show declining balances?	PivotTable + line chart
Credit risk	Which variables move with arrears (ie, payments that are overdue or unpaid when due) or default?	Correlation + distributions
Operations	Where are processing times unusually high? For what kind of transaction?	Histogram + outlier analysis
Fraud monitoring	Which transactions are atypical?	Patterns + anomaly flags
Data governance / Data quality	Where is information incomplete or inconsistent?	Missingness + quality checks

The data landscape in a bank

Most analysis starts from structured operational data — but meaning comes from context.

- **Core banking** — accounts · balances · products
- **Payments** — transactions · merchants · channels
- **CRM** — customers · campaigns · interactions
- **Risk systems** — exposure · ratings · arrears
- **External data** — rates · stock exchange · macro · geography

Source, operational data

ETL processing (Extract / Transform / Load) – usually overnight or in week-ends

Analysis-ready extract

`fx =ROWS(Table1)`

CustomerID	Date	Balance	Segment	Default
C1001	2026-08-31	12,500	Affluent	No
C1002	2026-08-31	8,200	Mass	No
C1003	2026-08-31	45,500	Affluent	No
C1004	2026-08-31	3,200	Mass	No
C1005	2026-08-31	165,000	Affluent	Yes
C1006	2026-08-31	9,700	Mass	No

A structured table ready to be analysed

Building a dataset in Excel: two starting points

A dataset can be created directly in Excel or imported from an external source. In both cases, the goal is a clean, structured, analysis-ready table.

1 IMPORT THE DATASET FROM OUTSIDE EXCEL

Typical external sources (maybe from an ETL process or ona real-time basis)
- CSV / TXT files
- Other Excel workbooks
- Databases
- Web / API sources via Power Query



Check before loading
- Delimiter and text encoding (UTF-8)
- Locale (decimal and date conventions)
- Inferred data types from the values (whenever data types are not declared in source data)
- Missing values and unexpected columns

OR

ANALYSIS-READY EXCEL TABLE

2 BUILD THE DATASET NATIVELY IN EXCEL

Typical sources
- Manual data entry
- Copy / paste from controlled sources
- Form-based or operational data entry

Customer_ID	Age	Region	Balance
CL001	34	North	5,240
CL002	41	West	8,120
CL003	29	South	3,410

Good practice
- One row = one observation
- One column = one variable

} tidy data

- Use clear headers and consistent data types
- Convert the range to an Excel Table (Ctrl+T)

Key point: regardless of the starting point, verify structure, data types, missing values and business meaning before starting EDA.

Chapter title



02 – Data & Types

Structured vs semi-structured vs unstructured data

Data Science works with many kinds of data. Excel is strongest with structured data, but analysts often receive all three forms.

STRUCTURED DATA

tabular data

Clear rows, columns and fields

Examples: Excel ranges/tables, CSV files, relational tables, banking transaction tables, Google sheets, etc
Typical fields: Customer_ID, Region, Balance, Date, Product_Count
Excel fit: excellent

Recognition: every row represents the same type of entity, and every column has a stable meaning.

SEMI-STRUCTURED DATA

Some organisation, but not fully tabular

Examples: JSON, XML, logs, email headers, API responses (there are also metadata)
Has labels or tags, but the shape may vary from record to record
Excel fit: possible after transformation or flattening

Recognition: records contain keys or tags, but not every record must contain the same fields.

UNSTRUCTURED DATA

No fixed table structure

Examples: PDFs, emails, contracts, images, audio, call recordings
Meaning exists, but structure is implicit rather than explicit
Excel fit: weak unless data is first extracted or summarised

Recognition: humans can understand the content, but software must work harder to extract variables.

Texts, images, videos, sounds, speech, 3Ds

Excel takeaway: structured data is the natural starting point for EDA in Excel; semi-structured and unstructured data usually need preparation before they can be analysed as tables.

Structured data: the backbone of everyday Data Science

Structured data has an explicit schema: rows are observations, columns are variables, cells are values.

Table1 · Customer snapshot

fx `=AVERAGE(Table1[Balance])`

CustomerID (primary key)	Age	Income	Balance	Segment
C1001	42	65,000	12,500	Affluent
C1002	35	54,000	8,200	Mass
C1003	61	92,000	45,500	Affluent
C1004	29	—	3,200	Mass
C1005	47	71,000	165,000	Affluent

This in Data Science is TIDY DATA

Rows = observations

One customer, account, loan, transaction or time period per row.

Columns = variables

Age, balance, product, segment, status, date, risk score...

Schema = meaning

Names, types, units, valid ranges, keys and business definitions.

Anatomy of a table

Understanding the grain of the table prevents many analytical mistakes.

Monthly customer snapshot

fx `=COUNTIFS(Table1[CustomerID],"C1001")`

CustomerID (PK)	Month (PK)	Balance	Products
C1001	Jan	11,800	3
C1001	Feb	12,100	3
C1001	Mar	12,500	3
C1002	Jan	8,700	2
C1002	Feb	8,450	2
C1002	Mar	8,200	2

Primary key / identifier

CustomerID + Month uniquely identifies each row.

Grain

One row = one customer in one month.

Risk of wrong grain

Averaging balances at the wrong level can double-count customers.

Observation, variable, value: three simple words, three different roles

A table becomes easier to reason about when these terms are explicit.

1

OBSERVATION (row)

A unit being analysed

Example: Customer C1003

2

VARIABLE (column)

A measured attribute

Example: Annual Income

3

VALUE (cell)

A concrete measurement

Example: €92,000

In Excel: rows usually represent observations, columns variables, and cells values.

What makes tabular data analysis-ready?

A clean table is a model of the business process, not merely a rectangular range.

One row, one observation

No merged cells; no subtotals inside the data area.

One column, one variable (a unique data type)

Do not mix currency, text and comments in the same field.

Consistent units

EUR vs thousands of EUR must be explicit.

Stable categories

“Retail”, “RETAIL”, “retail” should not be three categories.

Explicit missing values

Blank ≠ zero ≠ not applicable.

Traceable definitions

Every field has a business meaning and owner.

```
Tip: convert ranges to an Excel Table (Ctrl+T) before serious EDA.
```

Why convert a range to an Excel Table?

Before starting EDA, turn a normal cell range into a formal Excel Table (Ctrl+T).

Structured reference	Meaning
`Table1[Col1]`	Entire **Col1**
`Table1[@Col1]`	**Col1 in the current row**
`[@Col1]`	Same, when used inside Table1

NORMAL RANGE

Example: A1:F1000

ID	Age	Income	Bal.	Seg.
C01	42	65k	12.5k	A
C02	35	54k	8.2k	M
C03	61	92k	45.5k	A
C04	29	—	3.2k	M

Typical formula:
=AVERAGE(D2:D1000)

Ctrl+T

EXCEL TABLE

A structured object designed for analysis

- **Stable headers** — Column names remain explicit and usable.
- **Built-in filters** — Sort and filter every column immediately.
- **Automatic expansion** — New rows and formulas are included automatically.
- **Structured references** — Use readable formulas such as Table1[Balance].

```
=AVERAGE(Table1[Balance])
```

For EDA, an Excel Table makes the analysis more robust, readable and less error-prone.

The special behaviour of an Excel Table?

Here are the behaviours that set a table (Ctrl+T) apart from a plain range:

Structure and headers

- Headers must be text and unique: a number is converted to text, a duplicate name becomes "Name2".
- When scrolling down with the active cell inside the table, the headers replace the column letters: no need to freeze panes. **(just if the active cell belongs to the table)**
- Every header has a filter button (can be switched off in Table Design ▸ Filter Button); sorting and filtering always move entire rows.
- Table cells cannot be merged; a table cannot overlap another table or a PivotTable.

Automatic expansion

- Typing in the adjacent row or column extends the table on its own; Tab on the last cell creates a new row.
- New rows inherit formats, data validation, conditional formatting and column formulas: this is why the 12 rules in the file follow the table when customers are added.
- Inserting or deleting rows/columns inside the table (Insert Table Rows Above, etc.) does not affect the cells outside the table on the same sheet row.
- The handle in the bottom-right corner and Table Design ▸ Resize Table resize it manually.

Formulas

- Calculated columns: a formula entered in one cell fills the whole column and stays consistent; if a cell is edited by hand, Excel flags the exception.
- Structured references: `tblCustomers[Balance]`, `[@Balance]` (current row), `[#Headers]`, `[#Totals]`, `[#All]`; they adjust automatically as the table grows, whereas `$L$2:$L$401` stays fixed. This is why the A1-based counts in the Rules sheet do not update when rows are added, while `tblCustomers[...]` would.
- Important exception: structured references are not allowed in conditional-formatting rules (nor in data validation).
- Dynamic-array formulas cannot spill inside a table (#SPILL!).

Total row

- Table Design ▸ Total Row adds a row with a drop-down per column; the functions use `SUBTOTAL(109,...)`, which ignores filtered or hidden rows.

Formatting

- Table styles with banded rows/columns and highlighted first/last column; changing the style updates everything, and direct formatting overrides the style.
- Table Design ▸ Convert to Range removes the table while keeping its look; the style can be reset with Clear.

Integration

- The table name is a ready-made dynamic range for PivotTables, charts, data validation and Power Query (Data ▸ From Table/Range): on refresh they pick up the new rows without changing the source.
- Slicers are available directly on the table (Table Design ▸ Insert Slicer), not only on PivotTables.
- Remove Duplicates and the Table Design tab are always at hand.

Quick selection

- Black arrow on the top edge of a header: one click selects the column data, two clicks include the header; ~~arrow on the left edge for the row; top-left corner for the whole table~~.
- Ctrl+A once selects the data, twice includes the headers; Ctrl+Space selects the column, Shift+Space the table row.

Limits worth knowing

- With a table on the sheet, Subtotals (Data ▸ Subtotal), Custom Views and the legacy shared-workbook mode are unavailable.
- Group/Outline works only on sheet rows, not "inside" the table's logic.

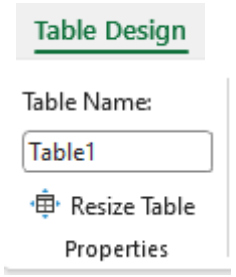
Where the table name appears?

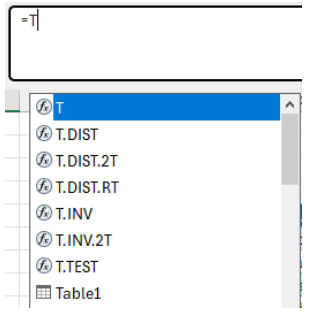
It doesn't appear anywhere on the sheet itself: the name is only visible in the interface, in three main places.

1. **Table Design tab** (in Excel 2010–2019 "Table Tools > Design"): it only appears when the active cell is inside the table. At the far left, in the Properties group, there is the "Table Name" box showing Table1. It is also the only place where we can rename it: in Name Manager the table name is listed (with a different icon from the other names), but the Edit and Delete buttons are disabled.
2. **Name Box** (to the left of the formula bar): opening the drop-down list we find Table1 among the names; selecting it, or typing it and pressing Enter, selects the table's data body. Conversely, if we select exactly the data body (one click on the arrow in the top-left corner of the table), the box shows Table1.
3. **In formulas**: typing `=Tab` makes Formula AutoComplete suggest Table1 with the table icon, and we can use it in structured references, for example `=SUM(Table1[Amount])`.

We then find it wherever the table is used as a source: in the "Table/Range" field when creating a PivotTable, as the query name in Power Query, in charts, in VBA with `ListObjects("Table1")`.

One useful detail: the name Table1 refers to the data body only, without the header row (Name Manager shows this in "Refers to"). To include the headers as well we use `Table1[#All]`.





How to convert an Excel Table back to a range

The data are preserved; only the Table structure is removed.

1. **Select the Table** — Click any cell inside the Excel Table.
2. **Open Table Design** — The Table Design tab appears on the Ribbon.
3. **Convert to Range** — Choose Table Design → Convert to Range.
4. **Confirm** — When Excel asks for confirmation, click Yes.

What changes?

Excel Table



Normal range

Data remain

Values and formulas are preserved.

Formatting

Table formatting usually remains.

Table features

Automatic expansion and structured Table behavior are removed.

Range → Ctrl+T → Excel Table → Table Design > Convert to Range → Range

Data types in Data Science, regardless Excel: meaning comes before formatting

The type determines which operations are meaningful — and which are nonsense.

Numeric

Measured quantities

Balance, age, income, interest rate

Categorical

A predefined number of values

Labels / groups

Segment, product type, branch

Boolean / binary

A special case of Categorical, with just 2 values

Two states

Default yes/no, active/inactive

Date / time

Time coordinates

Transaction date, month-end

Text

Free-form strings

Comments, descriptions, notes

Identifier

Keys, not quantities

CustomerID, IBAN, account number

A CustomerID that contains digits (eg, 4500) is still an identifier, not a numeric measurement.

Numeric data: discrete and continuous

Both are numeric, but the measurement scale and distribution may differ.

Discrete numeric variables

Countable values, often integers.

Examples: number of products, missed payments, complaints.

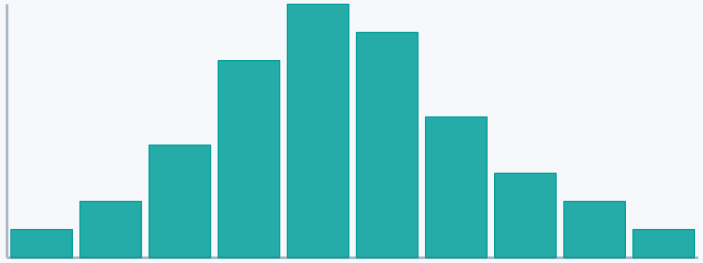
Continuous numeric variables (with decimal point)

Measurements on a scale.

Examples: balance, income, amount, duration, interest rate.



Discrete: Products per customer



Continuous: Account balance

```
=AVERAGE(Table1[Balance])      =MEDIAN(Table1[Balance])      =STDEV.S(Table1[Balance])
```

Categorical data: nominal, ordinal and binary

Categories can be labels, ordered levels or two-state flags.

Nominal

No intrinsic order

Region: North / South
Product: Card / Mortgage / Loan

Ordinal

Meaningful order

Risk grade: Low / Medium / High
Satisfaction: 1…5

Binary

Exactly two states

Default: Yes / No
Active: 1 / 0

Excel implications

- Use Data Validation lists to control allowed categories.
- Use PivotTables to compare counts, averages and rates across groups.
- Do not average arbitrary codes (e.g., Region 1, 2, 3).

Data types: qualitative and quantitative

A practical taxonomy for structured (tabular) data — with examples and software mappings.

VARIABLES IN TABULAR DATA

QUALITATIVE DATA

aka categorical

Values are labels or classes. **The set of possible values is often finite and relatively small.** Numeric operations such as sums or divisions are not meaningful.

NOMINAL

Unordered categories

No intrinsic ranking between values.

Examples: ZIP code, eye color, region, product type

ORDINAL

Ordered categories

Values have an order, but gaps are not numeric distances.

Examples: rating, risk class, customer tier

QUANTITATIVE DATA

aka numeric

Values represent quantities. Arithmetic and distances are meaningful. **The theoretical set of possible values may be very large or infinite.**

DISCRETE

Counts

Typically integer-valued observations.

Examples: # children, # transactions, # defects

CONTINUOUS

Measurements

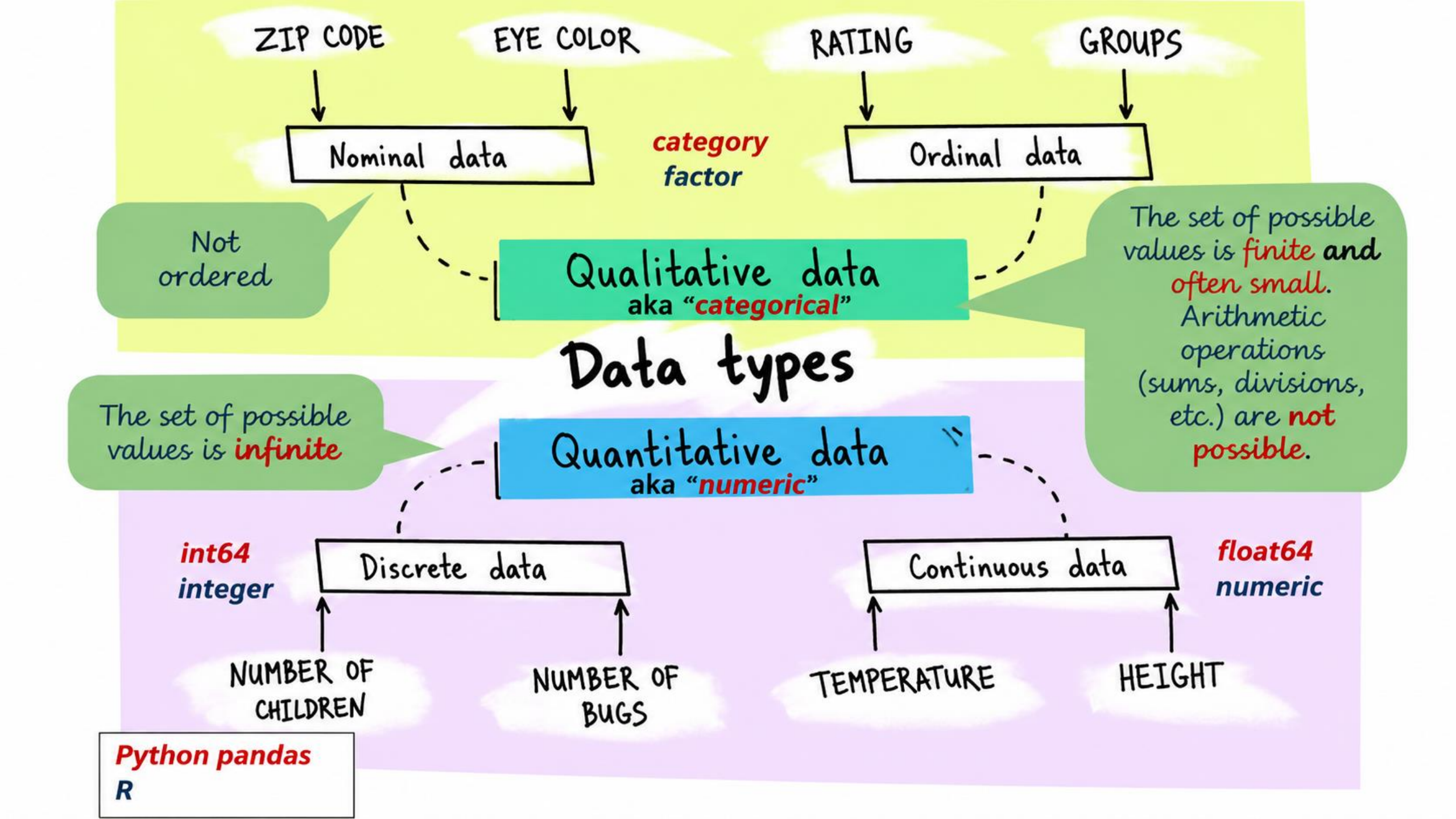
Can take any value within an interval, conceptually.

Examples: temperature, height, balance, interest rate

Software mapping in Excel See the next slide

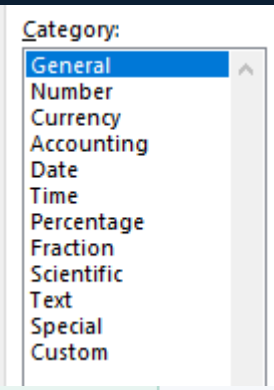
Important: **A numeric-looking code is not necessarily numeric data. ZIP code and Customer_ID are identifiers.**

Key idea: data type is about meaning and valid operations — not only about how a value is formatted in Excel.


ZIP CODE
EYE COLOR
RATING
GROUPS
Nominal data
category
factor
Ordinal data
Not ordered
Qualitative data
aka "categorical"
The set of possible values is finite and often small. Arithmetic operations (sums, divisions, etc.) are not possible.
Data types
The set of possible values is infinite
Quantitative data
aka "numeric"
int64
integer
Discrete data
Continuous data
float64
numeric
NUMBER OF CHILDREN
NUMBER OF BUGS
TEMPERATURE
HEIGHT
Python pandas
R

Statistical Data Types vs Excel Data Types

Excel stores values using technical cell types. The analyst assigns the statistical meaning.



Statistical data type (in general)	Meaning	Typical Excel format	Example
Nominal categorical	Labels with no inherent order	**Text**	Region = "Lombardy"
Ordinal categorical	Categories with a meaningful order	**Text or coded Number**	Low / Medium / High
Discrete numeric	Counts; integer-valued measurements	**Number (integer value)**	Num_Products = 4
Continuous numeric	Measurements on a continuous scale	**Number (decimal value)**	Balance = 5,823.47
Binary / Boolean	Two possible states	**TRUE/FALSE or 0/1**	Defaulted = FALSE
Date / Time	Temporal values	**Date/Time***	13/09/2026
Identifier	A code, not a quantity	**Usually Text**	Customer_ID = "CL00015"
Missing value	Unknown / not recorded	**Blank cell**	empty cell

** Excel stores dates/times internally as serial numbers, but formats them as dates/times.*

KEY IDEA

Excel data type ≠ statistical data type

EXAMPLE 1 — ORDINAL CODES

1 = Low 2 = Medium 3 = High

Excel sees numbers; the analyst sees ordered categories. An average may be mathematically possible but analytically questionable.

EXAMPLE 2 — IDENTIFIERS

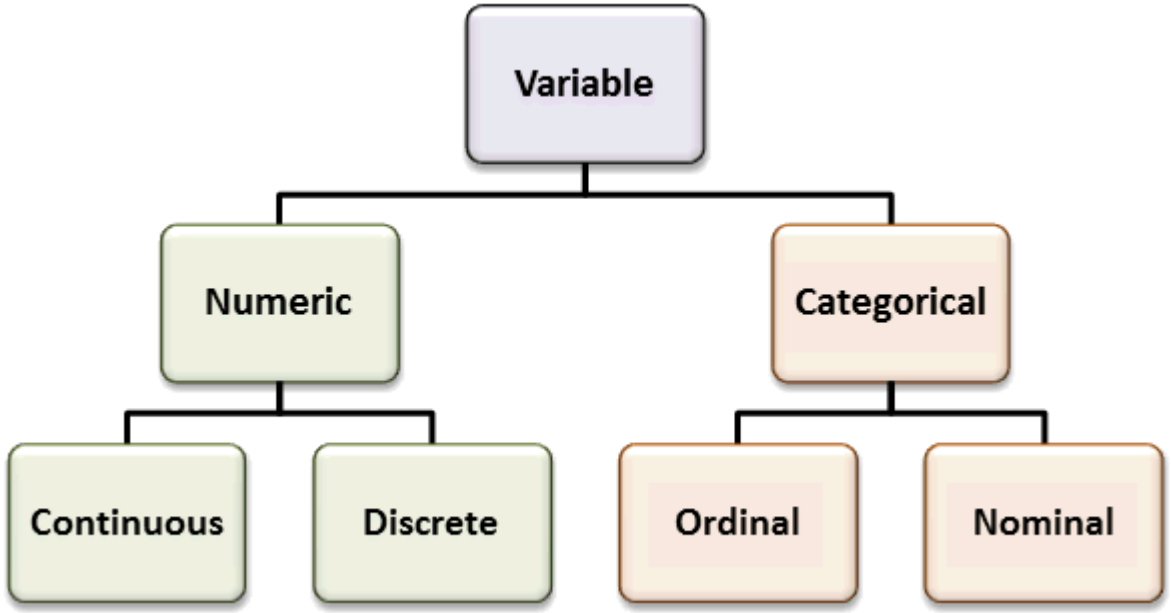
ZIP = 10121 Account = 000173

They look numeric, but they are labels. Store them as Text when leading zeros or code semantics matter.

Analyst rule: determine the variable's meaning first; only then decide the Excel representation and the valid operations.

Numerical vs Categorical Data Tyypes

Not every column that looks like a number should be treated as a number.



Dates, text and identifiers need special care

Not every column that looks like a number should be treated as a number.

Example	Semantic type	Appropriate analysis
2026-08-31	**Date**	Sort, group by month/quarter, compute intervals
C100245	**Identifier (key)**	Join / lookup key — never average it
0012345678	**Identifier stored as text**	Leading zeros matter
“Mortgage renewal”	**Text**	Search, classify, count terms
12.5%	**Numeric rate**	Store as 0.125; display as percentage

Formatting changes appearance; it does not automatically correct the underlying data type.

VLOOKUP and HLOOKUP in Banking

In Excel in italiano:

- VLOOKUP → CERCA.VERT
- HLOOKUP → CERCA.ORIZZ

Retrieve related information from a reference table using a lookup key.

VLOOKUP — Vertical Lookup

Search direction: Searches for the key in the FIRST COLUMN and returns a value from another column.

Banking example: Use Account Code C0004 to retrieve Product Name, Risk Category and Annual Fee.

Formula: =VLOOKUP(A2,$G$2:$J$6,2,FALSE)

Typical use: Product master data, branch codes, account types, customer categories.

HLOOKUP — Horizontal Lookup

Search direction: Searches for the key in the FIRST ROW and returns a value from another row.

Banking example: Use Rating Code C to retrieve the applicable Interest Rate and Credit Limit.

Formula: =HLOOKUP(B2,$F$2:$J$4,2,FALSE)

Typical use: Rate grids, rating bands, monthly/quarterly tables arranged horizontally.

Key idea: VLOOKUP = vertical reference table; HLOOKUP = horizontal reference table. FALSE requests an exact match.

- **VLOOKUP:**

Banking_VLOOKUP_Complete_Example.xlsx

"The table on the left represents bank transactions. Each row is a transaction and contains only the Account Code and the Transaction Amount.
The table on the right is a reference table that describes each Account Code: it tells us the corresponding Product Name, Risk Category, and Annual Fee.
VLOOKUP uses the Account Code as the key to connect the two tables and automatically add the correct product information to each transaction."

This table design (two separate tables) is required to storage space and avoid harmfull redundancy of data **(data table normalization)**

The table resulting from lookup

Account Code	Transaction Amount (EUR)	Product Name	Risk Category	Annual Fee (EUR)	Lookup Table			
C0001	1250	Current Account	Low	60	C0001	Current Account	Low	60
C0004	8200	Mortgage	Medium	250	C0002	Savings Account	Low	0
C0003	35000	Personal Loan	Medium	120	C0003	Personal Loan	Medium	120
C0002	5400	Savings Account	Low	0	C0004	Mortgage	Medium	250
C0005	980	Credit Card	High	45	C0005	Credit Card	High	45

Data Table (Transactions) | New Data (from the VLOOKUP)) | Lookup Table

Both the lookup tables are *reference tables* (by definition) – the first is organised by rows, the second by columns

- **HLOOKUP:**

Banking_HLOOKUP_Complete_Example.xlsx

"Credit ratings are arranged horizontally across the top of the table. HLOOKUP uses the customer's rating code to retrieve the corresponding interest rate, credit limit, or other lending conditions."

a less natural storing

The table resulting from lookup

Customer	Rating Code	Interest Rate	Credit Limit (EUR)	Horizontal Lookup Table				
Client A	A	3.5%	50000	A	B	C	D	E
Client B	C	6.0%	20000	3.5%	4.5%	6.0%	8.0%	10.0%
Client C	B	4.5%	35000	50000	35000	20000	10000	5000
Client D	D	8.0%	10000					
Client E	E	10.0%	5000					

Data Table | Lookup Table

aka, master data

"VLOOKUP is useful when reference data is organized vertically by rows, while HLOOKUP is useful when categories are arranged horizontally across columns."

VLOOKUP in Banking — Enriching Transactions with Product Information

Business scenario: the transaction table contains only an Account Code and a Transaction Amount. The Product Master table stores the business attributes associated with each Account Code. VLOOKUP uses the Account Code as a key to retrieve Product Name, Risk Category, and Annual Fee.

Account Code	Transaction Amount (EUR)
C0001	1,250.00
C0004	8,200.00
C0003	35,000.00
C0002	5,400.00
C0005	980.00

Account Code	Product Name	Risk Category	Annual Fee (EUR)
C0001	Current Account	Low	60.00
C0002	Savings Account	Low	0.00
C0003	Personal Loan	Medium	120.00
C0004	Mortgage	Medium	250.00
C0005	Credit Card	High	45.00

tidy data

Account Code	Transaction Amount (EUR)	Product Name	Risk Category	Annual Fee (EUR)
C0001	1,250.00	Current Account	Low	60.00
C0004	8,200.00	Mortgage	Medium	250.00
C0003	35,000.00	Personal Loan	Medium	120.00
C0002	5,400.00	Savings Account	Low	0.00
C0005	980.00	Credit Card	High	45.00

Formula pattern: =VLOOKUP(lookup_value, table_array, column_index, FALSE) Example: =VLOOKUP(A15,$E$7:$H$11,2,FALSE)

HLOOKUP in Banking — Retrieving Lending Conditions from a Rating Matrix

Business scenario: the customer table contains a Credit Rating for each customer. The Lending Conditions matrix stores ratings horizontally across the first row, with the corresponding interest rate and maximum credit limit underneath. HLOOKUP uses the rating as a key to retrieve those conditions.

Customer ID	Customer Name	Credit Rating
CL001	Northwind Ltd	A
CL002	Blue River SA	BBB
CL003	Greenfield SRL	AA
CL004	Metro Retail Ltd	BB
CL005	Sunrise Group	AAA

Metric	AAA	AA	A	BBB	BB
Interest Rate	3.2%	3.6%	4.2%	5.1%	6.7%
Maximum Credit Limit (EUR)	500,000	350,000	250,000	150,000	75,000

Non-tidy data

Customer ID	Customer Name	Credit Rating	Interest Rate	Maximum Credit Limit (EUR)
CL001	Northwind Ltd	A	4.2%	250,000
CL002	Blue River SA	BBB	5.1%	150,000
CL003	Greenfield SRL	AA	3.6%	350,000
CL004	Metro Retail Ltd	BB	6.7%	75,000
CL005	Sunrise Group	AAA	3.2%	500,000

Formula pattern: =HLOOKUP(lookup_value, table_array, row_index, FALSE) Example: =HLOOKUP(C15,$F$6:$J$8,2,FALSE)

1 : N Relationship Between a Lookup Table and a Data Table

One record in the lookup table can be related to many records in the data table



Product (Lookup Table)

Product_ID	Product_Name	Category	Unit_Price
P001	Current Account	Deposits	0.00
P002	Savings Account	Deposits	0.00
P003	Personal Loan	Loans	250.00
P004	Mortgage	Loans	500.00
P005	Credit Card	Cards	50.00

The lookup table contains **one row per product**, with descriptive information.

Sales (Data Table)

Sale_ID	Product_ID	Sale_Date	Quantity	Amount
S001	P001	2026-01-03	1	0.00
S002	P003	2026-01-05	1	250.00
S003	P002	2026-01-07	2	0.00
S004	P005	2026-01-10	1	50.00
S005	P003	2026-01-12	1	250.00
S006	P004	2026-01-15	1	500.00
S007	P002	2026-01-18	1	0.00

The **data table** contains **many rows**, each transaction refers to **one** product using **Product_ID (foreign key)**. The same product can appear in many rows.

VLOOKUP and HLOOKUP **join** two tables in a **1:N relationship** in one direction only, from the N side to the 1 side: each detail row is enriched with the attributes of its master, like a left join, provided the master key is unique. A special case is the **master-detail relationship**, where a detail can only exist if its master exists: Excel does not enforce this referential integrity, but a #N/A returned by the lookup pinpoints exactly the orphan details.

Common data type problems in Excel

Many “analysis errors” are actually type errors.

x Type issues hiding in plain sight

fx `=ISTEXT(C4)`

CustomerID	Date	Income	Rate	Balance
00125	31/08/26	65,000	4.25%	12,500
00126	08/31/26	54.000	0,0425	8,200
00127	text	92,000	4.10%	—
00128	31/08/26	71k	4.35%	165,000

Symptoms

Sort order looks wrong · SUM ignores values · dates do not group.

Checks

ISTEXT · ISNUMBER · DATEVALUE · VALUE · NUMBERVALUE.

Prevention

Define schema, use Tables, validation, Power Query types and controlled imports.

```
=ISNUMBER(C2)      =ISTEXT(C2)      =DATEVALUE(B2)
=NUMBERVALUE(C2)
```

Chapter title



03 – Exploratory Data Analysis (EDA)

Data Science: we always start from a “business problem”

EDA is a disciplined conversation with the dataset: structure → distributions → relationships → problems → hypotheses.

Example: How to sell more of your favorite product?

How many do we sell on average?
Is this sales increase significant?

What are the typical customers?



Who else will buy ?

What else did they buy?

How many will they buy?

What is the best price?
What if we add some new shops?

Fonte: Jos van Dongen

La tecnica ML di soluzione

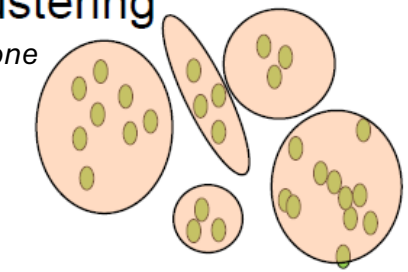
(1:1 posizionale con la slide precedente)

$$Y = f(reddito, zona) = k + w1xreddito + w2Xzona + reddito\text{^}2$$

Example: How to sell more of your favorite product?

Clustering

(segmentazione clienti)

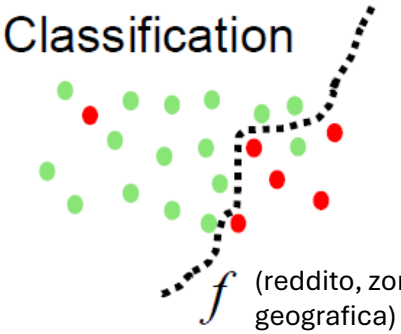


Reporting & Statistics

(nuovi termini: ML descrittivo, EDA)



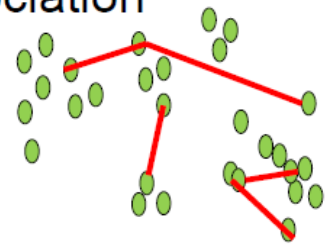
Classification



f (reddito, zona geografica)

Association

Nel marketing: Market Basket Analysis, cross-selling



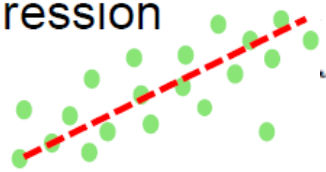


Operations Research



Analisi what-if

Regression



f(reddito)

Fonte: Jos van Dongen

Exploratory Data Analysis (EDA): learn the data before modelling it

EDA is a disciplined conversation with the dataset: structure → distributions → relationships → problems → hypotheses.

STRUCTURE

rows · columns · types · grain

DISTRIBUTIONS

centre · spread · shape

histograms

RELATIONSHIPS

correlation · groups · trends

DATA ISSUES

missing · outliers · inconsistencies

PATTERNS

segments · seasonality · anomalies

EDA output is not just charts

It is a set of findings: what the data represents, what looks suspicious, what relationships exist, and what should be investigated next.

```
EDA = ask → inspect → visualise → quantify → question → repeat
```

EDA in practice: one graphical example for each of the five lenses

Each lens asks a different question: how the table is organised, how values are distributed, how variables move together, where the problems are, and what broader regularities emerge.

STRUCTURE

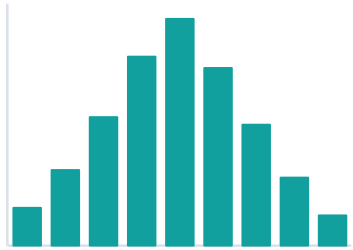
A table with fields and records

ID	Age	Region	Balan
C0	32	Nor	5,2
C0	41	We	8,1
C0	29	So	3,4
C0 04	55	Ea st	9,7 80

Example question:
Do we have one row per customer, clear column names and meaningful data types?

DISTRIBUTIONS

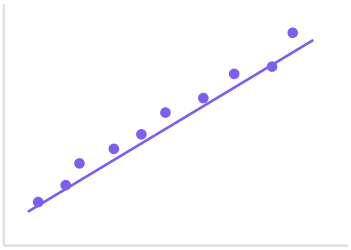
Histogram of a numeric variable



Example question:
How are transaction amounts distributed? Is the shape symmetric, skewed or multi-modal?

RELATIONSHIPS

Scatter plot between two variables



Example question:
Do income and account balance move together? A scatter plot reveals direction and strength.

DATA ISSUES

Missing values and outliers

Age	Income	Score
34	28,00	71
41	MV	69
39	31,50	70
36	999,999	72

Example question:
Where are the blanks, errors or extreme values that could distort averages, correlations, charts, and even predictive models?

PATTERNS

Trend / seasonality / anomaly



Example question:
Is there a repeated monthly pattern, an upward trend or a point that breaks the usual behaviour?

EDA = structure + distributions + relationships + data issues + patterns

A practical EDA workflow in Excel?

Use a repeatable sequence so important checks are not skipped.

1. Import / Table → 2. Check types → 3. Profile columns → 4. Summarise → 5. Plot → 6. Investigate issues

Import / Table

Convert the range to an Excel Table; preserve raw data.

Check types

Dates, numbers, categories, IDs, units.

Profile columns

Row counts, unique values, missing values, valid ranges.

Summarise

Mean, median, standard deviation, percentiles, category counts.

Plot

Histogram, box plot, scatter, line and bar charts.

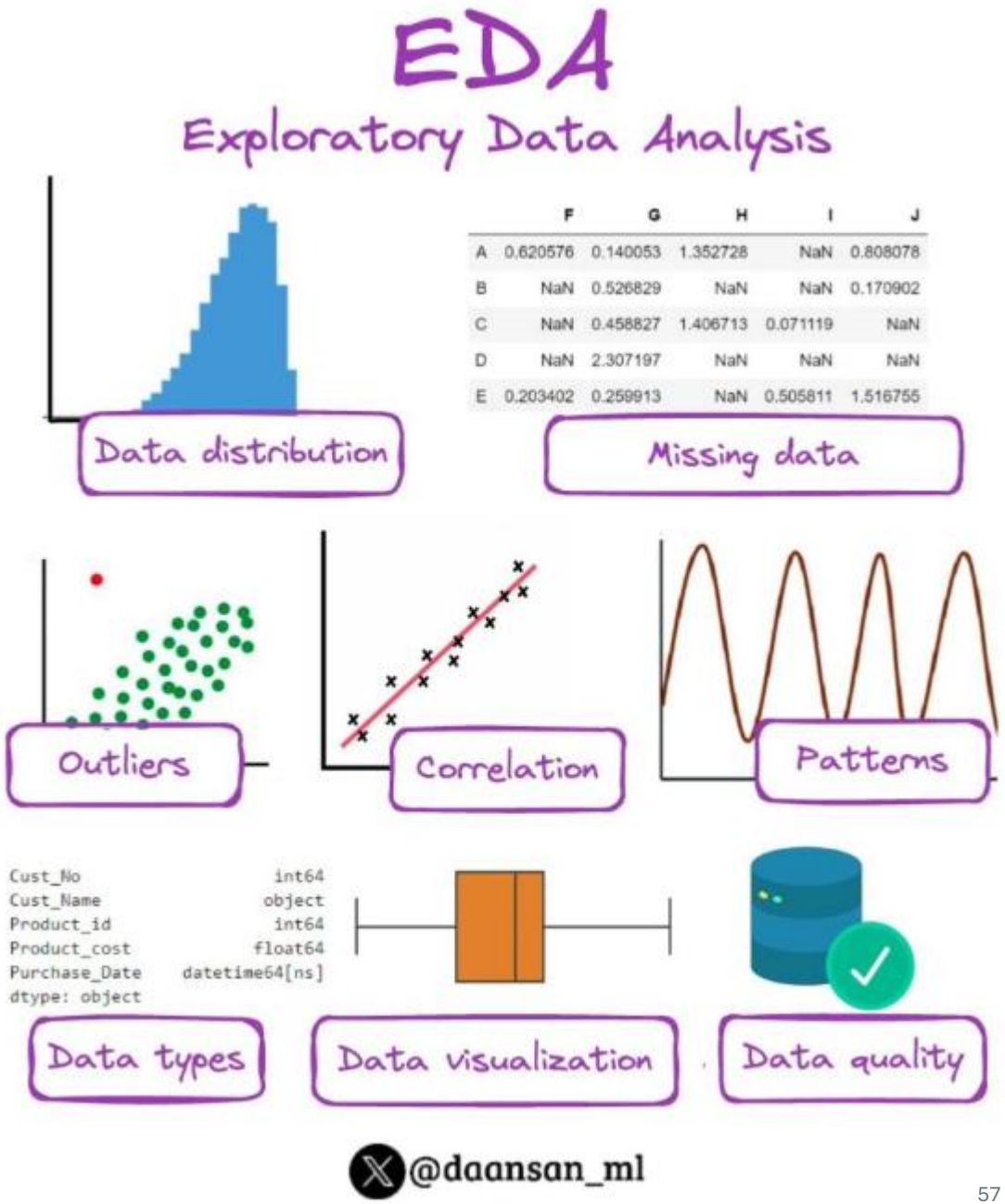
Investigate

Outliers, gaps, strange patterns, data-quality exceptions.

Best practice: keep “raw”, “clean”, “analysis” and “output” areas separate.

A better EDA workflow in Excel!

Use a repeatable sequence so important checks are not skipped.



Commentary on the "EDA – Exploratory Data Analysis" infographic

The infographic (by @daansan_ml) summarises the main activities of **Exploratory Data Analysis (EDA)**, the first phase of any data project. In EDA we examine a dataset before building models or drawing conclusions: we get to know its structure, check its quality and look for the patterns and anomalies that will guide the rest of the analysis. The infographic presents eight aspects, each with a small illustration.

1. Data distribution

The first panel shows a **histogram**, a chart that divides the values of a variable into intervals (bins) and draws a bar for each, with a height proportional to the number of observations in that interval. The histogram shows how the values are distributed: where they concentrate, how spread out they are and whether the shape is symmetric.

The example is **left-skewed** (negatively skewed): the bars rise gradually from the left, peak on the right and then drop sharply, so the long tail points to the left. It is the mirror image of what we typically see with income data, which are right-skewed, with most values low and a few very high ones. Recognising the shape matters because it decides which summary statistic to trust: with a skewed distribution, the **median** describes the typical value better than the mean.

2. Missing data

The second panel shows a table in which several cells contain **NaN** ("Not a Number"), the marker that pandas, the Python library for tabular data, uses for missing values. In EDA we need to find out **how many** values are missing, **where** they are (in which columns and rows) and, if possible, **why**. The reason matters: a value missing at random can be handled differently from one that is missing because of a systematic cause, such as customers who prefer not to declare their income. Depending on the case, we may remove the affected rows, fill the gaps with a suitable value (a process called **imputation**, for example with the median), or keep "missing" as information in its own right.

3. Outliers

The third panel is a **scatter plot** (one point per observation, with one variable on each axis) in which a single red point lies far from the green cloud. This is an **outlier**, an observation that differs markedly from the others. An outlier may be a data-entry error, which we should correct or remove, or a genuine but rare case, which we should keep and understand. EDA does not decide automatically: it flags the value so that we can investigate it.

4. Correlation

The fourth panel shows points aligned along an upward line. This is a **positive correlation**: as one variable increases, the other tends to increase too. Its strength is measured by the **correlation coefficient**, a number between −1 (perfect negative relationship) and +1 (perfect positive relationship), with 0 meaning no linear relationship. Two cautions are worth remembering: the coefficient only captures **linear** relationships, and **correlation does not imply causation**. Two variables may move together because a third factor influences both.

5. Patterns

The fifth panel shows a wave that repeats at regular intervals. This is a **periodic** or **seasonal pattern**, typical of **time series** (data recorded over time), such as sales that peak every December or branch traffic that rises every Monday. Identifying such regularities helps us explain the data and forecast future values.

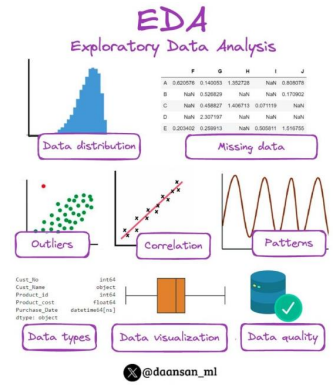
6. Data types

The sixth panel shows the output of the pandas `dtypes` attribute, which lists the **data type** of each column:

- `int64`: whole numbers;
- `float64`: decimal numbers;
- `object`: text (in pandas, strings are stored as the generic type *object*);
- `datetime64[ns]`: dates and times.

Checking types is one of the first steps of EDA, because a wrong type silently breaks later calculations. A date read as text cannot be sorted chronologically, and a number read as text cannot be summed.

The example also contains a subtle trap. `Cust_No` and `Product_id` are stored as `int64`, but they are **identifiers**, not quantities. Computing their mean or including them in a model as numbers would be meaningless. In EDA we should treat them as labels, for example by converting them to text or to the `category` type.



7. Data visualization

The seventh panel shows a **box plot**, a compact chart of a distribution. The box spans from the first quartile (Q1, the value below which 25% of observations fall) to the third quartile (Q3, 75%); the line inside it is the median; the lines extending from the box, called **whiskers**, reach the most extreme values that are not considered outliers. Points beyond the whiskers, usually more than 1.5 times the box height (the **interquartile range**) away from it, are drawn separately as potential outliers.

More generally, visualisation is the thread that runs through all of EDA: histograms, scatter plots, box plots and line charts often reveal in seconds what a table of numbers would hide.

8. Data quality

The last panel sums up the purpose of the whole process: making sure the data are **fit for use**. Data quality has several dimensions: **completeness** (no important values missing), **accuracy** (values that reflect reality), **consistency** (the same information recorded in the same way everywhere), **validity** (values within plausible ranges and formats) and **uniqueness** (no duplicate records). An annual income of $12, a customer aged 150 or two identical rows for the same customer are all quality issues that EDA should bring to light.

Overall assessment

The infographic is a good **checklist** and an effective visual summary for introducing EDA. A few remarks help put it in context:

- **The order is not the working order.** In practice, we usually start with **data types** and **data quality**, because every later analysis depends on them, and only then move to distributions, outliers and relationships.
- **The steps are interconnected, not separate boxes.** A box plot is used to find outliers, outliers affect correlation, and missing data affect every statistic we compute.
- **Some aspects are missing.** The infographic does not mention **duplicates**, the analysis of **categorical variables** (frequencies and counts of each category), or the distinction between **univariate** analysis (one variable at a time, like the descriptive statistics of Income and Age) and **bivariate** analysis (two variables together, like correlation).
- **EDA is iterative.** What we discover at one step often sends us back to an earlier one: finding an outlier may reveal a data-entry problem, which in turn raises questions about data quality in the whole column.

1. Import / Table

CSV files: simple tabular data in plain text

CSV stands for Comma-Separated Values: a lightweight format for storing and exchanging structured data.

customers.csv

```
Customer_ID,Age,Income,Balance,Default
CL0001,42,65000,12500,No
CL0002,35,54000,8200,No
CL0003,61,92000,45500,No
CL0004,29,45000,3200,Yes
```

Each line is a row. Separators divide fields into columns. The first row usually contains column names.

very convenient for import / export

Plain text

No formulas, charts, formatting or multiple worksheets — only values.

Tabular structure

Rows represent observations; fields represent variables.

Delimiter matters

Despite the name, CSV files may use commas, semicolons or other delimiters depending on locale and source.

CSV is a data format, not a spreadsheet workbook.

Why CSV is useful in Data Science

CSV is widely used as a simple interchange format between analytical tools, databases and programming environments.

Portable

Readable by Excel, Python, R, SQL tools, BI platforms and many other applications.

Transparent

The content can be inspected with a text editor: there is little hidden structure.

Easy to exchange

Useful for exporting datasets from systems and moving tabular data between tools.

Automation-friendly

Straightforward to read and write in scripts and data pipelines.

Lightweight

No workbook features are stored, so the format focuses on the data itself.

Good for datasets

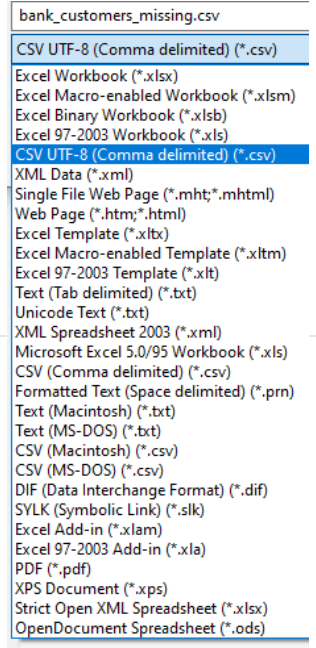
A natural representation for rows × columns used in EDA and Machine Learning.

Typical Data Science flow: source system → CSV → Excel / Python / R → analysis

CSV

CSV ↔ Excel: how to convert them in both directions

Import a CSV when you want to analyse it in Excel; export to CSV when you need a simple, portable data file.



1. Import / Table

CSV → Excel

1. In Excel, choose Data → From Text/CSV.
2. Select the .csv file.
3. Check delimiter, encoding and the previewed columns.
4. Choose Load to bring the data into Excel.
5. Save as .xlsx if you want to preserve Excel features.

Excel → CSV

1. Keep the worksheet you want to export active.
2. Choose File → Save As.
3. Select CSV UTF-8 (Comma delimited) (*.csv) when appropriate.
4. Save the file and confirm Excel's warnings.
5. Re-open or inspect the CSV if delimiter/encoding is critical.

Important: CSV stores one flat table only — formulas, formatting, charts, PivotTables and additional worksheets are not preserved.

CSV files across locales: when commas stop being simple

A CSV file is plain text. The delimiter, decimal symbol, dates and encoding may be interpreted differently by Excel depending on regional settings.

KEY IDEA **Never assume that "CSV" means the same parsing rules on every computer. Import settings must be explicit.**

1 What can change with locale?

- **Field delimiter** — Comma (,), semicolon (;), tab, etc.
- **Decimal symbol** — 3.75 in EN/US vs 3,75 in many EU locales
- **Thousands separator** — 25,900.50 vs 25.900,50
- **Date interpretation** — 03/04/2026 can mean Mar 4 or Apr 3
- **Text encoding** — UTF-8 avoids many accented-character problems

2 Example: the same data, different conventions

Source CSV created in an English/US environment

```
Customer_ID,Income,Rate,Region
C001,25900.50,3.75,Turin
C002,17200.00,4.10,Milan
```

Double-click / automatic opening

POSSIBLE RESULT

- all fields in one column
- decimal values split incorrectly
- numbers stored as text
- dates silently reinterpreted

Explicit import settings

SAFE RESULT

- delimiter set explicitly
- decimal locale set correctly
- columns typed as expected
- row/column counts validated

3 Safe Excel import workflow

1. **Data → From Text/CSV**
   Do not rely on double-click when locale is uncertain.
2. **Set delimiter explicitly**
   Choose comma, semicolon, tab, etc.
3. **Check File Origin / UTF-8**
   Preserve accented characters and symbols.
4. **Set locale for numbers & dates**
   Especially decimal comma/dot and date order.
5. **Validate before analysis**
   Check column count, data types, totals and sample rows.

For shared workflows: document the delimiter, encoding and locale.

CSV import in Excel: data type inference

Why the same CSV file can be interpreted differently across tools, locales and workflows

WHAT EXCEL DOES

CSV file (raw text values) → **Excel import engine scans sample values** → **Excel infers column data types** → **Worksheet result (numbers, dates, text)**

Typical inference examples

CSV values	Inferred as	Possible issue
"00123"	Number	Leading zeros lost
"13/09/2026"	Date	Locale ambiguity
"1,234"	Number/Text	Decimal vs thousands
"TRUE"	Logical/Text	Depends on parser
"NA" / blank / null	Missing/Text	May not mean the same thing
"CL00015"	Text	Usually OK

Key idea: a CSV file stores text separated by delimiters; data types are reconstructed during import.

Why this matters

The same column may be imported as text, number or date.
Wrong inference can distort sorting, filtering, formulas and charts.
IDs, ZIP codes and account numbers can be damaged by numeric conversion.
Cross-language imports may change decimal separators and date interpretation.

Safer Excel workflow

Use Data → From Text/CSV instead of double-clicking the file.
Review delimiter, file origin / encoding, locale, and detected data type.
Force Text for identifiers; verify dates, decimals and missing-value markers.

Takeaway: CSV files do not carry rich type metadata. Excel infers types during import, so analysts must review the import settings before trusting the data.

Inference of Data Types: how to interpret results

Excel infers column data types

I imported a csv file into Excel, and during the process I choose something like "Infer data types from the first 200 lines" (the default), but, now, if i look at cell formats I see always "general". Why?

Working out how to phrase the explanatory response.

Excel and Power Query use **two different concepts** that both sound like "type", and the dialog you saw only affects one of them.

1. Data type (Power Query level). When we import a CSV with *Data › From Text/CSV*, Excel uses **Power Query**, its import and transformation engine. The option "Data Type Detection: Based on first 200 rows" tells Power Query to look at those rows and assign a *data type* to each column: *Whole Number* for Income, Limit and Age, *Text* for Gender and Ethnicity, and so on. These types live inside the **query**, the saved recipe of import steps, not inside the cells.

2. Number format (cell level). What we see in *Format Cells › Number* is only a **display format**, meaning how a value is shown (thousands separator, decimals, currency symbol, date layout). It does not define what the value *is*. "General" means "no specific display rule, show the value as it is".

When Power Query loads its result into the sheet, it writes the actual values (real numbers for numeric columns, real text for text columns). It applies a display format only when the type needs one to be readable, typically *Date*, *Time* or *Date/Time*, because otherwise we would see serial numbers like 45123. For whole numbers and text there's nothing to format, so the cells stay "General". The detection worked. It just doesn't show up in that dialog.

How we can check that the types were really applied

In the sheet, numbers are right-aligned and text is left-aligned by default. In your screenshot, Income, Limit, Age and Balance are right-aligned (numbers), while Gender, Student and Ethnicity are left-aligned (text). We can also confirm with a formula such as `=ISNUMBER(C2)`, which should return TRUE, or `=TYPE(C2)`, which returns 1 for a number and 2 for text.

To see the types Power Query inferred, we can go to *Data › Queries & Connections*, double-click the query, and look at the icon in each column header of the Power Query Editor: **123** is whole number, **1.2** is decimal number, **ABC** is text, and a calendar icon is a date. The step called *Changed Type* in the *Applied Steps* panel on the right is exactly the result of the "first 200 rows" detection.

If we also want a display format

We can apply it normally with *Format Cells*, for example Number with a thousands separator for Income. In a table loaded by Power Query, formats we set manually are generally kept when we refresh the query, as long as the column keeps its name.

A side note on your file: `Column1` is the same as `ID`. It's almost certainly the row index written by the tool that exported the CSV (pandas does this with `to_csv` unless we pass `index=False`). We can remove it in Power Query with *Remove Columns*, or avoid it at export time.

First look: create a dataset profile

Before any deep analysis, answer a small set of basic questions.

Column profile

fx `=COUNTBLANK(Table1[Income])`

Field	Type	Missing	Unique	Min	Max
Age	Numeric	0	41	18	86
Income	Numeric	37	1,204	12k	420k
Segment	Category	0	4	—	—
Balance	Numeric	4	8,951	-1.2k	650k
Default	Binary	0	2	0	1

How to show the Total Row

1. Click any cell inside the table. This makes the **Table Design** tab appear in the ribbon (in some versions it's called **Table Tools → Design**).
2. In the *Table Style Options* group, tick **Total Row**.

Adding a total to another column

1. Click the cell in the Total Row directly under the column we want to total.
2. A small dropdown arrow appears on the right side of that cell. It is only visible while the cell is selected, which is why it's easy to miss.
3. Open the dropdown and choose the calculation we want: Sum, Average, Count, Max, Min, and so on.

How many rows?

ROWS, COUNTA or Table totals.

#rows <> null

How many unique values?

UNIQUE + ROWS (modern Excel) or PivotTable.

```
=ROWS(UNIQUE(credit[Income]))
```

What is missing? inconsistency

COUNTBLANK, ISBLANK, filters and conditional formatting.

TRUE/FALSE

```
=ROWS(Table1)    =COUNTA(Table1[CustomerID])
=COUNTBLANK(Table1[Income])
```

Descriptive statistics: centre, spread and shape

Summary statistics compress a distribution into a few interpretable numbers.

CV (Coefficient of Variation, defined as $\frac{\sigma}{\mu}$), is an ABSOLUTE measure of dispersion

Mean

measure of centrality

Average level

`=AVERAGE(range)`

Median

measure of centrality

Middle value

`=MEDIAN(range)`

Std. deviation

measure of relative dispersion (there are others)

Typical spread

`=STDEV.S(range)`

Min / Max

Observed range

`=MIN(range) / =MAX(range)`

Quartiles

25th / 50th / 75th percentiles

`=QUARTILE.INC(range,k)`

Important

Mean and standard deviation can be heavily affected by outliers. Always compare with median and quartiles.

The “Descriptive Statistics” option in Excel Analysis ToolPak can generate a compact summary table.

A selection trick in descriptive statistics

Selecting a table column as input for Descriptive Statistics

1. Open **Data → Data Analysis → Descriptive Statistics**.
2. Click in the **Input Range** box, the field where we specify which cells to analyse.
3. On the worksheet, click the **header cell** of the column we want to analyse (for example, the cell containing "Income").
4. Press **Ctrl+Shift+Down Arrow.** Excel extends the selection from the header down to the last filled cell in the column, so we don't need to scroll. In this shortcut, **Ctrl+Down Arrow** means "jump to the last filled cell", and adding **Shift** means "select everything along the way".
5. In the dialog box, tick **Labels in First Row**. This tells Excel that the first cell is the column name, not a data value, and it uses that name as the title of the output.
6. Choose where to place the results (**Output Range**, **New Worksheet Ply** or **New Workbook**), tick **Summary statistics**, and optionally **Confidence Level for Mean**. Then click **OK**.

Watch out for empty cells

The shortcut stops at the **first empty cell** it meets. If a value is missing in the middle of the column, the selection ends there and the statistics are computed on only part of the data, with no warning.

After pressing the shortcut, we check the range written in the Input Range box (for example `$D$1:$D$401`) and make sure it reaches the last row of the data. If it stops early, we press **Ctrl+Shift+Down Arrow** again to continue past the gap.

Alternative: typing the range

If we already know where the data ends, we can type the range directly into the Input Range box, for example `$D$1:$D$401`. To find the last used row quickly, we click any cell and press **Ctrl+End**: Excel moves to the last used cell of the sheet, and we read its row number.

Comparing portfolio risk: standard deviation vs coefficient of variation

When portfolios are recorded on very different numeric scales, standard deviation alone can mislead. CV makes relative risk comparable.

Standard deviation (SD)

Absolute measure of volatility.

- Same unit as the data
- Changes when the scale changes
- Not suitable for comparing portfolios measured on very different scales

SD = spread around the mean

Coefficient of variation (CV)

Relative measure of volatility.

- Unit-free
- CV = SD / Mean
- Suitable for comparing relative risk across different scales

$$CV = \frac{SD}{Mean}$$

Why SD can mislead

A larger SD may simply reflect a larger numeric scale, not higher relative risk.

Why CV helps

CV standardizes volatility by the mean, so portfolios on different scales become comparable.

Decision rule

Set an acceptable risk threshold on CV, for example 25%.

Teaching takeaway

Use SD to describe variability within one scale.
Use CV to compare relative risk across scales.

Example: three portfolios observed over one year

Portfolio	Mean return	Standard deviation	CV	Risk status
Portfolio A	0.8	0.16	20%	Acceptable
Portfolio B	8	1.6	20%	Acceptable
Portfolio C	80	24	30%	Not acceptable

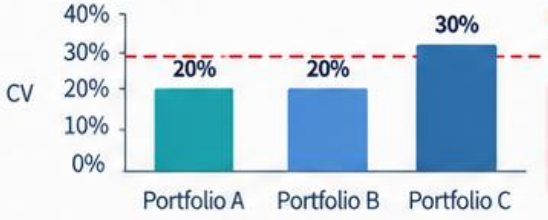
The SD values look very different because the scales are different. A and B actually have the same relative risk.

If you compare SD only...



Misleading across scales.

If you compare CV...



CV reveals which portfolio is relatively riskier.

Key message: standard deviation is an absolute measure of volatility; coefficient of variation is a relative measure and is better for cross-portfolio risk comparison.



Descriptive statistics of Income (Credit dataset, ISLR)

The Income column of the Credit dataset contains **400 observations**, one for each customer, with values expressed in dollars. The table below summarises the output of Excel's *Descriptive Statistics* tool.

Income	
Mean	40672.31
Standard Error	1776.185
Median	29715
Mode	23793
Standard Deviation	35523.7
Sample Variance	1.26E+09
Kurtosis	3.312689
Skewness	1.760822
Range	186622
Minimum	12
Maximum	186634
Sum	16268924
Count	400
Confidence Level(95.0%)	3491.851

Central tendency: where the "typical" value lies

The **mean** (the arithmetic average) is **$40,672**. The **median** (the middle value when the data are sorted, so that half of the customers earn less and half earn more) is **$29,715**. The **mode** (the most frequent value) is **$23,793**.

The mean is about $11,000 higher than the median. This gap is a first sign that a small number of very high incomes pull the average upwards. For this reason, the median is a better description of the typical customer's income than the mean.

The mode carries little information here. Income is a continuous variable, so almost every value is unique, and the mode simply reports a value that happens to appear at least twice. It may even point to a duplicated record worth checking.

Dispersion: how spread out the values are

The **standard deviation** (the typical distance of the values from the mean) is **$35,524**. The **sample variance** is its square, about 1,260,000,000, measured in squared dollars. It is hard to interpret directly and mainly serves as an input to other calculations. In the Excel output it appears as **1.26E+09**, which is **scientific notation**: "E+09" means "multiplied by 10 to the ninth power". Excel uses this format automatically when a column is too narrow to display the full number.

The **range** (maximum minus minimum) is **$186,622**, from a minimum of **$12** to a maximum of **$186,634**.

The standard deviation is almost as large as the mean. Their ratio, the **coefficient of variation**, is about 0.87 (35,524 / 40,672), which indicates very high variability: customers' incomes differ enormously from one another.

Shape of the distribution

The **skewness** (a measure of asymmetry) is **1.76**. A positive value means the distribution has a long tail to the right: most customers are concentrated at low-to-medium incomes, while a few earn much more. This is consistent with the mean being higher than the median, and it is the typical shape of income data.

The **kurtosis** is **3.31**. Excel's KURT function reports *excess* kurtosis, which assigns 0 to a normal distribution (rather than 3). A value of 3.31 therefore indicates **heavy tails**: extreme values are considerably more frequent than a normal distribution would predict.

Extreme values worth investigating

The **maximum** of $186,634 lies about 4.1 standard deviations above the mean ((186,634 − 40,672) / 35,524). It is a strong candidate for an **outlier**, a value far from the rest of the data. It is not necessarily an error, since high earners do exist, but it deserves a closer look.

The **minimum** of **$12** is much more suspicious. An annual income of twelve dollars is not realistic for a bank customer. It is more likely a data-entry error or a value recorded in a different unit (for example in thousands of dollars, which would mean $12,000). This value should be verified before any further analysis.

Precision of the mean: the confidence interval

Our 400 customers are a **sample** drawn from a larger **population**. The sample mean of $40,672 is therefore an estimate of the true population mean, which we do not know. The last two rows of the table tell us how precise that estimate is.

The **standard error** measures how much the sample mean would fluctuate if we drew many different samples from the same population. It is the standard deviation divided by the square root of the number of observations:

$$SE = \frac{s}{\sqrt{n}} = \frac{35{,}523.7}{\sqrt{400}} = \frac{35{,}523.7}{20} = 1{,}776.19$$

The row labelled **Confidence Level (95.0%)**, despite its name, reports the **margin of error**: the amount we add to and subtract from the mean to obtain the 95% confidence interval. It is the standard error multiplied by a **critical value** taken from **Student's t distribution** (a distribution similar to the normal, used when the population standard deviation is estimated from the sample). With 399 **degrees of freedom** ($n - 1$) and a 5% risk split equally between the two tails, the critical value is about 1.966:

$$\text{margin} = 1.966 \times 1{,}776.19 \approx 3{,}491.85$$

The **95% confidence interval** for the mean is therefore:

$$40{,}672.31 \pm 3{,}491.85 \;\rightarrow\; [37{,}180;\; 44{,}164]$$

We can say, with 95% confidence, that the average income of the population lies between **$37,180** and **$44,164**. More precisely: if we repeated the sampling many times and computed the interval each time in the same way, about 95% of those intervals would contain the true mean.

This interval concerns the **average** income, not individual customers. It does not mean that 95% of customers earn between $37,180 and $44,164; given the large standard deviation, most customers fall well outside that range.

Although the distribution is clearly skewed, with 400 observations the interval remains reliable thanks to the **central limit theorem**: in large samples, the sample mean is approximately normally distributed even when the underlying data are not.

The same margin of error can be obtained with the formula `=CONFIDENCE.T(0.05, 35523.7, 400)`, where 0.05 is the 5% risk corresponding to 95% confidence.

Practical takeaways

1. We should report the **median** alongside the mean when describing the typical customer's income, because the mean is inflated by a few very high values.
2. We should **inspect the extremes**, especially the $12 minimum, which is very likely an error, and the $186,634 maximum.
3. We should **visualise the distribution** with a histogram or a box plot to see the long right tail and the outliers directly.
4. If Income is used in a linear model, a **log transformation** (taking the logarithm of each value) is worth considering. It compresses the long right tail, makes the distribution more symmetric, and reduces the influence of the highest incomes on the results.



Descriptive statistics of Age (Credit dataset, ISLR)

The Age column of the Credit dataset contains **400 observations**, one for each customer, with values expressed in years. The table below summarises the output of Excel's *Descriptive Statistics* tool.

	Age
Mean	55.6675
Standard Error	0.862490338
Median	56
Mode	66
Standard Deviation	17.24980676
Sample Variance	297.5558333
Kurtosis	-1.06557969
Skewness	0.011495909
Range	75
Minimum	23
Maximum	98
Sum	22267
Count	400
Confidence Level(95.0%)	1.695593298

Central tendency: where the "typical" value lies

The **mean** (the arithmetic average) is **55.7 years**. The **median** (the middle value when the data are sorted, so that half of the customers are younger and half are older) is **56 years**. The **mode** (the most frequent value) is **66 years**.

The mean and the median are practically identical. This is a first sign that the distribution is **symmetric**: there is no group of extreme values pulling the average in one direction. Unlike skewed variables, here the mean is a good description of the typical customer's age.

The mode should not be over-interpreted. Age is recorded in whole years, and with 400 customers spread over about 75 possible ages, each age appears only a handful of times on average. The mode of 66 simply reflects which age happens to occur a few more times than the others; it lies ten years above the mean and does not describe a typical customer.

Dispersion: how spread out the values are

The **standard deviation** (the typical distance of the values from the mean) is **17.2 years**. The **sample variance** is its square, **297.6**, measured in squared years; it is hard to interpret directly and mainly serves as an input to other calculations.

The **range** (maximum minus minimum) is **75 years**, from a minimum of **23** to a maximum of **98**.

The **coefficient of variation** (standard deviation divided by the mean) is about 0.31 (17.25 / 55.67). This indicates moderate variability: customers' ages differ considerably, but far less, in relative terms, than variables such as income typically do.

Shape of the distribution

The **skewness** (a measure of asymmetry) is **0.01**, practically zero. The distribution is almost perfectly symmetric around its centre, which is consistent with the mean and the median being nearly equal.

The **kurtosis** is **−1.07**. Excel's KURT function reports *excess* kurtosis, which assigns 0 to a normal distribution. A clearly negative value indicates a **flat distribution with light tails**: compared with a normal distribution, there is no pronounced peak in the middle, and extreme values are rarer. As a reference point, a **uniform distribution** (where every value in the range is equally likely) has an excess kurtosis of −1.2. A value of −1.07 is close to it, suggesting that customers' ages are spread **fairly evenly** between the early twenties and the late nineties, rather than clustered around the mean.

Extreme values

To judge whether the extremes are unusual, we measure their distance from the mean in standard deviations, a quantity called the **z-score**:

- the **maximum** of 98 years has a z-score of about +2.5 ((98 − 55.67) / 17.25);
- the **minimum** of 23 years has a z-score of about −1.9 ((23 − 55.67) / 17.25).

A common rule of thumb flags as potential outliers the values with a z-score beyond ±3. Neither extreme reaches that threshold, so the Age column shows **no outliers**. Both values are also plausible: a 98-year-old customer is rare but possible, and the absence of customers younger than 23 may simply reflect the bank's customer base or its eligibility criteria.

Precision of the mean: the confidence interval

Our 400 customers are a **sample** drawn from a larger **population**. The sample mean of 55.7 years is therefore an estimate of the true population mean, which we do not know. The last two rows of the table tell us how precise that estimate is.

The **standard error** measures how much the sample mean would fluctuate if we drew many different samples from the same population. It is the standard deviation divided by the square root of the number of observations:

$$SE = \frac{s}{\sqrt{n}} = \frac{17.25}{\sqrt{400}} = \frac{17.25}{20} = 0.86$$

The row labelled **Confidence Level (95.0%)**, despite its name, reports the **margin of error**: the amount we add to and subtract from the mean to obtain the 95% confidence interval. It is the standard error multiplied by a **critical value** taken from **Student's t distribution** (a distribution similar to the normal, used when the population standard deviation is estimated from the sample). With 399 **degrees of freedom** ($n - 1$) and a 5% risk split equally between the two tails, the critical value is about 1.966:

$$\text{margin} = 1.966 \times 0.8625 \approx 1.70$$

The **95% confidence interval** for the mean is therefore:

$$55.67 \pm 1.70 \;\rightarrow\; [53.97;\ 57.36]$$

We can say, with 95% confidence, that the average age of the population lies between **54.0 and 57.4 years**. More precisely: if we repeated the sampling many times and computed the interval each time in the same way, about 95% of those intervals would contain the true mean.

This interval concerns the **average** age, not individual customers. It does not mean that 95% of customers are between 54 and 57 years old; with a standard deviation of 17 years, most customers fall well outside that range.

The interval is narrow, only about 3.4 years wide, because the variability of Age is moderate and the sample is reasonably large. The estimate of the mean is therefore quite precise.

The same margin of error can be obtained with the formula `=CONFIDENCE.T(0.05, 17.2498, 400)`, where 0.05 is the 5% risk corresponding to 95% confidence.

Practical takeaways

1. The **mean is a reliable summary** of the typical customer's age, since the distribution is symmetric and mean and median coincide.
2. There are **no outliers** to investigate: both extremes lie within a plausible range and within about 2.5 standard deviations of the mean.
3. We should **visualise the distribution** with a histogram to confirm its flat, nearly uniform shape, which the negative kurtosis suggests. A flat shape is not a problem, but it is worth knowing: the customer base covers all age groups fairly evenly rather than being concentrated around middle age.
4. If Age is used in a linear model, **no transformation is needed** to correct asymmetry or extreme values; the variable can be used as it is.



Descriptive statistics of Balance (Credit dataset, ISLR)

The Balance column of the Credit dataset contains **400 observations**, one for each customer. It records each customer's **average credit card balance**, that is, the amount of credit card debt outstanding, expressed in dollars. The table below summarises the output of Excel's *Descriptive Statistics* tool.

Balance	
Mean	520.02
Standard Error	22.99
Median	459.50
Mode	0.00
Standard Deviation	459.76
Sample Variance	211378.23
Kurtosis	-0.53
Skewness	0.58
Range	1999.00
Minimum	0.00
Maximum	1999.00
Sum	208006.00
Count	400.00
Confidence Level(95.0%)	45.19

Central tendency: where the "typical" value lies

The **mean** (the arithmetic average) is **$520**. The **median** (the middle value when the data are sorted, so that half of the customers have a lower balance and half a higher one) is **$460**. The **mode** (the most frequent value) is **$0**.

The mean is about $60 higher than the median. The gap is much smaller, in relative terms, than for a strongly skewed variable such as income, but it still indicates that higher balances pull the average slightly upwards.

Unlike other continuous variables, here the **mode is genuinely informative**. A mode of zero means that the single most common situation is a customer with **no outstanding balance**: someone who pays off the card in full or does not use it. This tells us that the distribution is a combination of two groups: a cluster of customers exactly at zero, and customers with a positive balance spread over a wide range. To measure how large the first group is, we can count the zeros with `=COUNTIF(range, 0)`.

Dispersion: how spread out the values are

The **standard deviation** (the typical distance of the values from the mean) is **$460**. The **sample variance** is its square, **211,378**, measured in squared dollars; it is hard to interpret directly and mainly serves as an input to other calculations.

The **range** (maximum minus minimum) is **$1,999**, from a minimum of **$0** to a maximum of **$1,999**.

The **coefficient of variation** (standard deviation divided by the mean) is about 0.88 (460 / 520). This indicates **high variability**: balances differ greatly from one customer to another, which is expected when the data include many customers at zero alongside others with substantial debt.

Shape of the distribution

The **skewness** (a measure of asymmetry) is **0.58**. A moderately positive value indicates a **mild right skew**: the distribution has a somewhat longer tail towards high balances. This is consistent with the mean being a little higher than the median.

The **kurtosis** is **−0.53**. Excel's KURT function reports *excess* kurtosis, which assigns 0 to a normal distribution. A moderately negative value indicates a distribution that is **somewhat flatter, with lighter tails**, than a normal one: values are spread over a broad range rather than tightly concentrated around the centre, and very extreme values are relatively rare.

Both measures should be read with care here. Because of the cluster of zeros, the distribution is not a single smooth shape but a **spike at zero** followed by a spread of positive values. Skewness and kurtosis summarise this mixture in a single number and may hide its structure. A **histogram** is the best way to see it.

Extreme values

To judge whether the extremes are unusual, we measure their distance from the mean in standard deviations, a quantity called the **z-score**:

- the **maximum** of $1,999 has a z-score of about +3.2 ((1,999 − 520) / 460);
- the **minimum** of $0 has a z-score of about −1.1 ((0 − 520) / 460).

A common rule of thumb flags as potential outliers the values with a z-score beyond ±3. The maximum **just exceeds** that threshold, so it is a **borderline outlier**. It is not necessarily an error, but it deserves a check. Its value, one dollar short of $2,000, is also worth noting: it could be linked to the customer's **credit limit**, and comparing it with the Limit column of the dataset would tell us whether this customer is close to the maximum allowed.

The **minimum** of $0 is not an anomaly but a legitimate and frequent value. From a data-quality perspective, however, we should confirm that these zeros are **true zeros** (no outstanding balance) and not **missing values recorded as 0**, a common problem in real datasets that would distort every statistic.

Precision of the mean: the confidence interval

Our 400 customers are a **sample** drawn from a larger **population**. The sample mean of $520 is therefore an estimate of the true population mean, which we do not know. The last two rows of the table tell us how precise that estimate is.

The **standard error** measures how much the sample mean would fluctuate if we drew many different samples from the same population. It is the standard deviation divided by the square root of the number of observations:

$$SE = \frac{s}{\sqrt{n}} = \frac{459.76}{\sqrt{400}} = \frac{459.76}{20} = 22.99$$

The row labelled **Confidence Level (95.0%)**, despite its name, reports the **margin of error**: the amount we add to and subtract from the mean to obtain the 95% confidence interval. It is the standard error multiplied by a **critical value** taken from **Student's t distribution** (a distribution similar to the normal, used when the population standard deviation is estimated from the sample). With 399 **degrees of freedom** ($n - 1$) and a 5% risk split equally between the two tails, the critical value is about 1.966:

$$\text{margin} = 1.966 \times 22.99 \approx 45.19$$

The **95% confidence interval** for the mean is therefore:

$$520.02 \pm 45.19 \;\rightarrow\; [474.83;\ 565.21]$$

We can say, with 95% confidence, that the average balance of the population lies between **$475 and $565**. More precisely: if we repeated the sampling many times and computed the interval each time in the same way, about 95% of those intervals would contain the true mean.

This interval concerns the **average** balance, not individual customers. It does not mean that 95% of customers have a balance between $475 and $565; with a standard deviation of $460 and many customers at zero, most customers fall well outside that range.

The **relative margin of error** (margin divided by mean) is about **±8.7%** (45.19 / 520.02). The average balance is therefore estimated with moderate precision, limited by the high variability of the data.

Although the distribution is not normal, with 400 observations the interval remains reliable thanks to the **central limit theorem**: in large samples, the sample mean is approximately normally distributed even when the underlying data are not.

The same margin of error can be obtained with the formula `=CONFIDENCE.T(0.05, 459.76, 400)`, where 0.05 is the 5% risk corresponding to 95% confidence.

Practical takeaways

1. We should **count the customers with a zero balance** and describe the two groups separately. The statistics of customers with a positive balance (for example with `=AVERAGEIF(range, ">0")`) give a clearer picture of how much those who carry a balance actually owe.
2. We should **verify that the zeros are genuine** and not missing values recorded as 0.
3. We should **check the maximum of $1,999**, a borderline outlier, and compare it with the customer's credit limit.
4. We should **visualise the distribution** with a histogram, which will show the spike at zero that the summary statistics cannot fully convey.
5. If Balance is the **target variable** of a model, as it often is in ISLR's regression examples, the cluster of zeros deserves attention. A plain linear regression may predict **negative balances** for some customers, which is impossible. A common alternative is a **two-part approach**: first predicting *whether* a customer carries a balance, then *how much* among those who do.

CREDIT DATASET (ISLR) · 400 CUSTOMERS

What income, age and balance tell us about our customers

Business implications of three descriptive analyses

link to the PDF

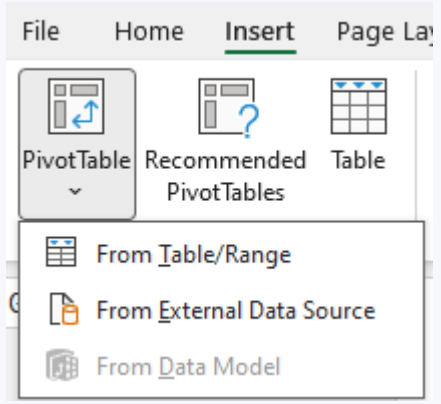
PivotTables: fast multivariate EDA without writing formulas

PivotTables help answer “how does a metric differ across groups?”

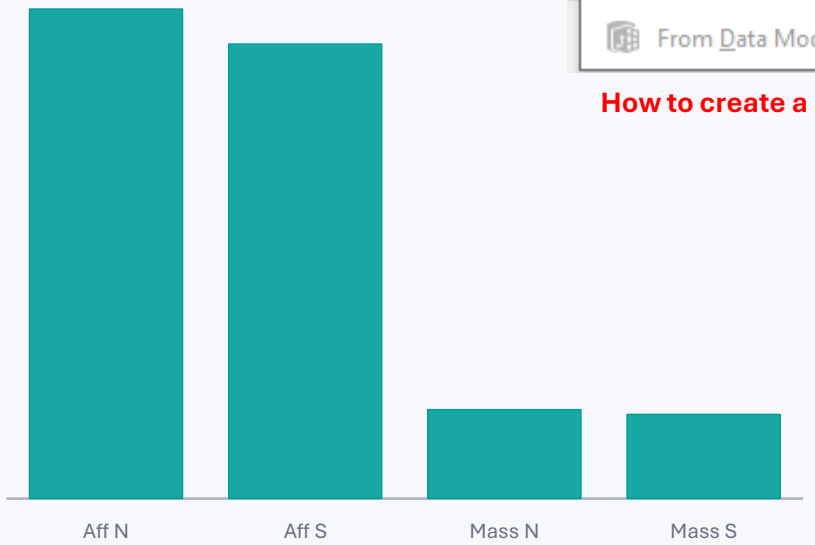
PivotTable output

fx Average of Balance / Count of CustomerID

Segment	Region	Customers	Avg Balance	Default %
Affluent	North	1,240	46,200	0.8%
Affluent	South	980	42,900	1.1%
Mass	North	4,650	8,400	2.9%
Mass	South	4,210	7,950	3.4%



How to create a pivot table



Average balance by segment & region

PivotTables are a bridge between raw tables and clear business questions.

The easiest way: select the whole table and choose columns afterwards

A PivotTable works best when its source is the **whole table**. We don't pick individual columns at this stage: once the PivotTable is created, the **PivotTable Fields** pane (the panel on the right listing every column of the source) lets us tick or drag only the columns we need. Selecting the full table is therefore the recommended choice even if we only plan to use one or two columns.

Option 1: let Excel detect the range

1. Click **Cancel** to close this dialog.
2. Click **any cell inside the data**, for example A2.
3. Go to **Insert → PivotTable**.

Excel automatically detects the whole block of data and writes its range in the *Table/Range* box. The box is empty in our screenshot because the selected cell was outside the data when we opened the dialog.

Option 2: if the data is formatted as an Excel Table

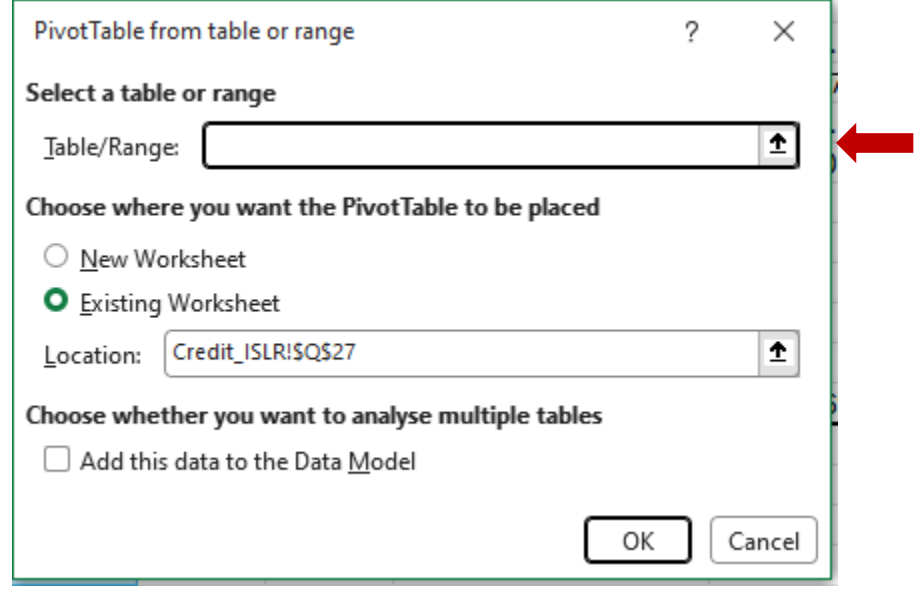
We can type the **table's name** in the *Table/Range* box, for example `Table1`. To find the name, we click a cell in the table and look at **Table Design → Table Name** (top left of the ribbon). This is the most robust option: if rows are added later, the PivotTable source grows with the table automatically after a refresh.

Option 3: select the range with the keyboard

1. Click in the **Table/Range** box.
2. On the sheet, click the **top-left header cell** of the data (for example A1).
3. Press **Ctrl+Shift+Right Arrow** to extend the selection to the last column, then **Ctrl+Shift+Down Arrow** to extend it to the last row.

This avoids scrolling. As before, these shortcuts stop at the first empty cell, so we check that the range shown in the box (for example `Credit_ISLR!$A$1:$L$401`) really covers all the data.

We should avoid **Ctrl+Shift+End** here. It selects up to the last used cell of the whole sheet, and our sheet already contains other output (the descriptive statistics near column Q), which would end up inside the source range.



Selecting one column or a few columns

If we really want to restrict the source:

- **One column**: click its header cell and press **Ctrl+Shift+Down Arrow**. The header must be included, because the PivotTable uses it as the field name.
- **Several adjacent columns**: click the first header, **Shift+click** the last header, then press **Ctrl+Shift+Down Arrow**.
- **Non-adjacent columns** (for example A and D) are **not allowed**: the source of a PivotTable must be a single contiguous block. In this case we select the whole table and pick the columns in the Fields pane.

Two further settings in this dialog

Location: *Existing Worksheet* with `Credit_ISLR!$Q$27` places the PivotTable on the same sheet, starting at Q27. We need to make sure there is enough empty space to the right of and below that cell, otherwise the PivotTable may overlap other content. When in doubt, **New Worksheet** is safer.

Add this data to the Data Model: we tick this box if we want the **Distinct Count** option, which counts the unique values of a column. Without it, that option does not appear among the value field settings.

	A	B	C	D	E	F
1	transaction_id	date	month	branch	category	amount
2	TRX0001	02/01/2026	01/01/2026	Rome EUR	Deposit	497984
3	TRX0002	02/01/2026	01/01/2026	Rome EUR	Card payment	39909
4	TRX0003	02/01/2026	01/01/2026	Naples Central	Deposit	221919
5	TRX0004	02/01/2026	01/01/2026	Naples Central	Withdrawal	51648
6	TRX0005	03/01/2026	01/01/2026	Bologna East	Card payment	11149
7	TRX0006	03/01/2026	01/01/2026	Turin North	Withdrawal	24801
8	TRX0007	03/01/2026	01/01/2026	Naples Central	Card payment	18484
9	TRX0008	03/01/2026	01/01/2026	Turin North	Transfer	962292
10	TRX0009	04/01/2026	01/01/2026	Naples Central	Card payment	11991
11	TRX0010	06/01/2026	01/01/2026	Naples Central	Deposit	210733
12	TRX0011	06/01/2026	01/01/2026	Naples Central	Card payment	44171
13	TRX0012	06/01/2026	01/01/2026	Bologna East	Card payment	21941
14	TRX0013	07/01/2026	01/01/2026	Rome EUR	Transfer	553727
15	TRX0014	08/01/2026	01/01/2026	Milan Central	Card payment	20412
16	TRX0015	08/01/2026	01/01/2026	Rome EUR	Transfer	709886
17	TRX0016	08/01/2026	01/01/2026	Milan Central	Withdrawal	80894
18	TRX0017	08/01/2026	01/01/2026	Naples Central	Deposit	307581
19	TRX0018	08/01/2026	01/01/2026	Bologna East	Deposit	11331
20	TRX0019	09/01/2026	01/01/2026	Turin North	Card payment	41831
21	TRX0020	09/01/2026	01/01/2026	Bologna East	Card payment	46771
22	TRX0021	09/01/2026	01/01/2026	Naples Central	Transfer	334079
23	TRX0022	10/01/2026	01/01/2026	Milan Central	Card payment	16276
24	TRX0023	10/01/2026	01/01/2026	Naples Central	Withdrawal	60557
25	TRX0024	10/01/2026	01/01/2026	Bologna East	Deposit	340287

`customer_transactions.csv` one row = one operation made by a customer at a bank branch

800	5	12	4
transactions	branches	months of 2026	transaction types

`transaction_id`
Unique code of the row, e.g. TRX0001
Identifier, not aggregated

`date`
Day of the operation, e.g. 2026-01-02
Timeline filter

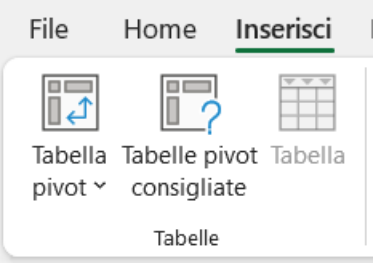
`month`
Year and month as text, e.g. 2026-01
Key: columns

`branch`
Milan Central, Turin North, Rome EUR, Naples Central, Bologna East
Key: rows

`category`
Deposit, Withdrawal, Transfer, Card payment
Key: filter or slicer

`amount`
Euro value, always positive, e.g. 4979.84. Transfers up to 10,000; card payments under 500
Measure: values

The colors group the columns by the role they play when we build the pivot. Teal Key: rows marks the keys, meaning the columns we group by, such as branch, month and transaction type: they end up in the Rows, Columns or Filters areas. The date column is teal too Timeline filter , because it also selects which operations we look at, but we don't drag it into an area: we use it through Insert Timeline, the sliding bar that filters by years, quarters or months. Coral Measure: values marks the measure, the amount of money each operation moves, the only numeric column we add up in the Values area. Gray Identifier, not aggregated marks the identifier, the code that labels each customer operation but makes no sense to sum or group, so we leave it out of the pivot.



In the Rows area we put branch: each branch becomes a row of the table.

In the Columns area we put month: each month becomes a column, from 2026-01 to 2026-12.

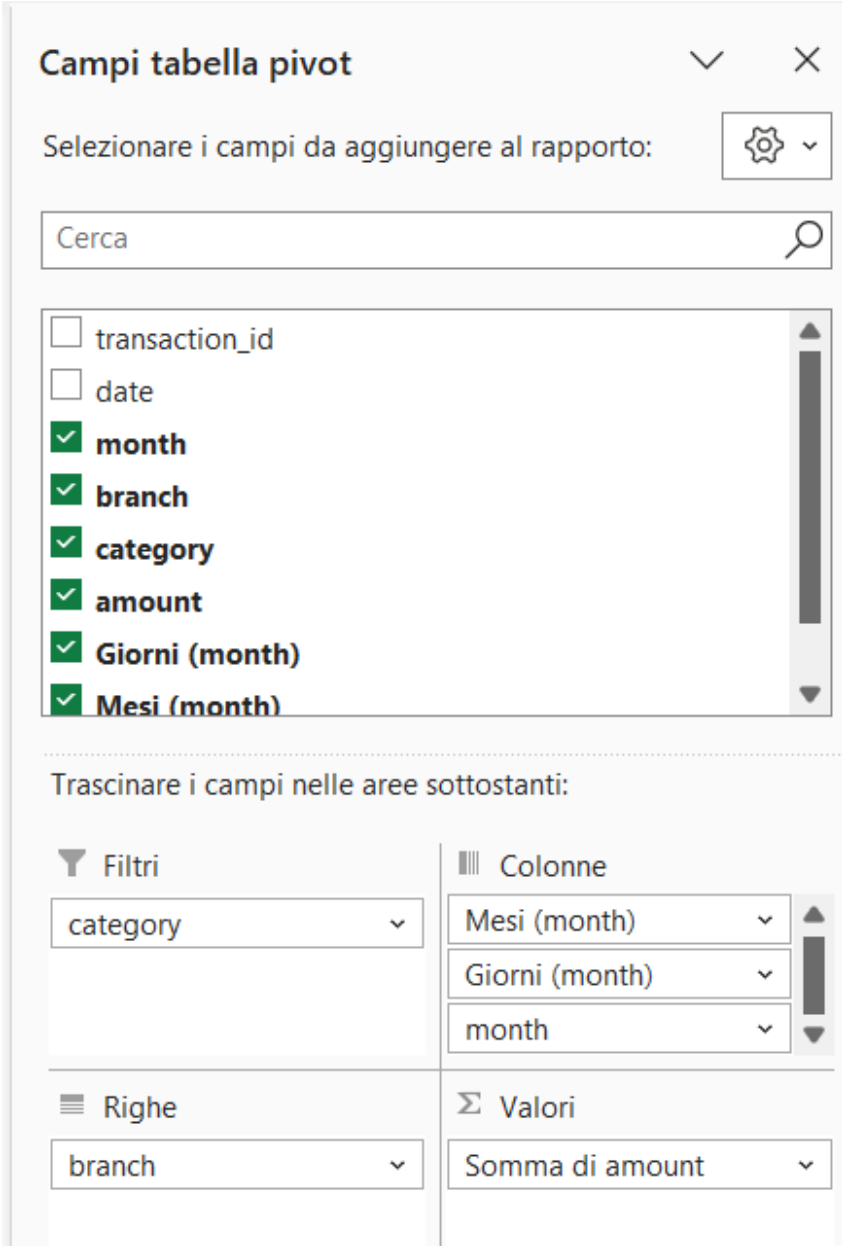
In the Values area we put amount, checking that it shows as Sum of amount. That way each cell shows how much money the customers of that branch moved in that month.

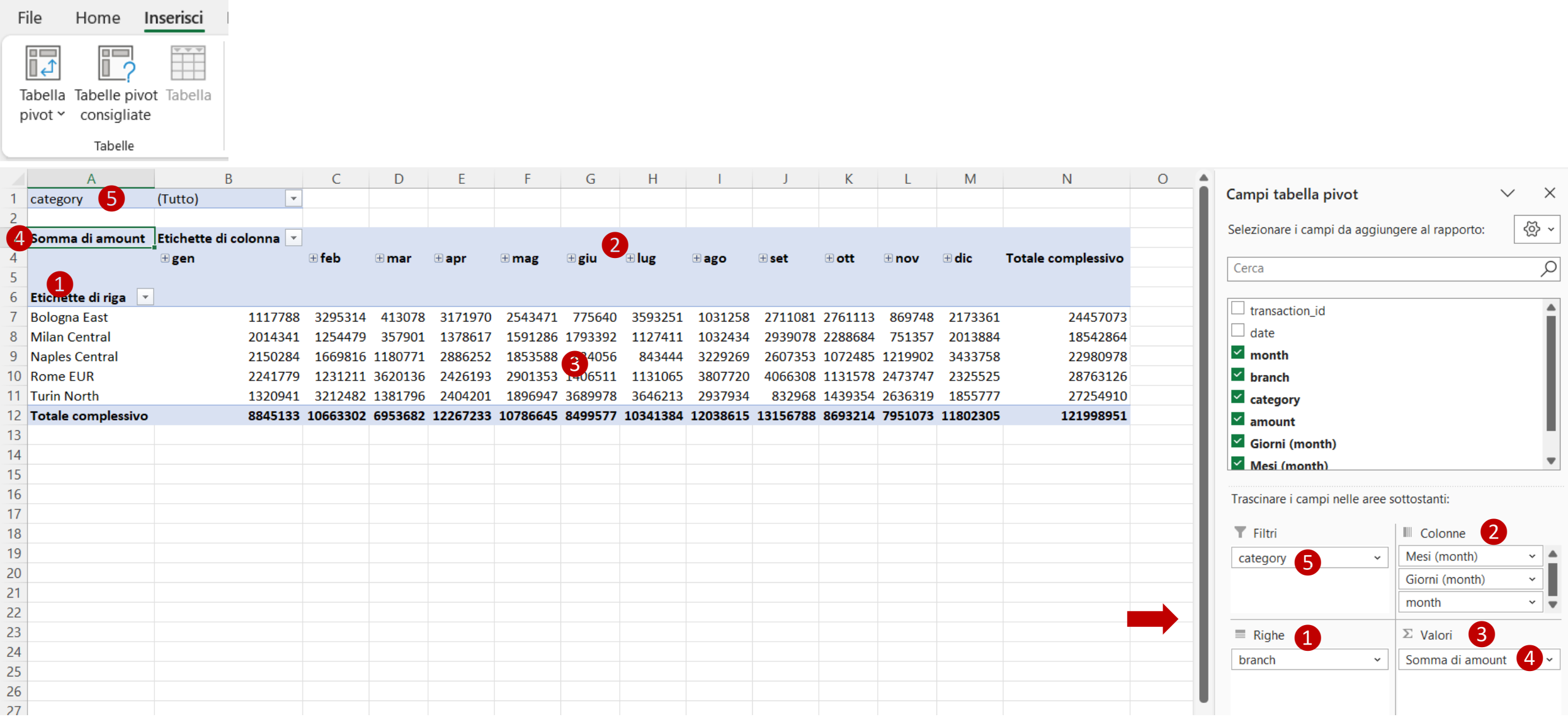
In the Filters area we put category: from the drop-down menu above the table we can choose to see only withdrawals, only transfers, and so on. As an alternative to the filter, we can use a slicer on category, the clickable buttons we saw earlier.

The date column doesn't go into any area: we use it through PivotTable Analyze, Insert Timeline, the sliding bar that filters by years, quarters or months.

We leave the transaction_id column out of the pivot, because it's just the row's code and it makes no sense to sum or group it.

A useful variation for the classroom: if we move category from Filters to Columns, in place of month, we get a branch-by-transaction-type table, which shows right away which branch has more deposits and which has more card payments. And if we add amount to Values a second time, set to Count, we also see how many operations there are, not just how much money they move.




File
Home
Inserisci
Tabella pivot
Tabelle pivot consigliate
Tabella
Tabelle
category
(Tutto)
Somma di amount
Etichette di colonna
Etichette di riga
gen
feb
mar
apr
mag
giu
lug
ago
set
ott
nov
dic
Totale complessivo
Bologna East 1117788 3295314 413078 3171970 2543471 775640 3593251 1031258 2711081 2761113 869748 2173361 24457073
Milan Central 2014341 1254479 357901 1378617 1591286 1793392 1127411 1032434 2939078 2288684 751357 2013884 18542864
Naples Central 2150284 1669816 1180771 2886252 1853588 843444 3229269 2607353 1072485 1219902 3433758 22980978
Rome EUR 2241779 1231211 3620136 2426193 2901353 1131065 3807720 4066308 1131578 2473747 2325525 28763126
Turin North 1320941 3212482 1381796 2404201 1896947 3689978 3646213 2937934 832968 1439354 2636319 1855777 27254910
Totale complessivo 8845133 10663302 6953682 12267233 10786645 8499577 10341384 12038615 13156788 8693214 7951073 11802305 121998951
Campi tabella pivot
Selezionare i campi da aggiungere al rapporto:
Cerca
transaction_id
date
month
branch
category
amount
Giorni (month)
Mesi (month)
Trascinare i campi nelle aree sottostanti:
Filtri
category
Colonne
Mesi (month)
Giorni (month)
month
Righe
branch
Valori
Somma di amount
1
2
3
4
5
Value Field Settings...
Sum of Income

Excel Pivot Tables: Deterministic Multidimensional Reporting

Useful for summarising data — but not for inferential statistical analysis

EXPLORE
SUMMARISE
FIND INSIGHTS
GO FURTHER

1 What Pivot Tables Do

Pivot tables summarise detailed rows through aggregation.
The same input data always produce the same output.

Sum | Count | Average | Minimum | Maximum

They organise measures across dimensions such as Branch, Month, and Category.

Detailed Data (transactions)

Branch	Month	Category	Amount
North	Jan	Sales	250
North	Jan	Services	120
South	Jan	Sales	340
South	Feb	Sales	280
East	Feb	Services	160
...	...	...	...

Pivot Table

Summarised View (by Branch and Category)

Branch	Sales	Services	Total
North	250	120	370
South	620	160	780
East	410	230	640
Total	**1,280**	**510**	**1,790**

2 Why They Are Not Statistical-Probabilistic Analysis

Pivot Tables provide

- Deterministic calculations
- Descriptive summaries
- Multidimensional reporting

Pivot Tables do NOT provide

- Probability models
- Hypothesis tests
- Confidence intervals
- Prediction
- Estimates of uncertainty

So calling pivot tables statistical or probabilistic data analysis would be a stretch.

3 Pivot Tables and OLAP

Both pivot tables and OLAP use the same multidimensional idea: analyse measures across multiple dimensions.

Almost the same vocabulary

Dimensions	→	fields we group by (Branch, Month, Category)
Measures	→	values we aggregate (e.g. Amount)
Slice	→	filter one dimension
Dice	→	combine multiple filters
Drill-down / Roll-up	→	move between Month, Quarter, Year
Pivot	→	move fields between Rows and Columns

Excel pivot tables can connect to OLAP cubes, and Power Pivot is a small multidimensional model inside Excel.

4 Important Nuance: They ARE Descriptive Statistics

Pivot tables can do more than simple totals and averages. Excel also offers Standard Deviation and Variance, so they can describe both the centre and the spread of the data.

But they cannot generalise beyond the data shown. They do not tell us whether a difference is reliable or just due to chance.

Describing data is not the same as inferring from data.

5 Why They Can Mislead

- An average based on 8 rows may look as authoritative as one based on 8,000 rows.
- Aggregated totals can hide subgroup patterns.
- In extreme cases, totals can reverse subgroup trends (Simpson's paradox).

6 Best Way to Frame Pivot Tables

Detailed data (all rows) → **Pivot table summary** (aggregated view) → **Identify where to look next** (patterns and questions) → **Statistical analysis explains why** (tests, models, uncertainty)

Pivot tables are an excellent first exploratory step: they help us see patterns and organise data.
But they are not the analysis that explains causality, uncertainty, or statistical significance.

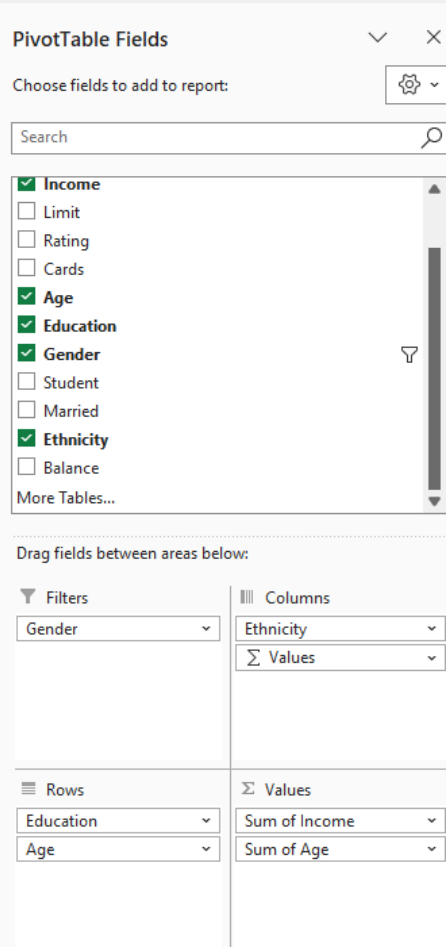
PIVOT TABLE IN EXCEL

Pivot tables are well suited to **tabular data in long format**, meaning one row per observation, with at least one categorical column to use as a grouping key and one numeric column to aggregate. Our file is the textbook example: branch and category are the keys, amount is the measure.

They work poorly, on the other hand, when **the data is already aggregated**, when categories have thousands of distinct values, or with free text and unstructured data. And they don't handle relationships between multiple tables, unless we move to the Data Model with Power Pivot.

PivotTables: applied to the usual Customer dataset

Gender	Female								
	Column Labels								
	African American		Asian		Caucasian		Total Sum of Income	Total Sum of Age	
Row Labels	Sum of Income	Sum of Age	Sum of Income	Sum of Age	Sum of Income	Sum of Age			
5					**34142**	**47**	**34142**	**47**	
47					34142	47	34142	47	
6					**36597**	**116**	**36597**	**116**	
37					89	37	89	37	
79					36508	79	36508	79	
7			**99317**	**121**	**143668**	**183**	**242985**	**304**	
23					24314	23	24314	23	
38					31335	38	31335	38	
45					57337	45	57337	45	
57			80616	57			80616	57	
64			18701	64			18701	64	
77					30682	77	30682	77	
8	**303581**	**244**	**213295**	**127**	**127264**	**123**	**644140**	**494**	
35	27794	35					27794	35	
58	180682	58					180682	58	
59					54319	59	54319	59	
60			32916	60			32916	60	
64					72945	64	72945	64	
67			180379	67			180379	67	
70	54663	70					54663	70	
81	40442	81					40442	81	
9	**67937**	**199**	**39827**	**109**	**127967**	**227**	**235731**	**535**	
28			4957	28			4957	28	
34			19537	34			19537	34	
40					27369	40	27369	40	
47			15333	47			15333	47	
52	5953	52					5953	52	
55					33657	55	33657	55	
63					36934	63	36934	63	



Sorting, filtering and conditional formatting are EDA tools

Simple spreadsheet operations often reveal data issues faster than complex methods.

Filter & inspect

fx `=SORT(Table1[Balance],,-1)`

Customer	Income	Balance	DaysPastDue
C1001	65,000	12,500	0
C1002	54,000	8,200	0
C1003	92,000	45,500	12
C1004		3,200	0
C1005	71,000	165,000	97
C1006	58,000	9,700	0

Sort

Find top/bottom values and suspicious extremes.

Filter

Inspect one segment, branch, category or exception at a time.

Conditional formatting

Highlight blanks, duplicates, thresholds and unusual values.

Freeze panes / Table headers

Keep context while scanning many rows.

Tip: “show me the 20 largest values” is often a very good EDA question.

Conditional Formatting in Excel: make important patterns visible

Conditional Formatting changes how cells look when rules are met — helping analysts spot exceptions, thresholds and data-quality issues quickly.

WHAT IT IS

A visual rule applied to cells based on their values or on a formula.

Examples:
- values above a threshold
- blank or duplicate cells
- Data Bars and Color Scales
- Icon Sets for status/risk

It changes appearance, not the underlying value.

EXAMPLE

Flag unusual account balances

Customer_ID	Balance	Status
C001	1,200	OK
C002	850	OK
C003	15,900	HIGH
C004		MISSING
C005	670	OK
C006	22,300	HIGH

Exceptions become visible without changing the dataset.

HOW TO USE IT

Home → Conditional Formatting

Main rule types:
- Highlight Cells Rules / Top-Bottom
- Data Bars / Color Scales / Icon Sets
- New Rule / Use a formula

Formula-based rules

=ISBLANK($M2) → missing value

=$M2>10000 → threshold

Common Data Science uses

Missing values | **Outliers** | **Thresholds** | **Duplicates** | **Trends / magnitude**

Takeaway: Conditional Formatting is a visual diagnostic layer — useful for seeing patterns quickly, but it does not clean or modify the data.

The dialog classifies rules **by criterion** (how the test is defined), the menu **by effect and by use** (what we want to obtain and how quickly).

Conditional Formatting in Excel: the rule types

The cross-reference table (btw the two methods: dialog and menu)

Type in the dialog	Corresponding menu item	Extra options versus the menu
Format all cells based on their values	Data Bars, Colour Scales, Icon Sets	thresholds as number/percentile/formula, 2-colour scale, bar only
Only format cells that contain	Highlight Cells Rules (Greater Than, Between, Equal To, Text that Contains, A Date Occurring)	Not Between, blanks/no blanks, errors/no errors
Only format top or bottom ranked values	Top/Bottom Rules (Top/Bottom 10 Items, 10 %)	any N
Only format values that are above or below average	Top/Bottom Rules (Above/Below Average)	1, 2 or 3 standard deviations above/below the mean, "equal or above"
Only format unique or duplicate values	Highlight Cells Rules ▸ Duplicate Values	—
Use a formula to determine which cells to format	none: only through New Rule…	anything that can be written as a formula

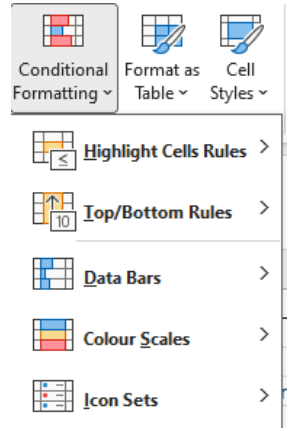
There are two ways to define a NEW conditional formatting:

1. In the «Manage Rules» **dialog** (in the bottom of conditional formatting) – the best to me



Select a cell within the table

2. In the top **menu** of conditional formatting



Conditional Formatting - Examples

Two files ready.

Workbook `Credit_ISLR_conditional_formatting.xlsx` — two sheets:

- **Customers**: the 400 customers in the table `tblCustomers` (A1:L401) with 12 rules, one for each category of the Conditional Formatting menu:
  - Highlight Cells: *Greater Than* 10,000 on Limit, *Duplicate Values* on Limit (border only, so it coexists with the fill), *Between* 18–30 on Age, *Equal To* "Yes" on Student
  - Top/Bottom: *Above Average* on Education, *Top 10 Items* on Balance
  - *Data Bars* on Income, *Color Scale* on Rating, *Icon Set* (traffic lights with numeric thresholds, reversed order) on Cards
  - Formulas (New Rule ▸ Use a formula): `=B2>AVERAGE($B$2:$B$401)+2*STDEV($B$2:$B$401)` on Income, `=$L2=0` and `=AND($I2="Yes",$L2>1000)` on the whole row
- **Rules**: for each rule the menu path, the dialog settings, the equivalent formula, the format applied and the count of matching cells computed with live formulas (COUNTIF, LARGE, SUMPRODUCT…).

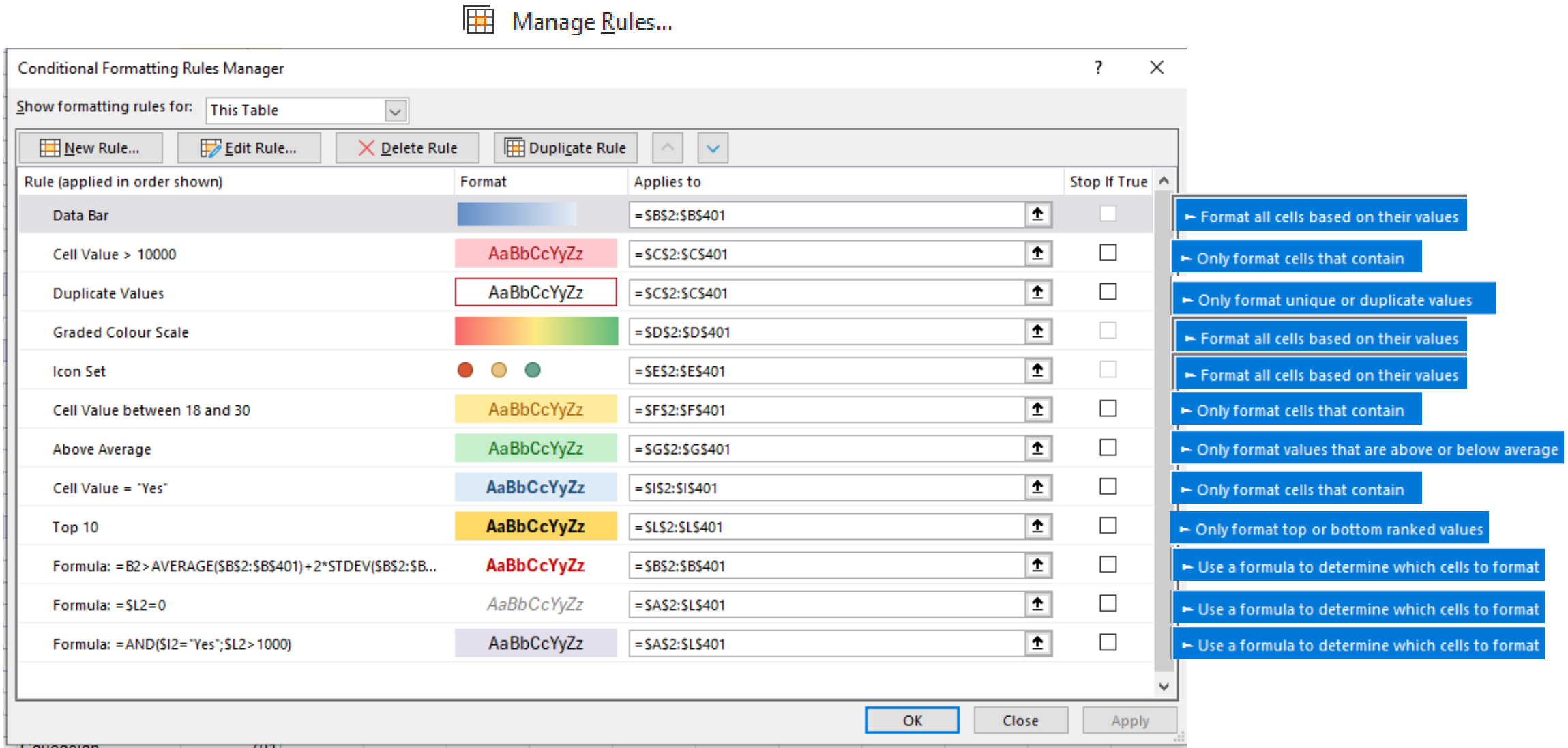
A comprehensive example

Guide `Conditional_Formatting_Guide_Excel2024.md` (in English, "we" form): importing the CSV with Power Query or Ctrl+T, how conditional formatting works (selection, `$` references, priority), the 12 operational steps, the Rules Manager, a summary table and common pitfalls.

Three notes:

- Two clean-ups on the CSV: the first unnamed column (a duplicate of ID) was dropped and the leading space in `" Male"` was removed.
- The data in the file is embedded, not linked through a query: the Power Query connection cannot be recreated from outside Excel, but the steps to build it are in the guide (option A).
- The rule formulas use A1 references because conditional formatting does not accept structured references; the z-score rule is saved with `STDEV` (equivalent to `STDEV.S`) for compatibility.

Conditional Formatting - Examples



Conditional Formatting - Examples

A possible NEW rule to check for Missing Values

Built-in rule — New Rule… ▸ *Only format cells that contain* ▸ in the first drop-down, instead of *Cell Value*, we choose **Blanks** (or *No Blanks*, and next to them *Errors* / *No Errors* for #N/A and the like). It has no item in the galleries, so it is only visible from New Rule. Internally Excel stores it as `=LEN(TRIM(B2))=0`, so it also catches cells that contain only spaces.

Formula rule, more flexible, applied to the table body A2:L401:

- `=ISBLANK(A2)` → truly empty cells only; a cell holding an empty string `""` (typical after an import or a formula) is *not* caught;
- `=LEN(TRIM(A2))=0` → empty cells, `""` and spaces only;
- `=OR(A2="",A2="NA",A2="N/A",A2="-",A2="?")` → "disguised" missing values stored as text placeholders;
- `=AND(A2<>"",NOT(ISNUMBER(B2)))` on numeric columns → text where a number is expected (e.g. "n.a." imported into Income);
- `=COUNTBLANK($A2:$L2)>0` → highlights the **entire row** when at least one value is missing (COUNTBLANK counts both empty cells and `""`).

The Credit dataset has no missing values, so on the current file these rules would colour nothing; for a demo we delete a few cells and the formatting appears immediately. If you like, I can add to the workbook a rule 13 (orange fill on empty cells) and a rule 14 (row with at least one missing value) and update the Rules sheet and the guide.

Chapter title



04 – Probability & Statistics

Random variables: a bridge from data to probability

A random variable represents a quantity whose observed value depends on an uncertain outcome.

Experiment / process

A customer may or may not default within 12 months.

Random variable X

X = 1 if default occurs
X = 0 otherwise

Observed value

For one customer: X = 0

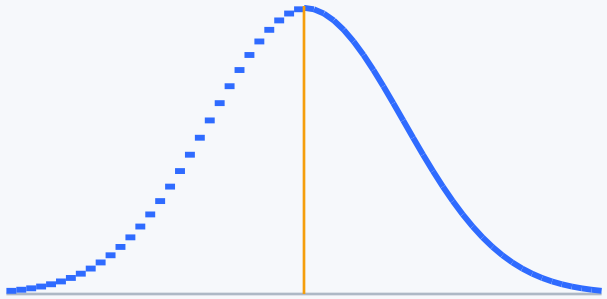
Other examples

Transaction amount · number of late payments · account balance next month · time until loan closure · daily loss.

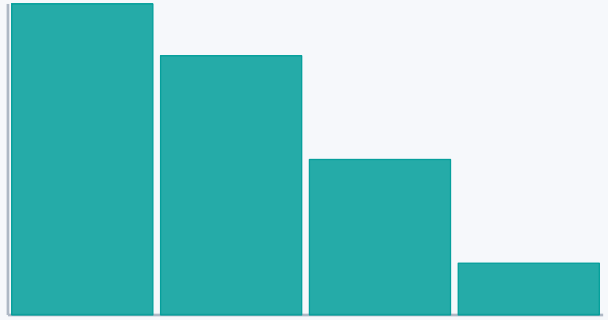
Random does not mean “without structure”. It means the exact outcome is uncertain before observation.

Probability distributions: the “shape” of uncertainty

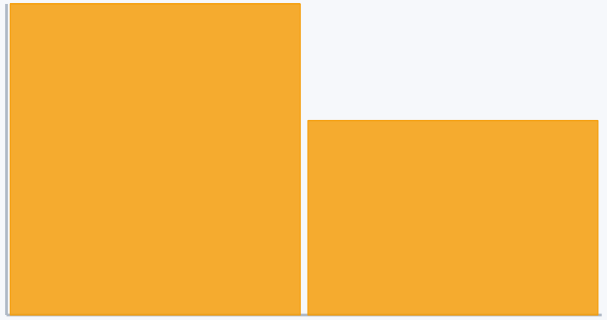
A distribution describes which values a random variable can take and how likely they are.



Continuous distribution



Discrete distribution

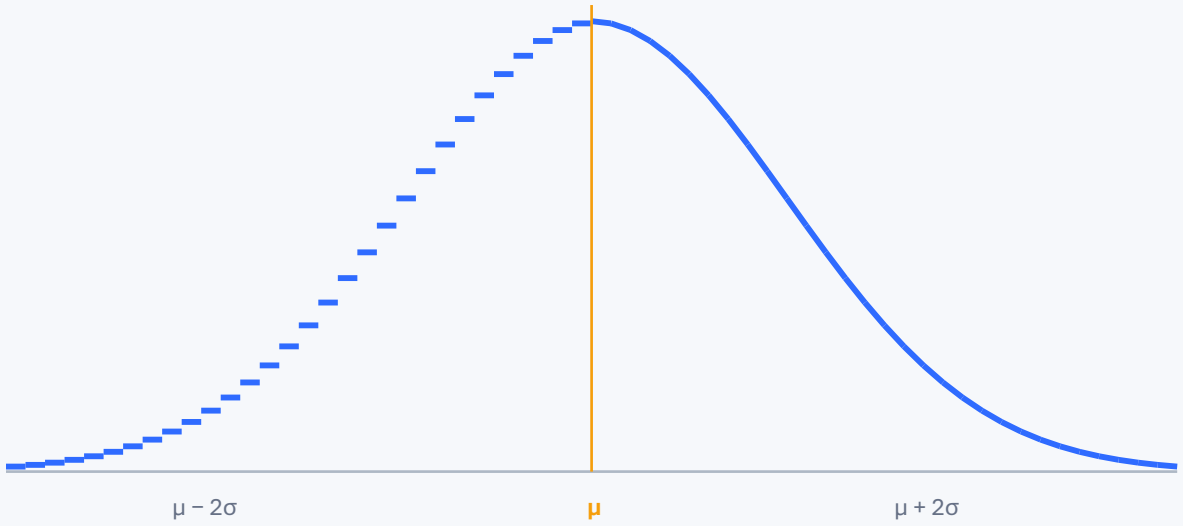


Binary distribution

- Distributions help us reason about typical values, variability and tail risk.
- The empirical distribution comes from observed data; a theoretical distribution is a model.
- EDA asks whether the observed shape is symmetric, skewed, multi-modal, heavy-tailed or irregular.

The Normal distribution: useful model, not universal law

Many measurement errors and aggregate quantities are approximately bell-shaped — but financial data often has skew and heavy tails.



Parameters

Mean μ controls location; standard deviation σ controls spread.

EDA question

Does the histogram look roughly symmetric and bell-shaped?

Banking caution

Losses, balances and transaction amounts are often skewed and may have heavy tails.

```
Excel examples: =NORM.DIST(x, mean, stdev, FALSE)   |   =NORM.INV(p, mean, stdev)
```

Bernoulli and Binomial: natural distributions for “event / no event”

Binary outcomes appear everywhere in banking: default, fraud flag, churn, approval, response.

Bernoulli random variable

One trial with two outcomes.

X = 1 with probability p
X = 0 with probability 1 − p

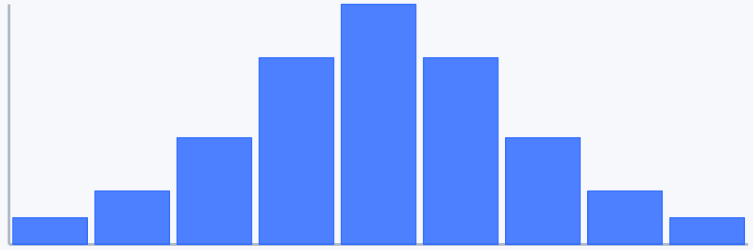
Binomial random variable

Counts how many “successes” occur in n comparable Bernoulli trials.

Example: number of defaults among 100 similar loans.



Bernoulli: one customer



Binomial: count across many customers

Probability vs statistics: opposite directions of reasoning

Probability starts from a model; statistics starts from data.

Probability

MODEL / assumptions



Possible data / outcomes

Statistics

OBSERVED DATA



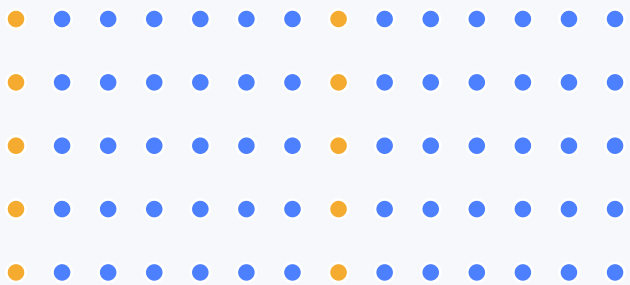
Model / parameters / inference

EDA sits mainly on the statistics side: use observed data to learn what is plausible, typical and suspicious.

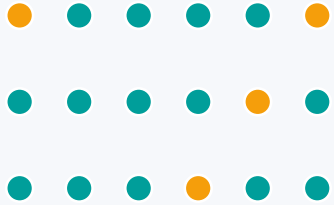
Sampling and uncertainty: a sample is not the population

Any estimate from a sample varies from sample to sample. Statistical thinking quantifies that uncertainty.

Population



Sample



Estimate

Sample mean
Sample default rate
Sample correlation
... each has uncertainty.

- Larger, representative samples usually reduce uncertainty.
- Biased samples can be precise but wrong.
- Always ask: “how was this data selected?”

Chapter title



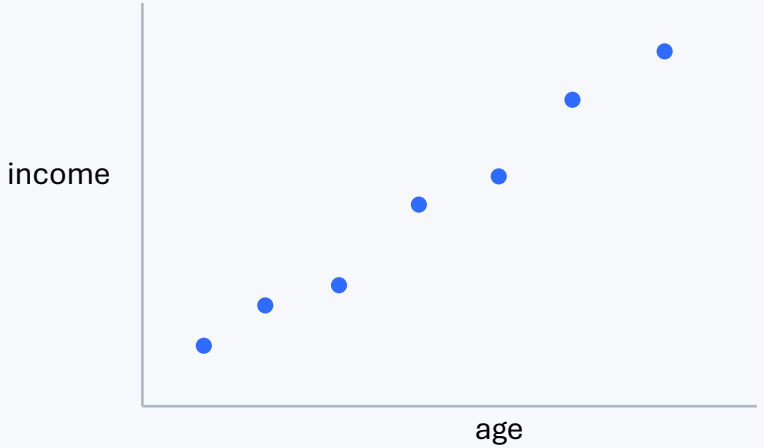
05 - Relationships

One of the main goals in EDA is just finding relationships among variables (the columns of your dataset)

Correlation: how two numeric variables move together

Correlation quantifies the strength and direction of a linear relationship.

Positive correlation



Negative correlation



Little linear correlation



r > 0

High values of X tend to occur with high values of Y.

r < 0

High values of X tend to occur with low values of Y.

r ≈ 0

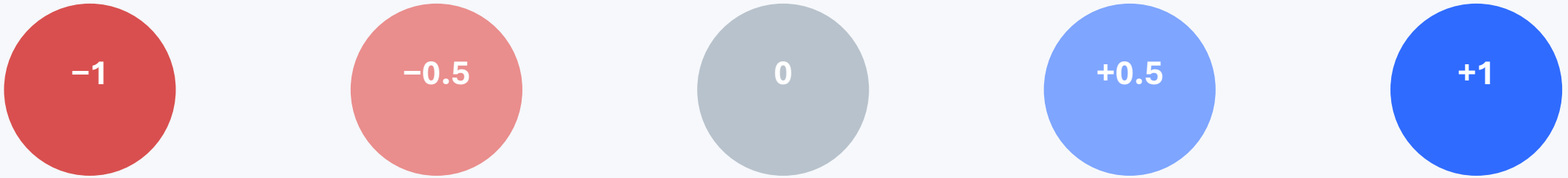
No strong linear relationship — but nonlinear structure may still exist.

The Pearson correlation coefficient

Pearson r is bounded between −1 and +1 and measures linear association.

`CORREL` computes the **Pearson product-moment correlation coefficient** — the standard linear correlation, identical to Excel's `PEARSON` function (the two exist for historical reasons and return the same value):

Excel has no built-in Spearman or Kendall. For **Spearman's rank correlation** we rank both columns first and then correlate the ranks:



Perfect negative | Moderate negative | No linear relationship | Moderate positive | Perfect positive

Scale-free

Changing EUR to thousands of EUR does not change r.

Linear

r can miss curves, thresholds and other nonlinear patterns.

Sensitive to outliers

One extreme point can dramatically change the result.

Interpret the coefficient together with a scatter plot.

Correlation in Excel: CORREL + scatter plot

A number is useful; a number plus a plot is much safer.

x Income vs Balance

fx =CORREL(Table1[Income],Table1[Balance])

Customer	Income	Balance
C1001	65,000	12,500
C1002	54,000	8,200
C1003	92,000	45,500
C1004	45,000	3,200
C1005	71,000	40,000
C1006	58,000	9,700




```
=CORREL(Table1[Income], Table1[Balance])
```

Always check units, missing rows and outliers before interpreting r.

Correlation matrix: many pairwise relationships at once

A correlation matrix is a compact map of linear associations among numeric variables.

	Age	Income	Balance	Products	Arrears
Age	1.00	0.28	0.21	0.15	0.05
Income	0.28	1.00	0.63	0.36	-0.18
Balance	0.21	0.63	1.00	0.42	-0.31
Products	0.15	0.36	0.42	1.00	-0.12
Arrears	0.05	-0.18	-0.31	-0.12	1.00

What to look for

Large absolute values |r| reveal strong linear association.

Diagonal

A variable is perfectly correlated with itself: r = 1.

Symmetry

corr(X,Y) = corr(Y,X), so the matrix mirrors across the diagonal.

```
Excel: Data Analysis → Correlation   or   multiple =CORREL(...) formulas
```

Correlation matrix: how to compute it with the Data Analysis Toolpak

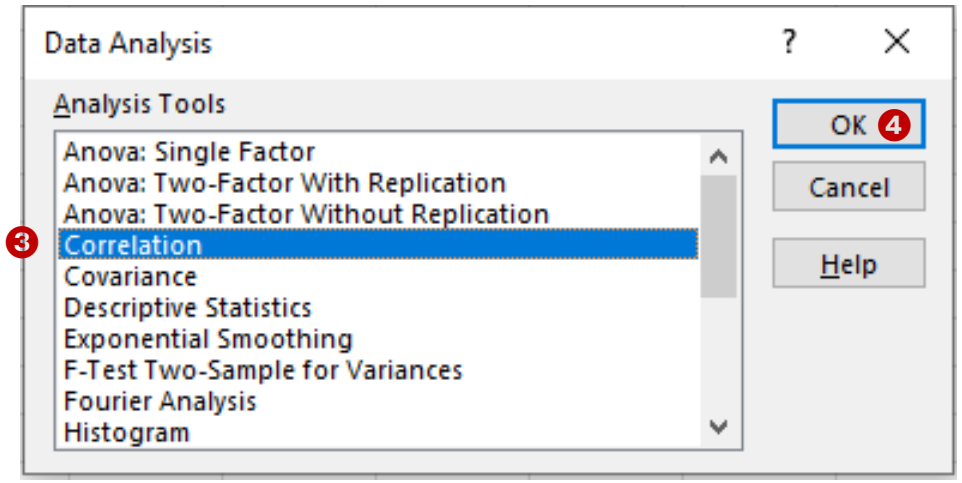
Correlation Matrix here computed on the Credit_ISLR dataset

1 Data

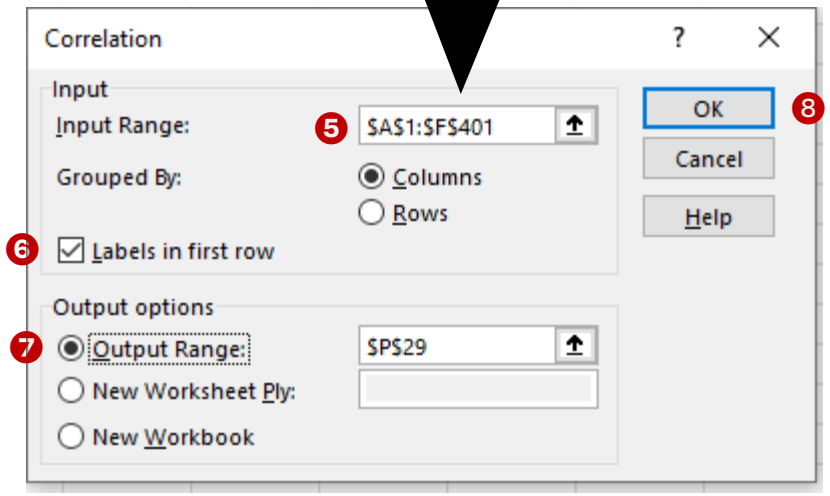
2 Data Analysis

Solver

Analyze

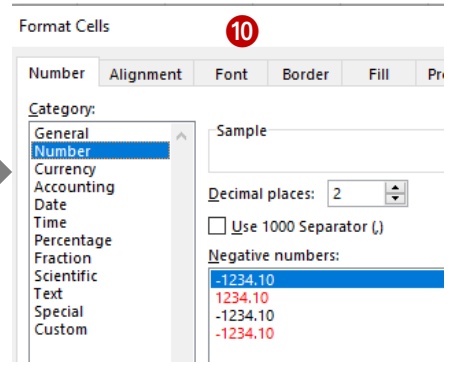


A unique rectangular range, The correlation function is quite old and doesn't accept separate intervals



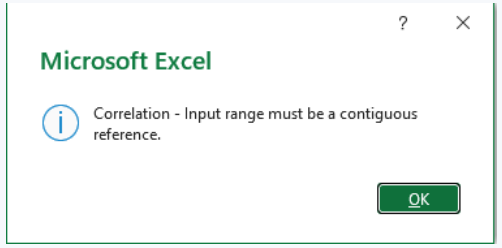
9 Select the whole matrix

	Income	Limit	Rating	Cards	Age	Balance
Income	1					
Limit	0.707152	1				
Rating	0.703455	0.99688	1			
Cards	-0.01221	0.010231	0.053239	1		
Age	0.123258	0.100888	0.103165	0.042948	1	
Balance	0.432625	0.861697	0.863625	0.086456	0.001835	1



	Income	Limit	Rating	Cards	Age	Balance
Income	1.00					
Limit	0.71	1.00				
Rating	0.70	1.00	1.00			
Cards	-0.01	0.01	0.05	1.00		
Age	0.12	0.10	0.10	0.04	1.00	
Balance	0.43	0.86	0.86	0.09	0.00	1.00

Correlation matrix: how to compute it with the Data Analysis Toolpak



The typical mistake with the Data Analysis Toolpak

Yes. This is a **limitation of the Correlation tool in Excel's Analysis ToolPak.**

The error message:

> **"Correlation – Input range must be a contiguous reference."**

means that the ToolPak **does not accept non-adjacent columns** as a single Input Range.

For example, suppose you want to calculate correlations among:

`Income`, `Limit`, `Rating`, `Age`, `Education`, `Balance`

If these columns are separated by other columns in the original Excel Table, you cannot select them using **Ctrl + Click** and use that selection directly in:

Data → Data Analysis → Correlation

Ctrl-A within the table (to fully select it), extend selection to the headers too, then copy and paste

Solution: create a temporary contiguous table

In a new area of the worksheet, or preferably in a new worksheet, copy the numerical variables you want to analyze and place them **next to each other:**

Income	Limit	Rating	Age	Education	Balance
14891	3606	283	34	11	333
106025	6645	483	82	15	903
...	...	...	...	...	...

Then select the entire **contiguous range** and go to:

Data → Data Analysis → Correlation

Set:

Grouped By: `Columns`
Labels in First Row: ✓

Excel will generate a **correlation matrix**, such as:

	Income	Limit	Rating	Age	Education	Balance
Income	1	...	...	...	...	...
Limit	...	1	...	...	...	...
Rating	...	...	1	...	...	...
Age	...	...	...	1	...	...
Education	...	...	...	...	1	...
Balance	...	...	...	...	...	1

Alternative: use the CORREL function

If you only need the correlation between **two variables**, there is no need to create a temporary table.

For example:

```excel
=CORREL(Table1[Age],Table1[Income])
```

Here, **the columns do not need to be adjacent.** `CORREL()` simply receives the two arrays whose Pearson correlation coefficient has to be calculated.

So the key distinction is:

Analysis ToolPak → Correlation: requires one **contiguous rectangular range.**

`CORREL()` **function:** can work directly with **non-adjacent columns.**

Correlation is not causation

A strong relationship can be real and useful without proving that X causes Y.

Common cause

A third variable drives both X and Y.

Example: season affects both heating usage and energy bills.

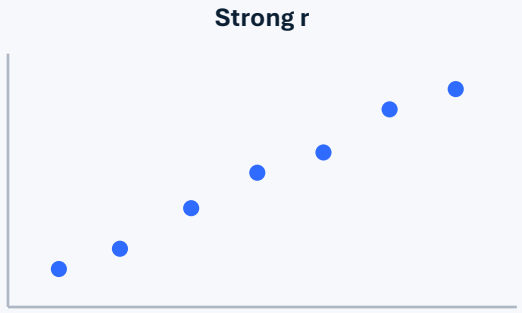
Reverse direction

Y may influence X rather than X influencing Y.

Correlation alone cannot determine direction.

Spurious relationship

With many variables, some correlations appear by chance.



Does X cause Y?

Not answered by correlation alone

Need additional evidence

Domain knowledge · time ordering · experiments · causal designs.

Correlation in Anti Money Laundering

Real-life Italian data in 2015

Suspicious transaction report (STR)

X = ML Crimes / Bank Branch

Y = STR / Bank Branch



	X	Y
X	1.00	
Y	0.68	1.00

AML activities are effective. Why.

Correlation and Data Visualization Start with a Business Problem

Plots and correlations are tools to answer business questions, not the starting point.

From business questions to better decisions.



1 Business Problem



2 different business problems

Sales are falling. Customer churn is increasing.

2 Business Question



What factors move with sales or churn?

3 Data Selection



Choose relevant variables and metrics.

4 Correlation & Visualization



Use plots and correlation to explore relationships in the data.

5 Insight / Decision



Identify patterns, explain relationships, support action.



Never start with the chart. Start with the business need.

Visualizations and correlations are tools to answer business questions, not the starting point.

Example: Understanding rising defaults

A third business problem



Business problem
Why are defaults rising?



Look at income, debt, arrears, age



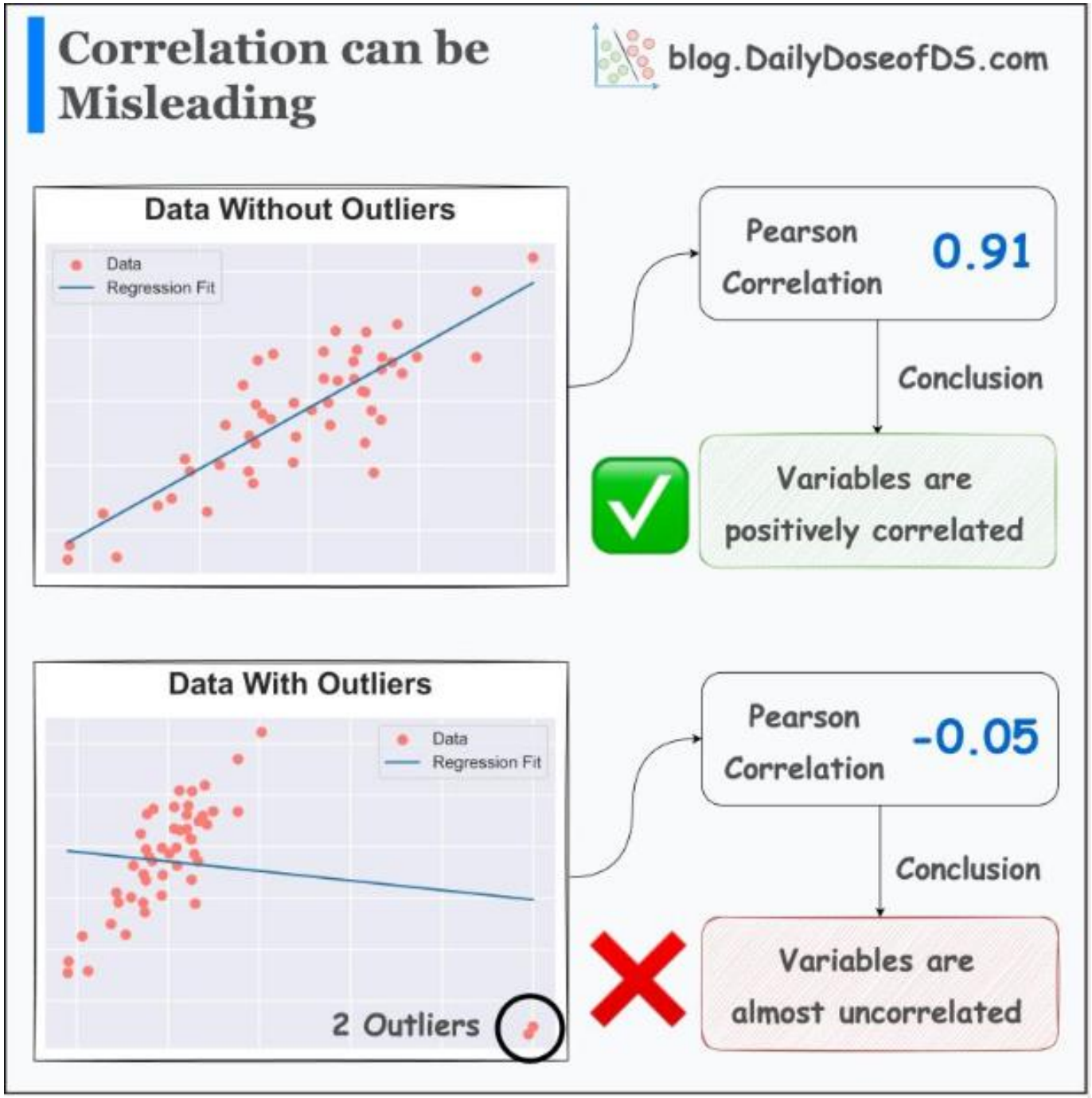
Use plots and correlation



Find the most informative relationships.

Correlation: typical business questions in banks

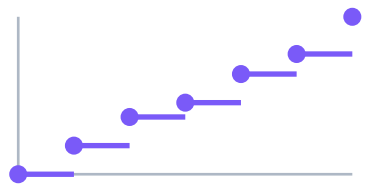
- **Portfolio diversification** — to assess how different assets move together and identify diversification opportunities.
- **Market risk management** — to understand how market variables (e.g. interest rates, FX rates, equity prices) are related and how shocks may propagate.
- **Credit risk assessment** — to measure dependencies between borrowers, sectors, or geographic regions and identify concentration risk.
- **Stress testing** — to estimate how different risk factors may behave together under adverse market scenarios.
- **Liquidity risk management** — to analyse relationships between cash flows, funding sources, deposits, and market conditions.
- **Fraud detection** — to identify unusual relationships between transactions, accounts, merchants, or customer behaviours.
- **Anti-money laundering (AML)** — to detect patterns or dependencies across transactions, accounts, counterparties, and entities that may warrant further investigation.
- **Customer behaviour analysis** — to understand relationships between products and customer behaviours, e.g. credit card usage and loan applications.
- **Interest rate risk management** — to analyse the relationship between interest rates, bond prices, deposits, and lending activity.
- **Forecasting and modelling** — to identify variables that are statistically related and may be useful as inputs to predictive models.
- **Trading and hedging** — to assess whether two instruments tend to move together and therefore may be useful for hedging purposes.
- **Operational risk** — to identify relationships between operational incidents, business units, processes, or external factors.


Correlation can be Misleading
blog.DailyDoseofDS.com
Data Without Outliers
Data
Regression Fit
Pearson Correlation
0.91
Conclusion
Variables are positively correlated
Data With Outliers
Data
Regression Fit
2 Outliers
Pearson Correlation
-0.05
Conclusion
Variables are almost uncorrelated

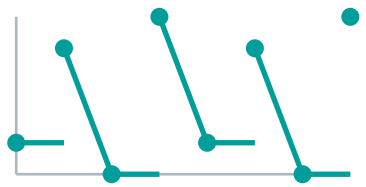
Patterns in data: what EDA is trying to discover

A pattern is a repeatable structure in the data that helps explain, predict or segment observations.

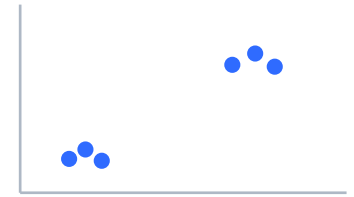
Trend



Seasonality



Clusters



Association



Anomaly



- Patterns are hypotheses until validated — they may reflect sampling, missingness or data quality problems.
- In banking, patterns often differ across segments, products, branches and time periods.
- A useful EDA habit: ask “what would make this pattern disappear?”

```
Pattern ≠ proof. It is a structured clue that guides the next question.
```

What is a “pattern”?

Pattern recognition means finding recurring structures inside data that may initially look random.

Look at this set of numbers and letters:

- knsxoi543y77778efh8rfv097777d
- j39e8389f7777lmjbh8ihrz7777
- knsxoi543y77778efh8rfv097777
- 39e8389f7777lmjbh8ihrz>7777

Do you notice anything unusual?

Patterns are repeated structures



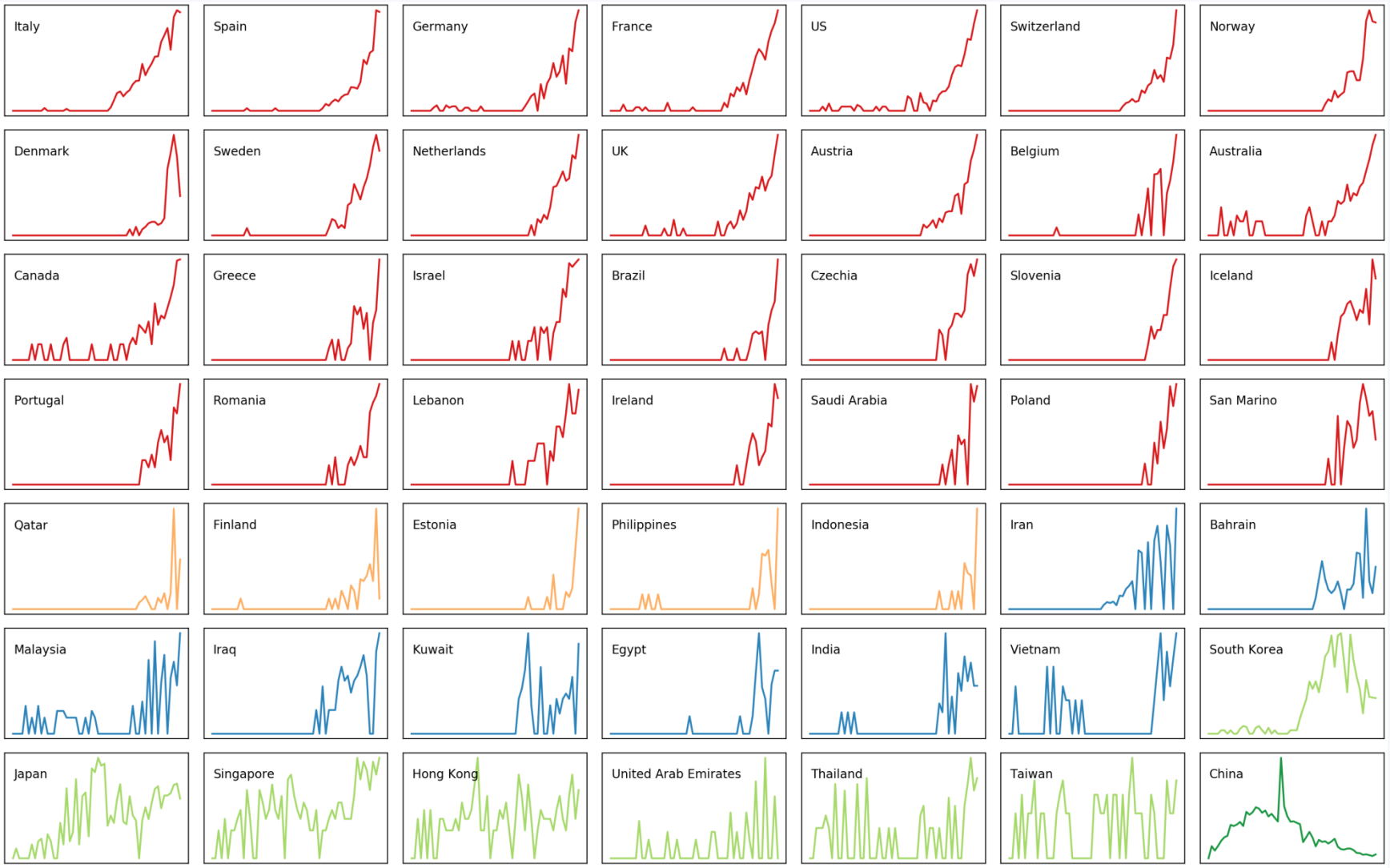
The same idea applies to data: repeated forms, relationships, sequences or behaviours can be discovered and described.

Exactly. After every 10 apparently random letters and numbers, the digit **“7” is repeated four times** — the pattern **“7777”**. Your brain has recognised a regular structure inside data that initially looks **random**. This process is called **pattern recognition(one of a business analyst’s task)**

Pattern recognition = identifying recurring models, structures or regularities within raw data.

Not yet pre-processed data

Patterns



5 different patterns in Covid-19 growth (by colour)

Chapter title



06 – Visualisation

Choose the plot that matches the question

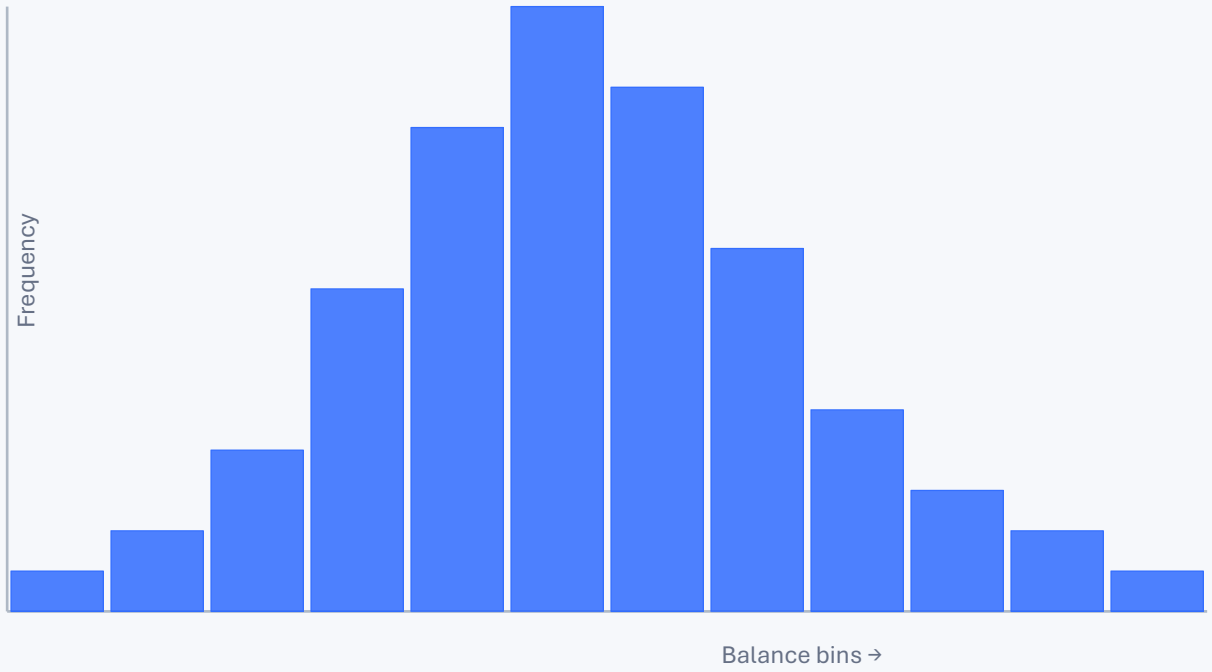
Visualisation is part of reasoning, not decoration.

Distribution / frequencies of one numeric variable	**Histogram / Box plot**
Relationship between two numeric variables	**Scatter plot**
Metric over time(eg. for time series)	**Line chart**
Compare categories	**Bar / Column chart**
Missingness / quality exceptions	**Conditional formatting / table**

Avoid charts that hide the data structure. Pie charts, 3D effects and decorative clutter are often poor analytical choices.

Histogram: see the distribution, not only the average

A histogram groups numeric observations into bins and shows how frequently values occur.



Question: “What does the distribution of customer balances look like?”

Reveals

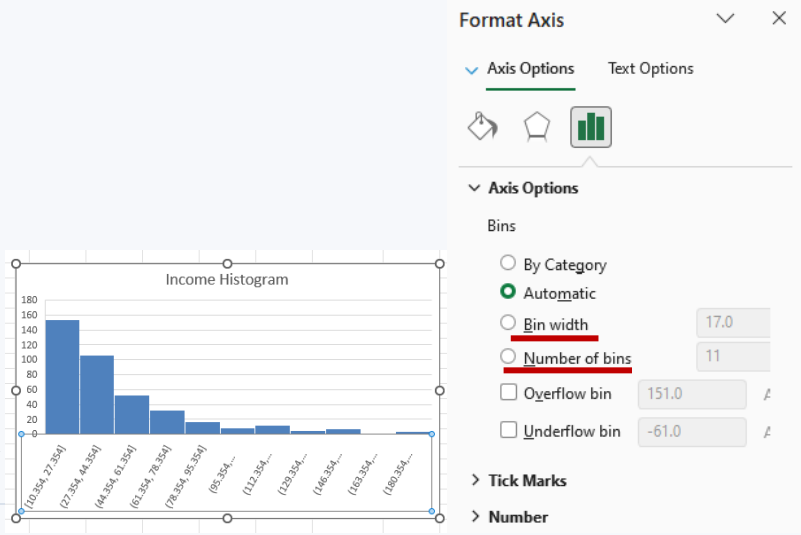
Skewness · multiple modes · gaps · tails · unusual peaks.

Excel

Insert → Statistical Chart → Histogram, or Data Analysis → Histogram.

Watch the bins

Too few bins hide structure; too many bins create noise.



Histogram: why in business?

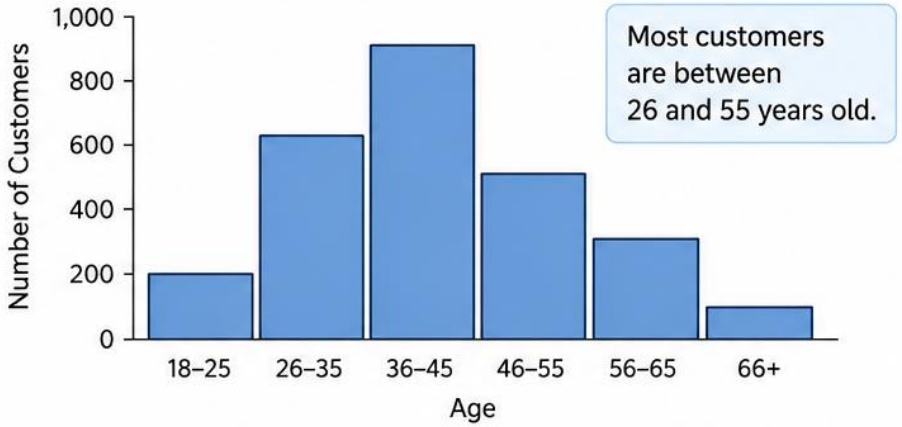
Two Business Needs for a Histogram

A histogram shows the distribution of a numeric variable, helping you understand the pattern, spread and frequency of values.

1 Understand the Distribution of a Key Metric

A histogram helps you see the overall shape of the data (e.g. typical values, variability, skewness) to support business decisions.

Customer Age Distribution

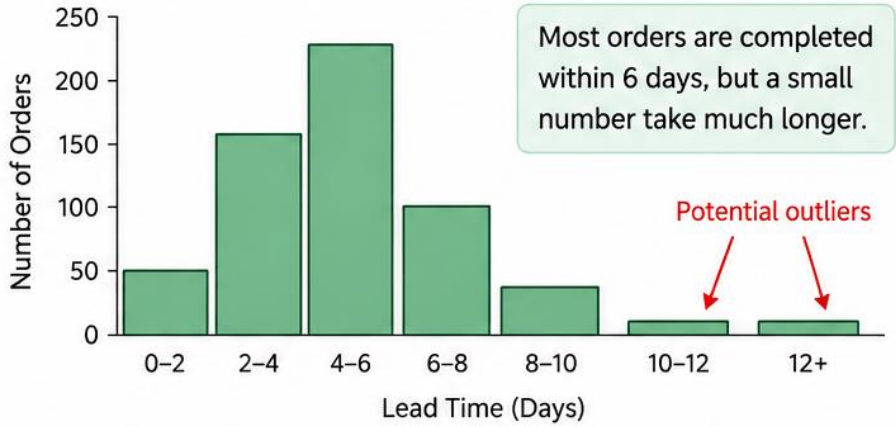


Business outcome: Use this insight to tailor marketing campaigns, product offerings or communication strategies to the main customer segments.

2 Monitor Process Performance and Detect Outliers

A histogram shows how frequently values occur, helping you identify unusual values, variability and whether the process meets the desired range.

Production Lead Time Distribution



Business outcome: Identify and investigate delayed orders, remove bottlenecks and improve operational efficiency.

Box plot: compact view of centre, spread and outliers

The box spans Q1 to Q3, the line is the median, whiskers show the non-outlier range, and points may mark outliers.



Strength

Very fast comparison across customer segments, products or branches.

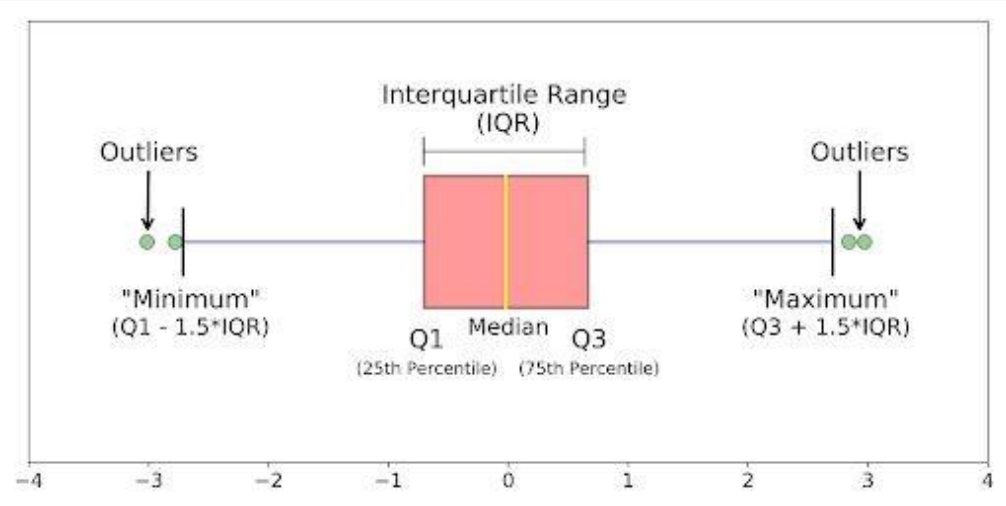
Robust centre

Median and quartiles are less sensitive to extreme values than the mean.

Excel

Insert → Statistical Chart → Box & Whisker.

Box plot: the classical semantics



Potential outliers might be due to data-entry mistakes; otherwise, we have to keep them. Deleting them is professional malpractice



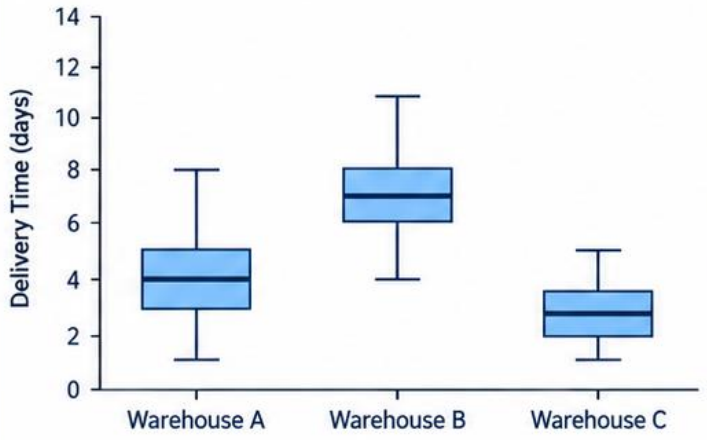
Box plot: why in business?

Two Business Reasons to Use a Box Plot

1 Compare distributions across groups

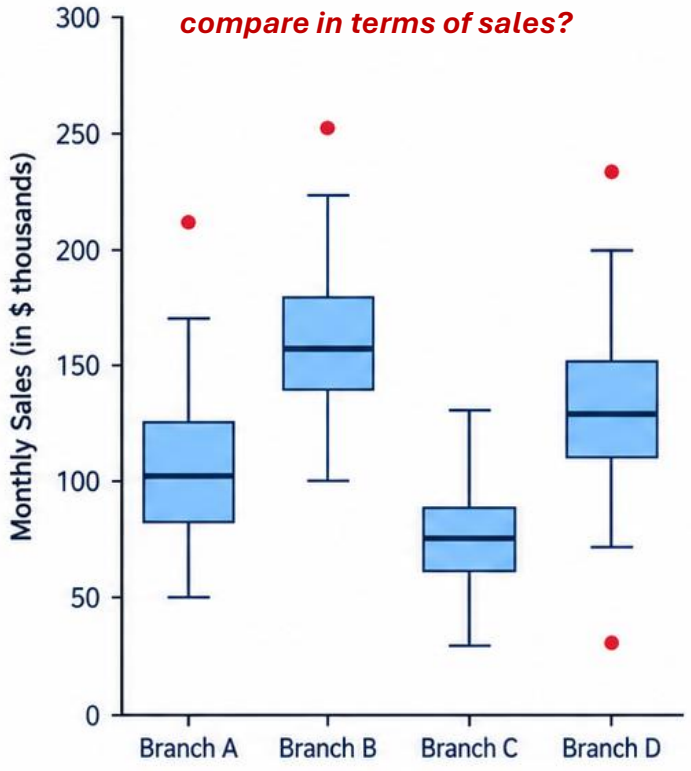
A box plot helps compare the median, spread, and variability across categories such as branches, product lines, teams, or customer segments.

Example: Delivery Time by Warehouse



Box Plot

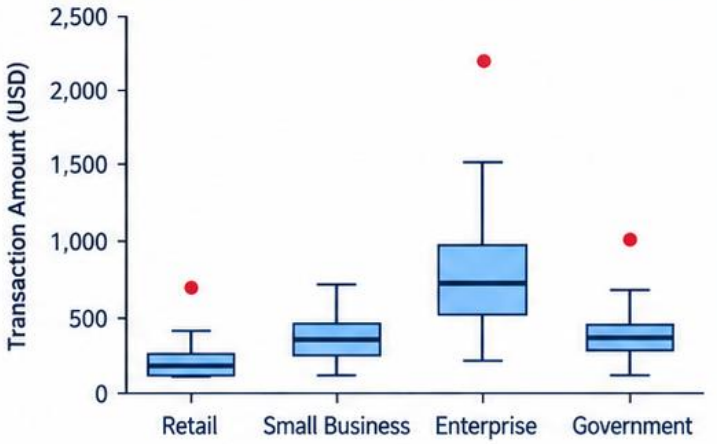
Business question: ***how branches compare in terms of sales?***



2 Detect outliers and unusual variation

A box plot helps identify unusual values, extreme cases, and groups with unstable performance, for example suspicious transactions, defective batches, unusually slow service times, or abnormal store results.

Example: Transaction Amount by Customer Segment



A box plot summarizes the median, quartiles, spread, and potential outliers.

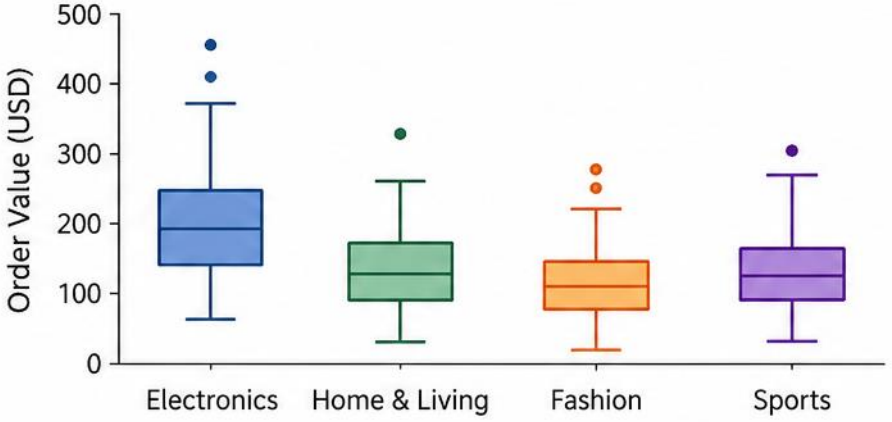
Compared Box plots: why in business?

Two Business Needs for a Comparative Boxplot

1 Compare Performance Across Groups

A comparative boxplot helps identify differences in distribution (median, variability, outliers) between groups, such as product lines, regions or customer segments, to support data-driven decisions.

Order Value by Product Category

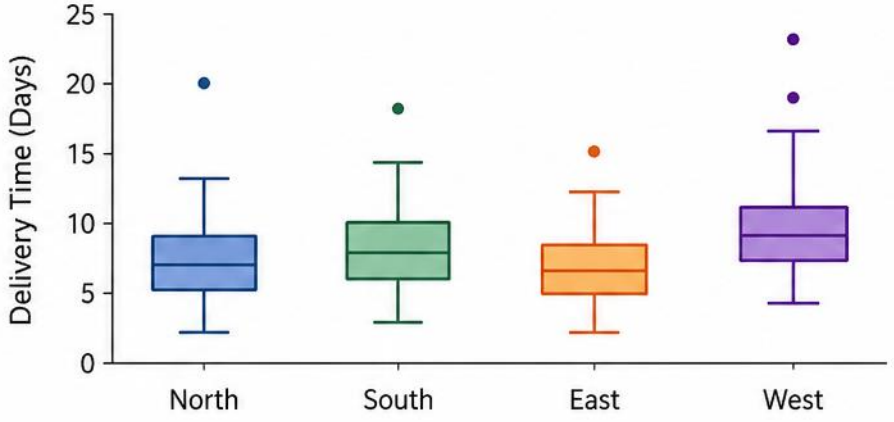


Business outcome: Understand which product categories generate higher order values and which show more variability, to optimize pricing, inventory and marketing strategies.

2 Assess Consistency Across Locations

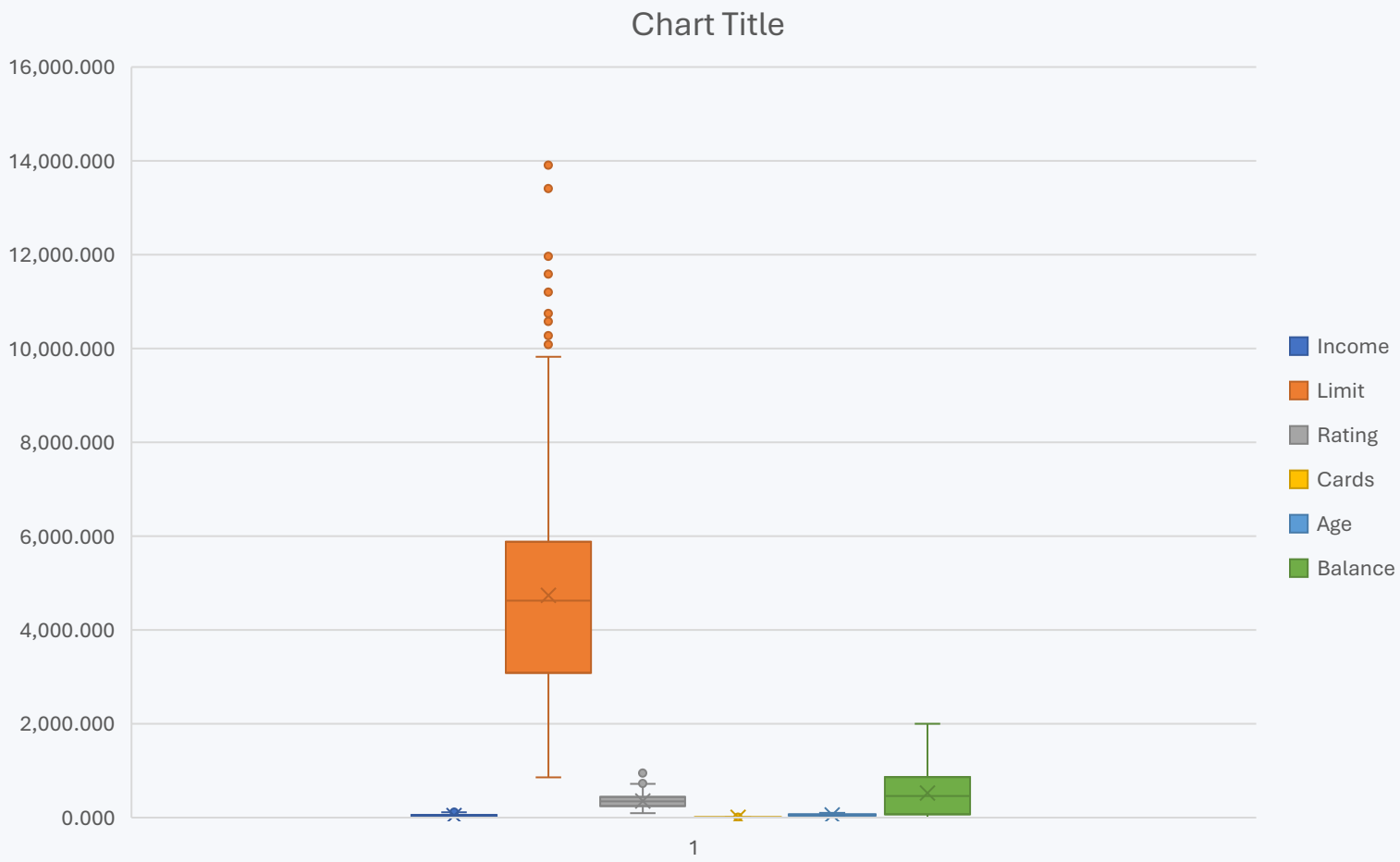
A comparative boxplot allows you to evaluate and compare the spread and central tendency of a metric across different locations (e.g. stores, branches, or regions), highlighting inconsistencies or outliers that may require attention.

Delivery Time by Region



Business outcome: Identify regions with longer or more variable delivery times to improve operational efficiency and ensure a consistent customer experience across locations.

Compared Box plots: why in business?



The columns have very different scales

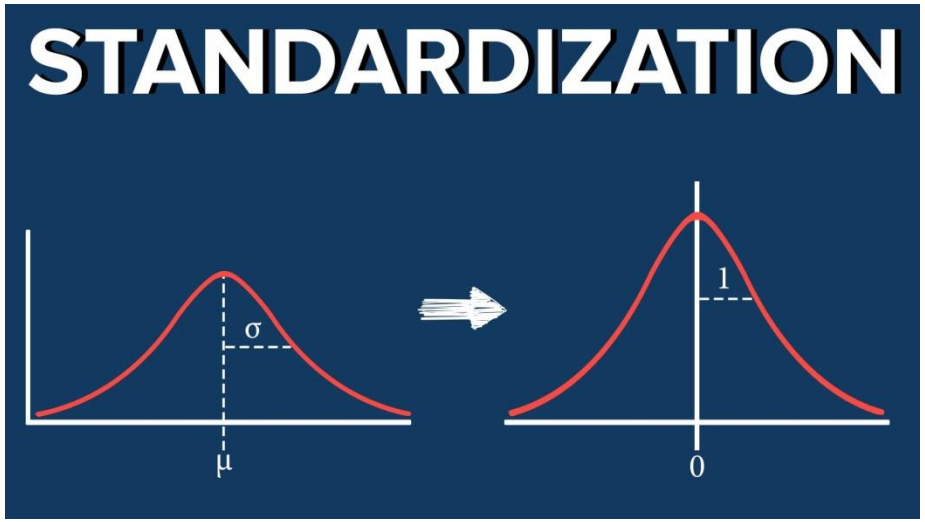
We cannot coompare boxplots in the Credit_ISLR dataset (numeric columns) as the Limit boxplot dominates over the others (which are non even understandable).

I's a TYPICAL situation in Data Science.
What we have to do?

STANDARDISATION

Box plots: required to compare unevenly distributed data

Two typical useful techniques to compare unevenly distributed data by boxplots → this way, we can then appreciate differences



Normalization Formula

$$X_{normalized} = \frac{(X - X_{minimum})}{(X_{maximum} - X_{minimum})}$$

Standardising a numeric column in Excel (Z-score)

Standardisation expresses each value as a number of standard deviations from the mean.

CONCEPT

Z-score formula

$$z = (x - \mu) / \sigma$$

Interpretation

z = 0 → value is at the mean
z = +1 → one standard deviation above
z = −1 → one standard deviation below

EXCEL FORMULA

For values in A2:A101

```
=(A2-AVERAGE($A$2:$A$101))
/STDEV.S($A$2:$A$101)
```

Copy the formula down the column.

EXAMPLE

Annual income (k€)

Original	Z-score
20	−1.26
40	−0.63
60	0.00
80	+0.63
100	+1.26

When standardisation is useful

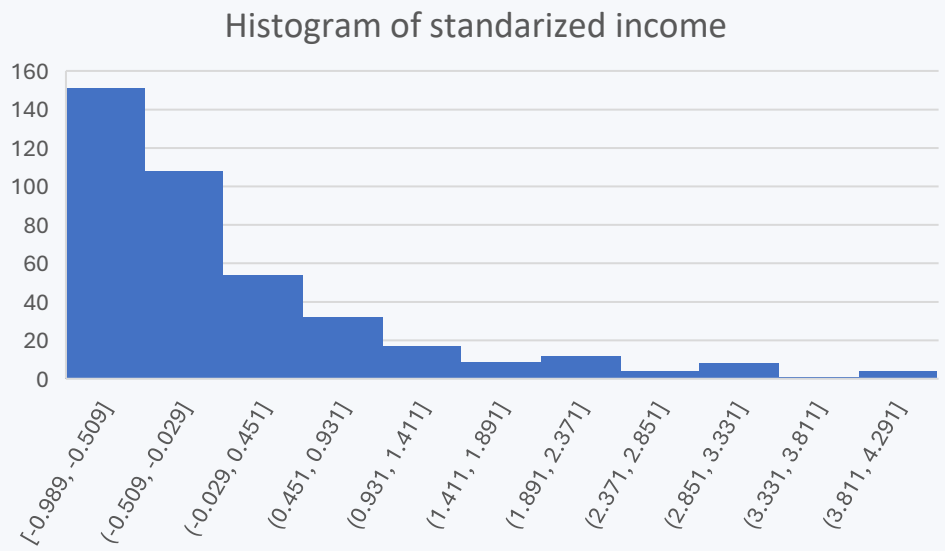
- Variables have very different units or scales
- Distance-based methods are used (e.g. k-NN, k-means)
- You want a common scale centred around zero

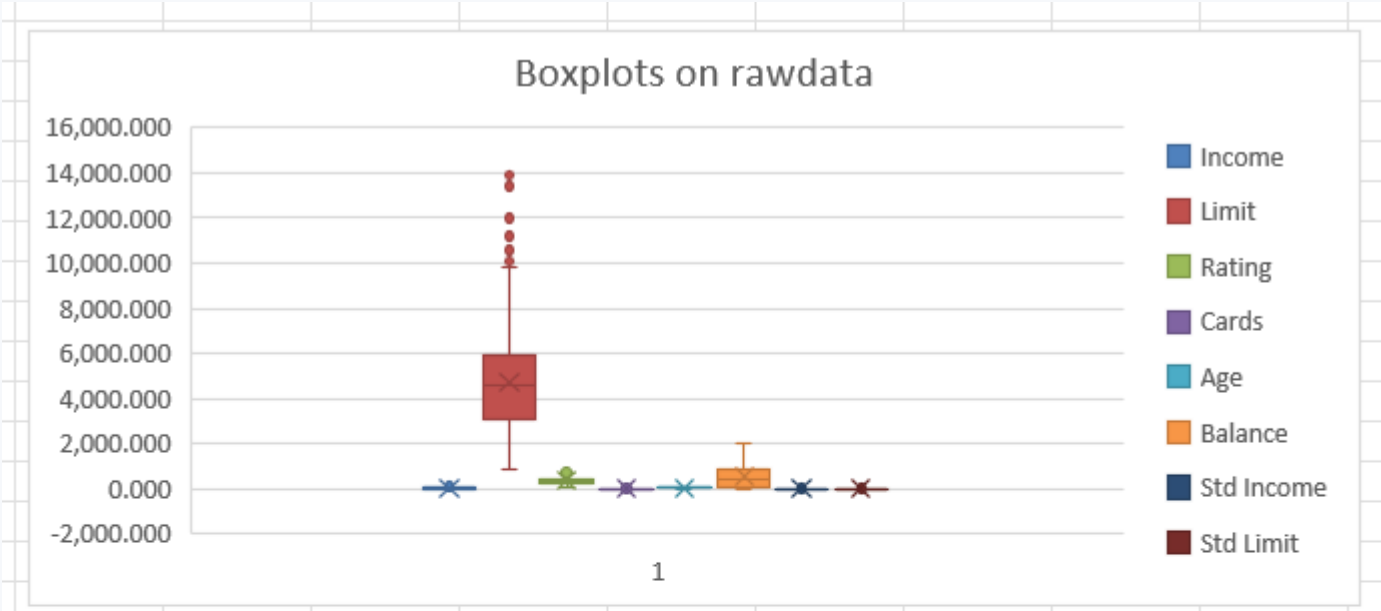
Important

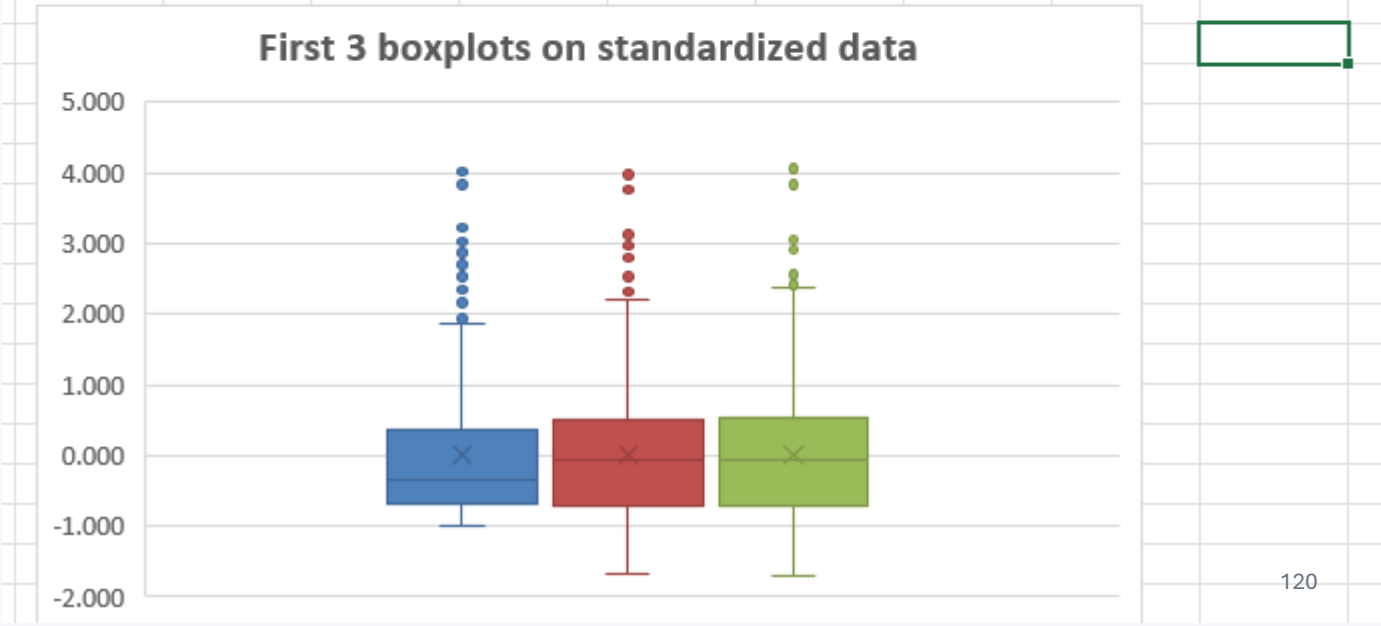
Standardisation does not force values into a fixed range. Values below −3 or above +3 are possible and may highlight unusual observations.

Takeaway: standardisation changes the scale, not the relative ordering of the observations.

Boxplot comparison on standardized data







Apart from box comparison, preliminary standardisation of raw data is required with many data science models

Normalising a numeric column in Excel (Min–Max)

Min–Max normalisation rescales a variable to a fixed interval, most commonly 0 to 1.

CONCEPT

Min–Max formula

$$(x - \min) / (\max - \min)$$

Result

minimum → 0
maximum → 1
all other values → proportionally between 0 and 1

EXCEL FORMULA

For values in A2:A101

```
=(A2-MIN($A$2:$A$101))
/(MAX($A$2:$A$101)-
MIN($A$2:$A$101))
```

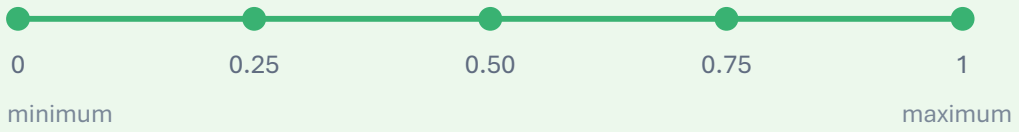
For a 0–100 scale, multiply the result by 100.

EXAMPLE

Customer score

Original	0–1 scaled
20	0.00
40	0.25
60	0.50
80	0.75
100	1.00

How to read the new scale



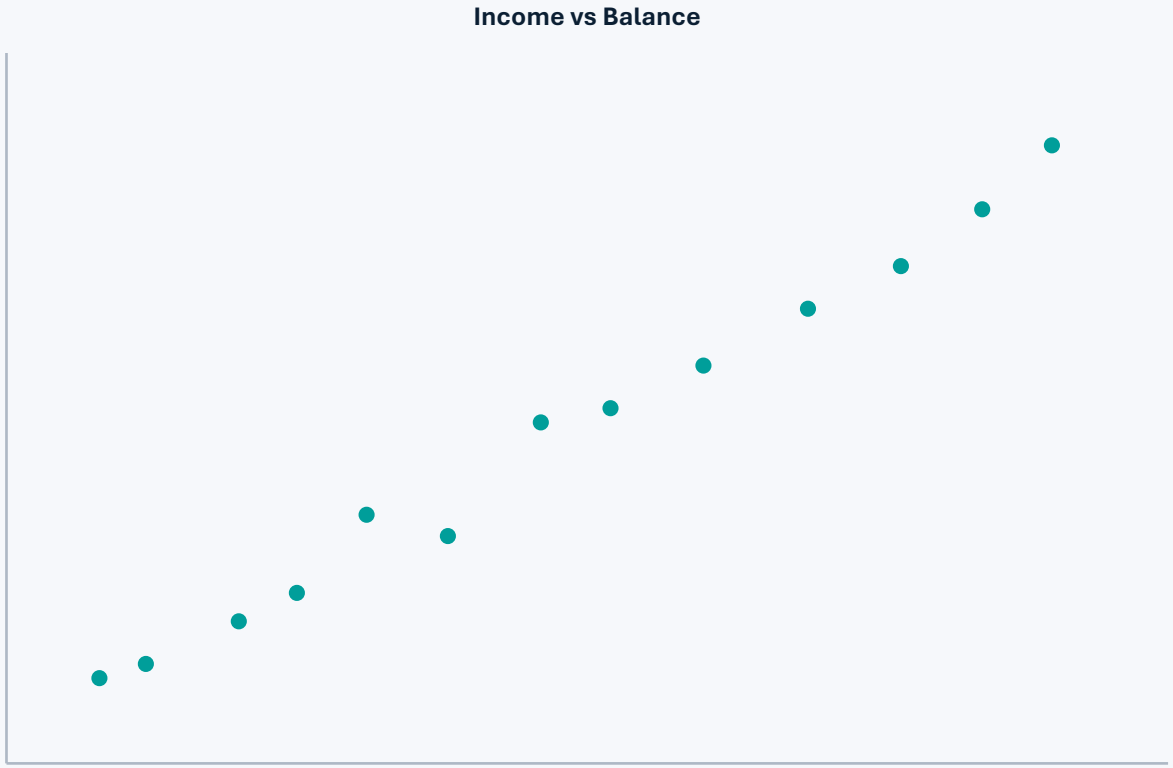
Watch out for outliers

Min–Max normalisation is highly sensitive to extreme minimum or maximum values. One outlier can compress most observations into a very narrow part of the 0–1 interval.

Takeaway: normalisation maps values to a chosen range; standardisation centres and scales them around the mean.

Scatter plot: the essential chart for relationships

Every point is one observation positioned by two numeric variables.



Look for

Direction · strength · curvature · groups · isolated points.

Add meaning

Colour points by segment or branch; use labels only for key exceptions.

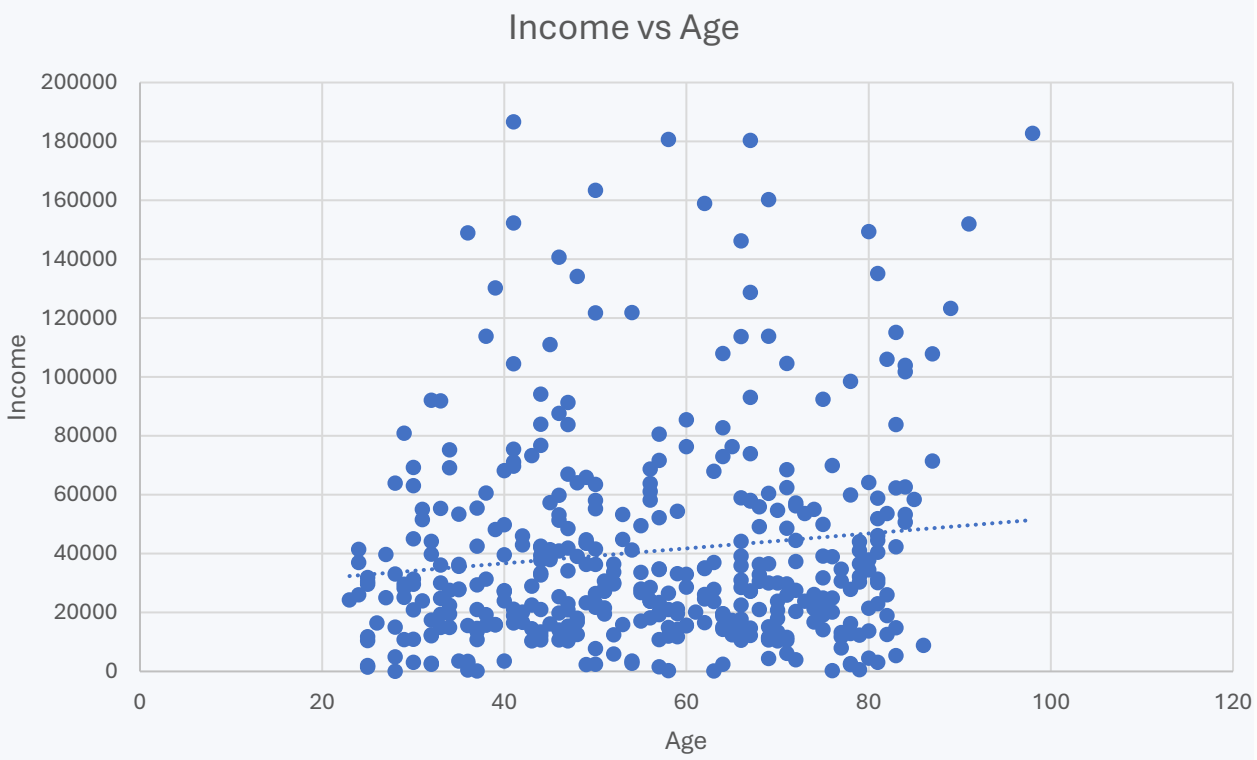
Excel

Insert → Scatter. Add a trendline only when it supports the question.

```
Pair with =CORREL(...) – never interpret the coefficient without seeing the points.
```

Scatter plot: the essential chart for relationships

The steps of the non-trivial procedure to create a scatter plot



How to Create an Age vs Income Scatter Plot in Excel

1. Select the **Age** column in the Excel Table. **Age will be used as the X variable and therefore displayed on the horizontal (X) axis.** Move the mouse pointer to the top edge of the `Age` header until a **black downward arrow** appears, then click.
2. Go to **Insert → Scatter (X, Y) → Scatter with Only Markers**. Choose the first Scatter chart type: points only, with no connecting lines.
3. Excel will create an initial chart. **Right-click the chart → Select Data**.
4. Remove any series automatically created by Excel by selecting them and clicking **Remove**.
5. Click **Add** and define the new series:
   - **Series name:** `Age vs Income`
   - **Series X values:** select all the values in the **Age** column, excluding the header.
   - **Series Y values:** select all the values in the **Income** column, excluding the header.
6. Confirm by clicking **OK → OK**.
7. Check that the resulting scatter plot has:
   - **Age** on the X-axis
   - **Income** on the Y-axis
   - A single series of points
8. Add the axis titles. Select the chart and go to **Chart Design → Add Chart Element → Axis Titles**. Select both **Primary Horizontal** and **Primary Vertical**.
9. Enter the axis titles:
   - **Horizontal (X) axis:** `Age`
   - **Vertical (Y) axis:** `Income`
10. Set the chart title to **Income vs Age**.

Each point represents **one customer (one row of the Excel Table)**:

$$(\text{Age}, \text{Income})$$

For example, a customer with **Age = 34** and **Income = 14,891** is represented by the point:

$$(34,\ 14891)$$

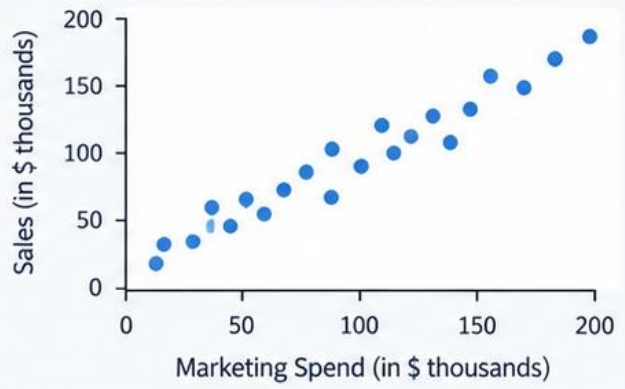
Scatter plot: why in business?

Two Business Reasons to Use a Scatter Plot

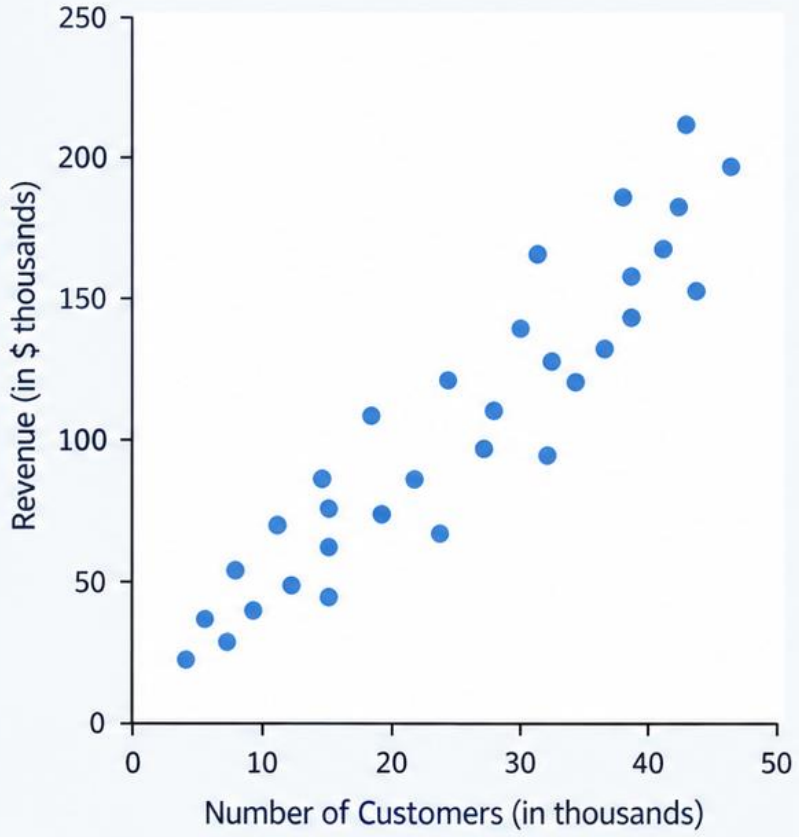
1 Discover relationships between variables

A scatter plot helps reveal whether two variables move together, for example sales vs marketing spend, income vs loan balance, or age vs premium.

Example: Sales vs Marketing Spend



Scatter Plot



2 Detect outliers and unusual cases

A scatter plot helps spot abnormal observations, such as suspicious transactions, unusual customers, underperforming stores, or faulty measurements.

Example: Transaction Amount vs Customer Age



Each dot represents one observation (customer, store, product, month, or transaction).

Line chart: patterns across time

Use line charts when order matters — especially for time series.



Best for

Trend · seasonality · regime shifts · before/after comparisons.

Banking example

Monthly deposits, delinquency rate, call volume or transaction counts.

Caution

Do not connect unordered categories with a line.

```
Excel: put a real date in the X-axis whenever possible.
```

Line chart: why in business?

Two Business Needs for a Line Chart

A line chart shows how a metric changes over time, helping you identify trends, patterns and turning points.

1 Track Performance Over Time

A line chart helps you monitor a key business metric over time, understand the trend and assess whether you are meeting your goals.



Business outcome: Evaluate business performance, identify growth (or decline) and make informed decisions (e.g. adjust marketing, sales or resources).

2 Detect Seasonality and Plan Ahead

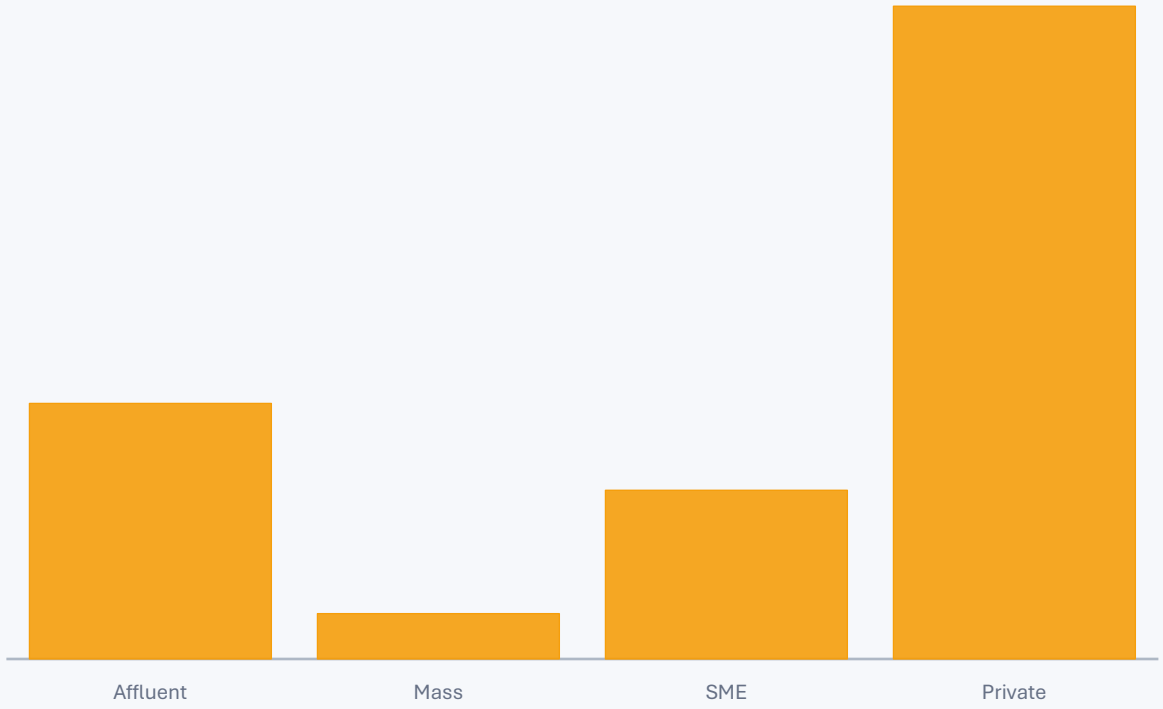
A line chart helps you identify recurring patterns (e.g. seasonal demand) so you can plan inventory, staffing and budget more effectively.



Business outcome: Anticipate seasonal demand, optimize inventory and staffing, and reduce costs while improving customer service.

Bar and column charts: compare categories clearly

The baseline should normally start at zero because bar length encodes magnitude.



Use for

Counts, averages, totals and rates by category.

Good practice

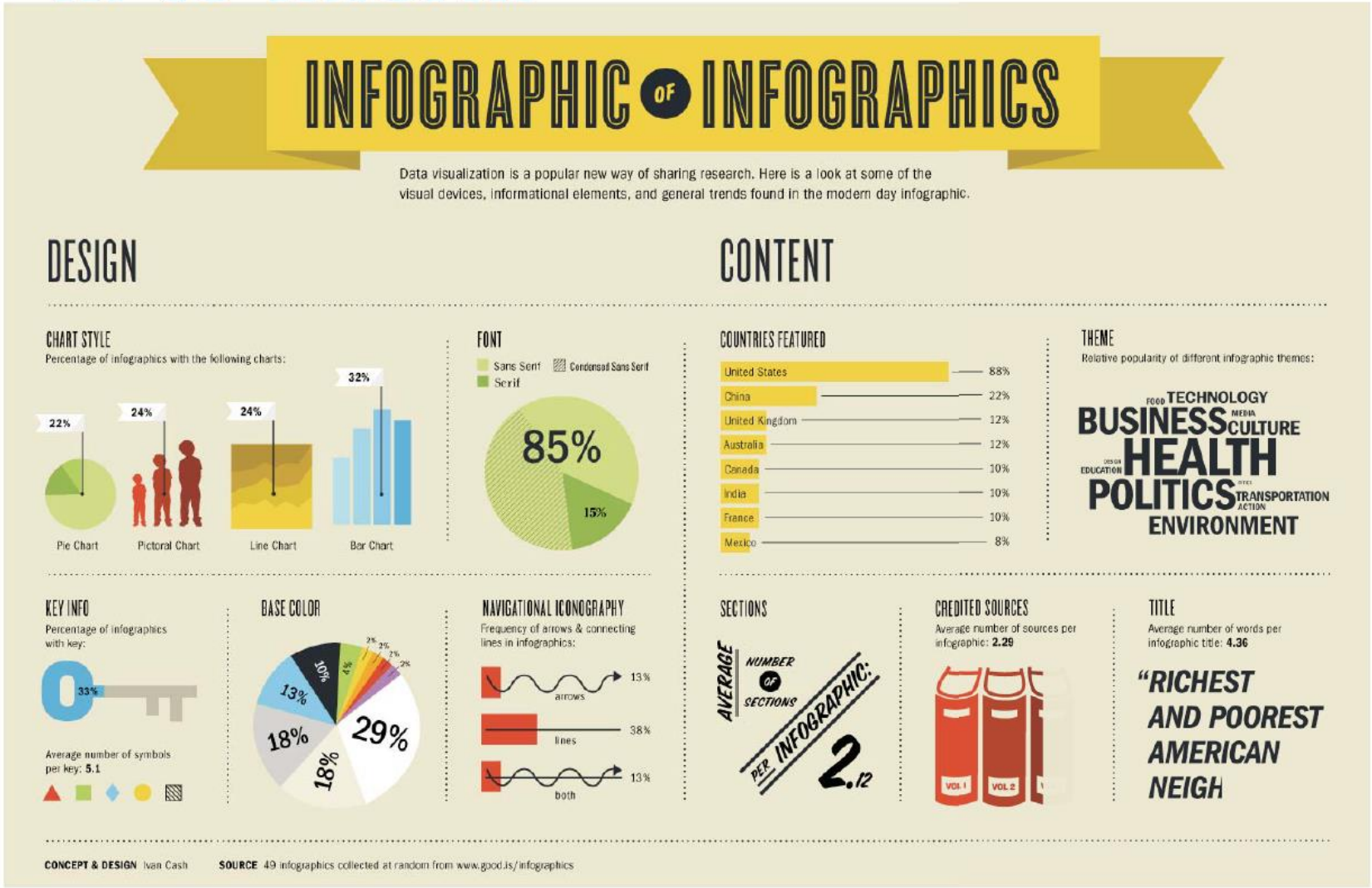
Sort bars when ranking matters; keep category labels readable.

Avoid

3D bars, too many categories and truncated baselines.

Question: “Which customer segment has the highest average balance?”

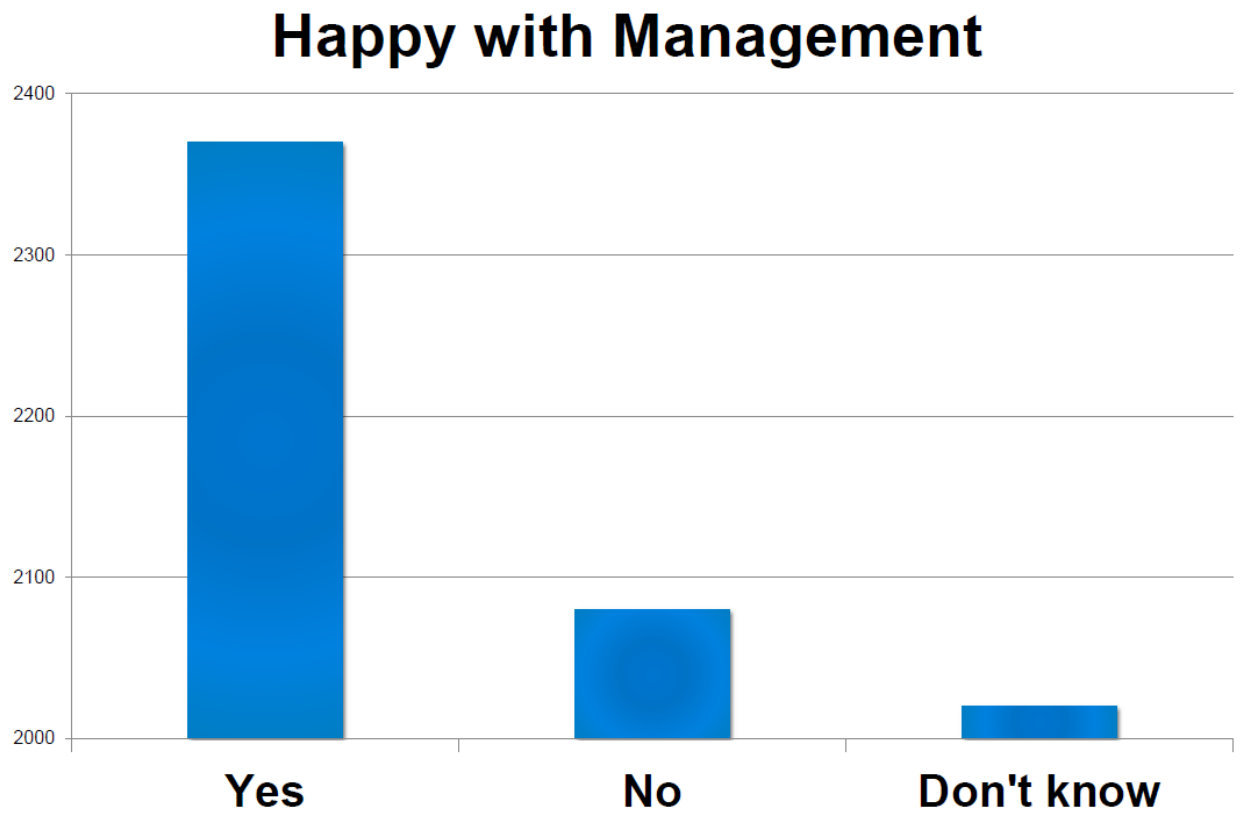
... to Inform



Presentazione di
licrosoft PowerPoi

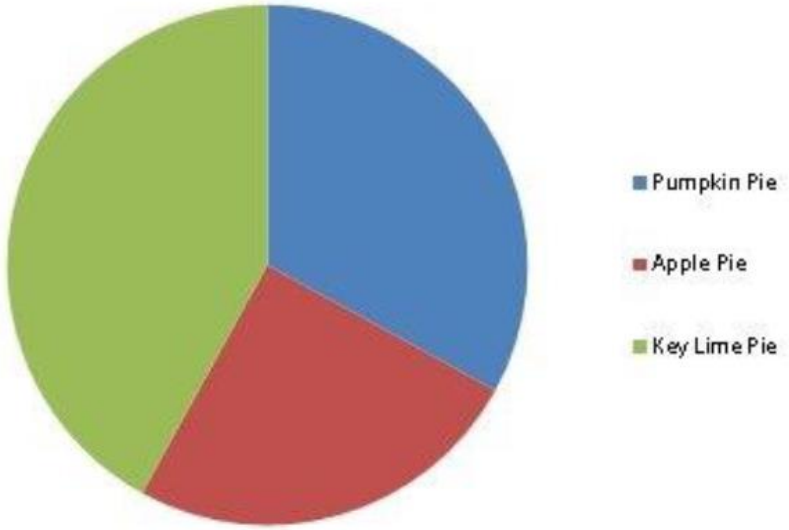
Fonte: Jos van Dongen

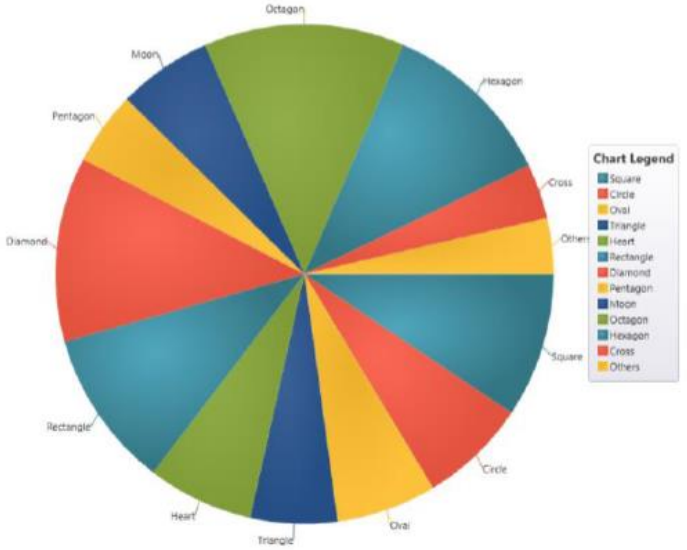
...to Lie



Fonte: Jos van Dongen

Be Careful with Pie Charts...







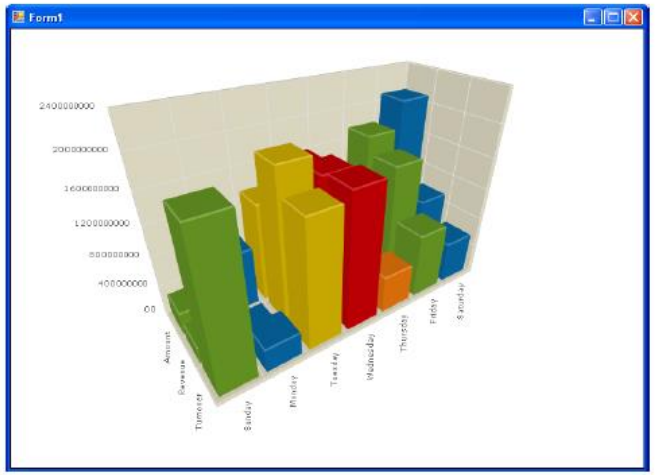
Total = 100%
<=5 categories
Part to whole
Rough indication

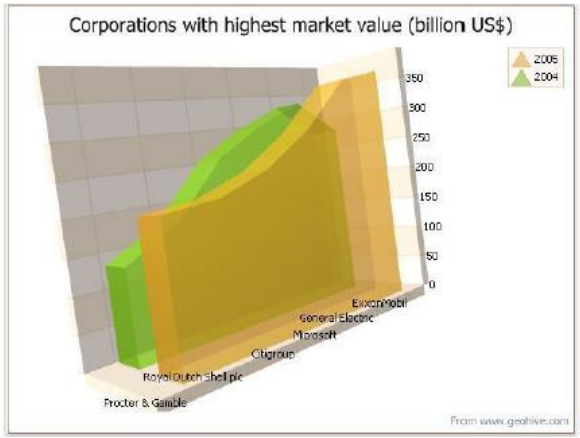


Even in this case, the analyst shoul pay a lof of attention to the mix of colours (do they match?)

Fonte: Jos van Dongen

And Never, EVER, use 3D!



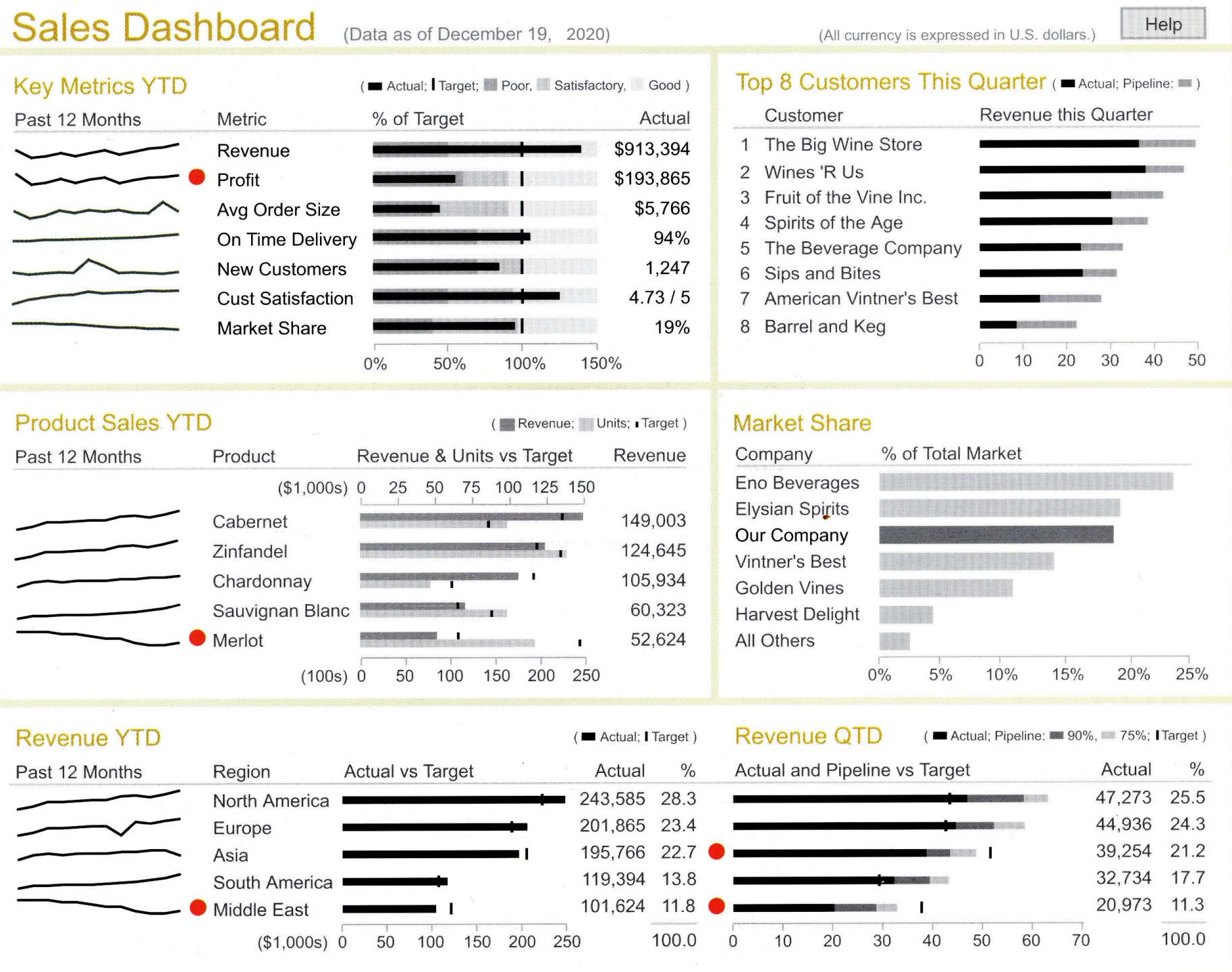






This is MALPRACTICE in Data Visualization

Fonte: Jos van Dongen



The most frequent dashboard

The information:

- Sales revenue (*revenue*)
- Sales pipeline *revenue* (i.e., broken down by probability level)
- Profit (*profit*)
- Customer satisfaction
- Top 10 customers
- Market share percentage (*market share*)

For each one:

- Level of aggregation?
- Unit of measure?
- Contextual information?
- Graphical tools?
- Comparison with other measures?
- Positioning?

Scannerizzata a 1200 dpi

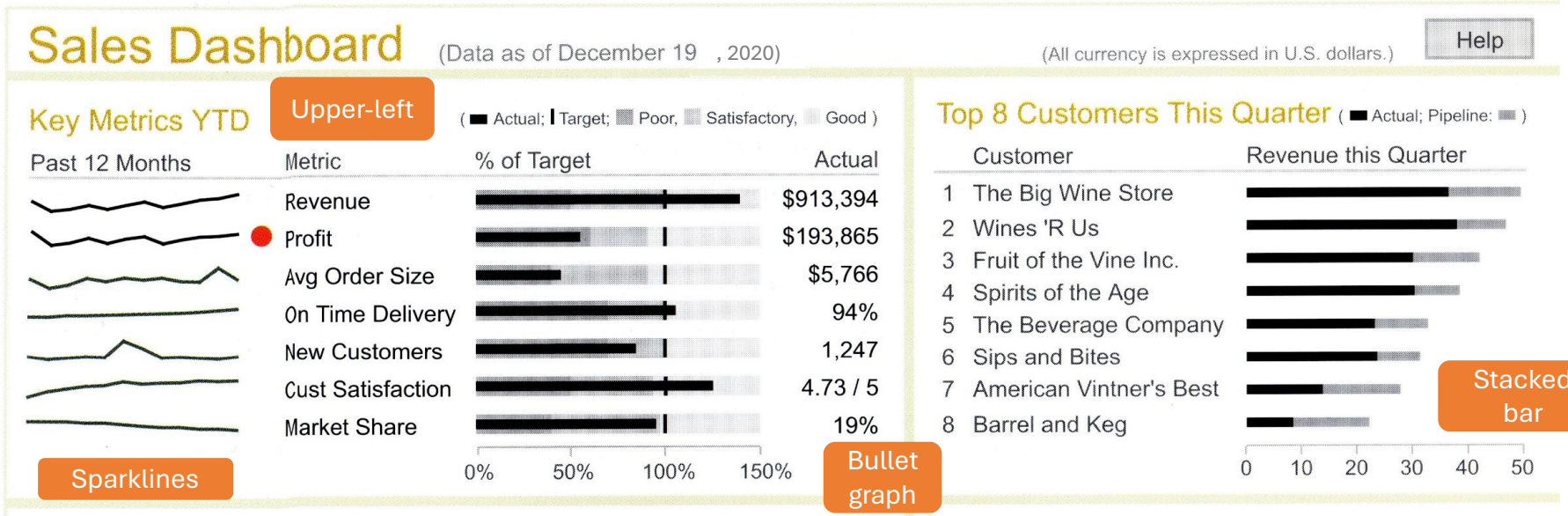
Definition of *year-to-date*

What colours?

What fonts?

What plots?

Images?



NOTE:

Use colors sparingly

A lot of information in little space

Text and graphics

Stacked bar

Reduction of *non-data pixels*

No contextual help or instructions

Conditional Formatting

The baseline should normally start at zero because bar length encodes magnitude.



Link to the video

Data Validation

Strictly speaking, this topic is NOT data visualization. However, while Data validation and conditional formatting in Excel are separate tools, you can easily connect them to create dynamic, interactive spreadsheets.



Excel Custom Data Validation (Use formulas to check for text, numbers & length)
340K visualizzazioni 8 anni fa

Link to the video

Chapter title



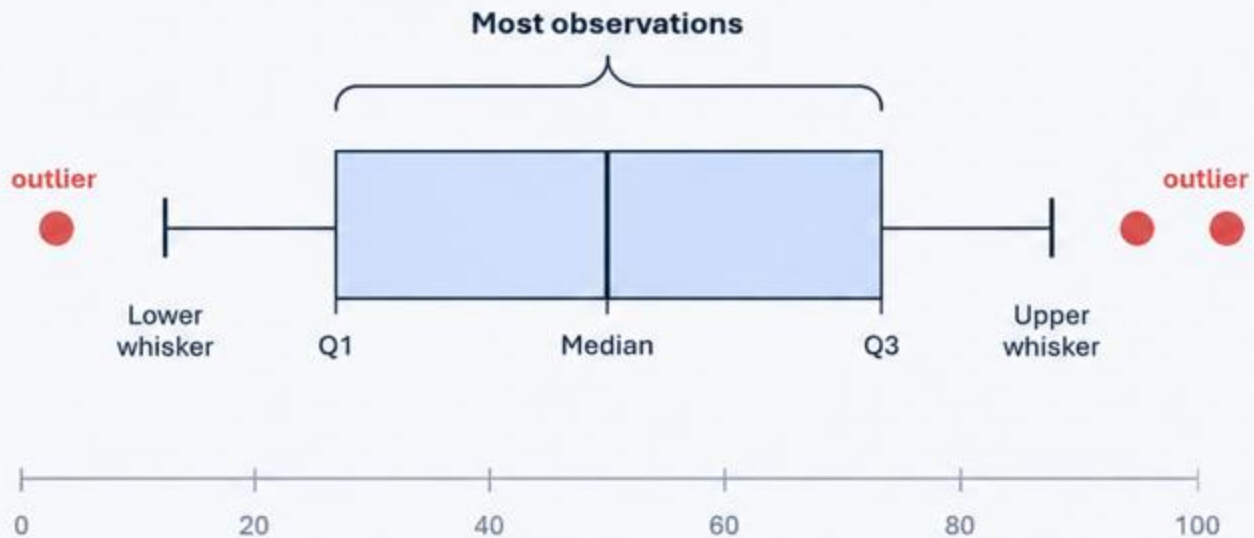
07 – Data issues

Outliers Identification and Management

Outliers in one dimension: detect extremes with a boxplot

For a single variable, outliers are unusually low or high values compared with the bulk of the distribution.

Typical rule: flag values below Q1 – 1.5×IQR or above Q3 + 1.5×IQR



Data error

Typing mistake, wrong unit, duplicate entry.

Valid rare case

Exceptional customer, unusual transaction, uncommon event.

Process change

New policy, migration, market shock or operational shift.

Interesting signal

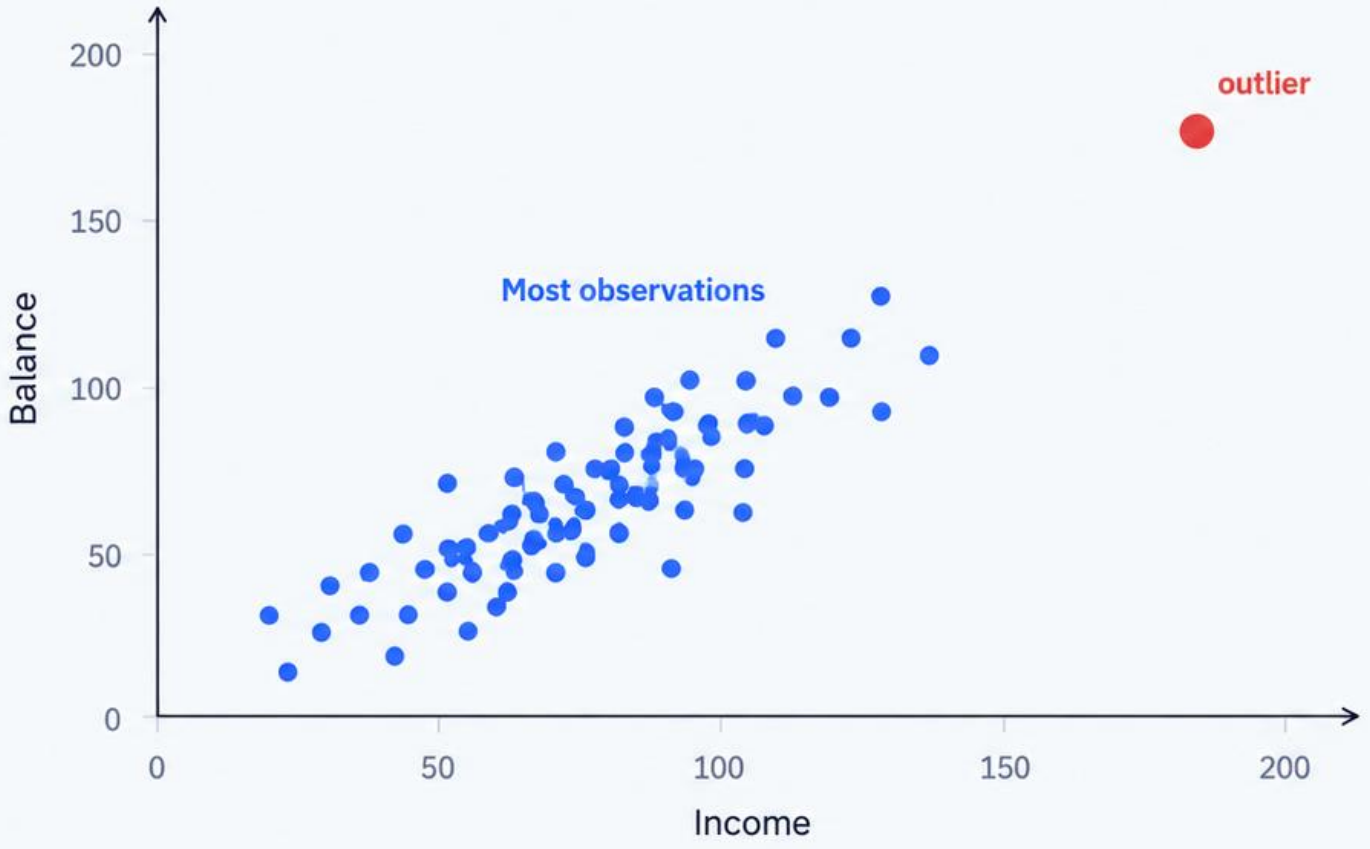
Potential fraud, risk event or business opportunity.

Never remove an outlier without understanding why it is there.

Outliers in two dimensions: detect unusual cases in bivariate space

With two variables together, a point may look acceptable on each variable separately but still be unusual in combination.

Typical idea: flag points far from the main pattern in 2D space



Data error

Wrong value, wrong unit, duplicate record, input error.

Valid rare case

Exceptional customer, unusual combination of values.

Process change

New policy, market change, operational shift or new segment.

Interesting signal

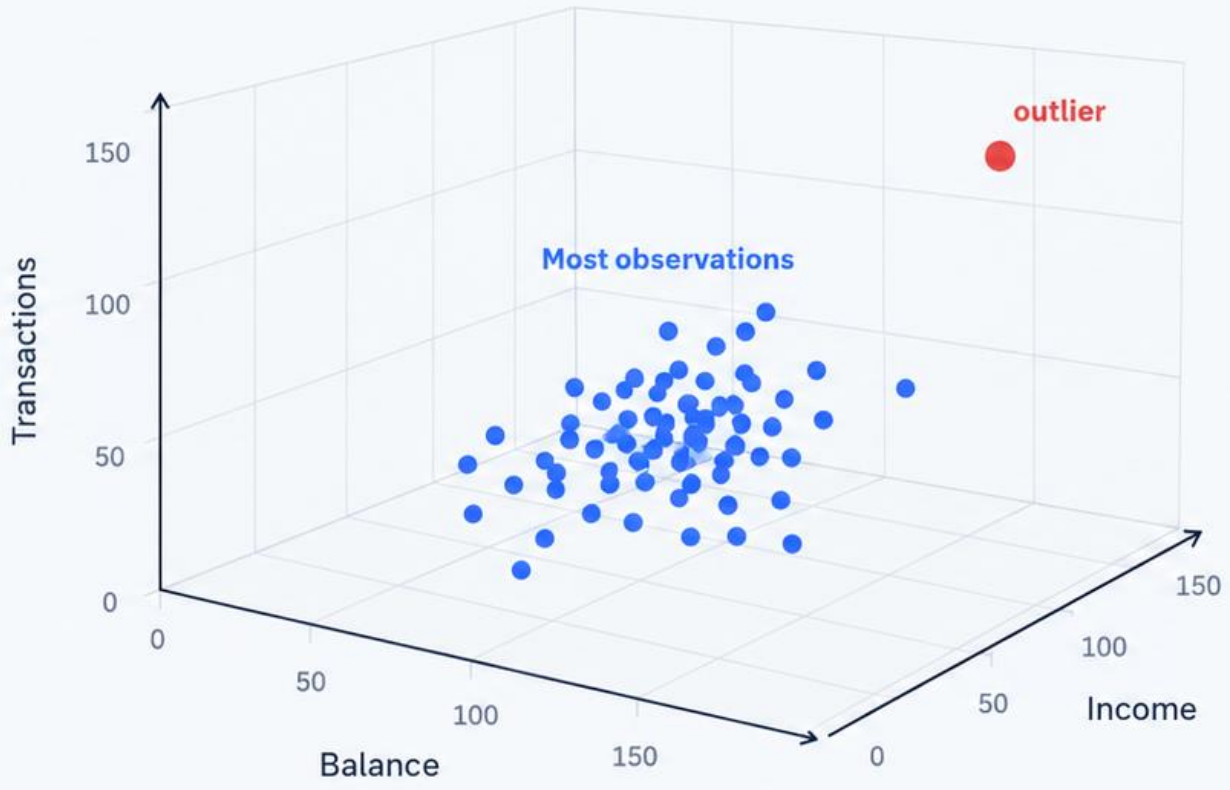
Potential fraud, anomaly or business opportunity to investigate.

Never remove a bivariate outlier before understanding why it is unusual.

Outliers in three dimensions: detect unusual cases in multivariate space

With several variables together, a point may look normal on each variable separately but still be unusual in combination.

Typical idea: flag points far from the main cloud in 3D space



Data error

Wrong unit, duplicate record, sensor/input issue.

Valid rare case

Exceptional customer profile or uncommon combination of values.

Process change

New policy, migration, operational shift or new segment.

Interesting signal

Potential fraud, anomaly or business opportunity to investigate.

Never remove a multivariate outlier before understanding why it is unusual.

Detecting outliers with the IQR rule

The interquartile range is robust and works well as a first-pass rule.



IQR rule

Q1 = 25th percentile
Q3 = 75th percentile
IQR = Q3 − Q1
Flag values below Q1 − 1.5×IQR or above Q3 + 1.5×IQR (the boxplot rule)
Another rule: Flag values below Average − 2×std or above Average + 2xstd (on standardized data).

```
Q1: =QUARTILE.INC(range,1)
```

```
Q3: =QUARTILE.INC(range,3)
```

```
IQR: =Q3-Q1
```

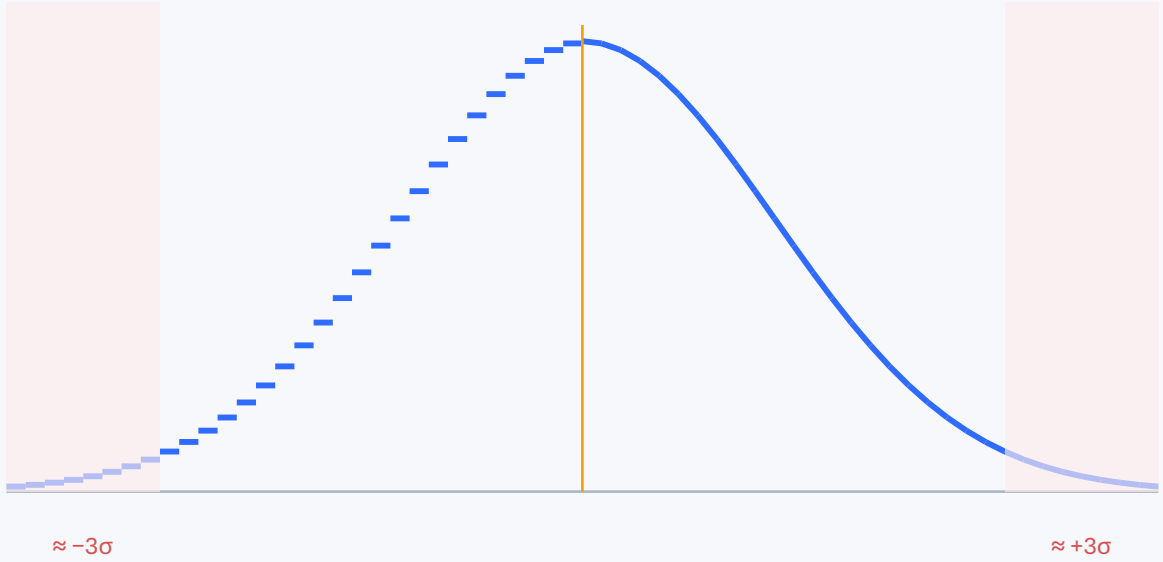
```
Lower = Q1-1.5*IQR
```

```
Upper = Q3+1.5*IQR
```

A statistical flag is a prompt to investigate, not a business verdict.

Detecting outliers with z-scores

A z-score expresses how many standard deviations an observation lies from the mean.



Formula

z = (x − mean) / standard deviation

```
=(B2-AVERAGE($B$2:$B$101))/STDEV.S($B$2:$B$101)
```

Caution

z-scores are sensitive to outliers and assume the mean/std are meaningful for the distribution.

For strongly skewed banking data, the IQR rule is often safer as a first screen.

Handling outliers: preserve information and business meaning

The right treatment depends on why the value is extreme.

Validate

Check source system, unit, sign, duplicate and business event.

Correct

Fix confirmed errors using traceable rules.

Keep

Valid extreme cases may be essential to risk analysis.

Transform

Log scale can reduce skew for positive amounts.

Cap / Winsorise

Limit extreme influence when methodologically justified.

Segment / model separately

Different populations may need different treatment.

Document every exclusion or transformation — especially in regulated analytical processes.

Finding Outliers in Excel

Several methods, one goal: **identify unusual values** that may need investigation

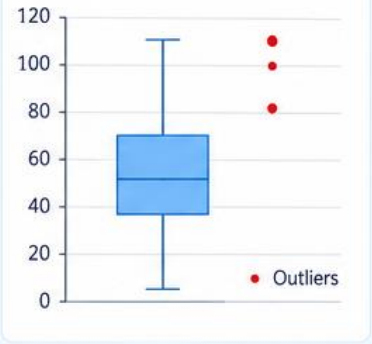
over just ONE column

1 Visual inspection

Box plot or scatter plot

Outliers often appear as isolated points in a box plot or scatter plot.

Box plot (example)



In Excel

1. Select the data.
2. Go to Insert → Statistical Chart → Box & Whisker.
3. Look for points outside the whiskers (these are potential outliers).

Available in Excel 2016 and later (Insert → Insert Statistic Chart).

2 Z-score method

Values far from the mean

An observation is an outlier if its z-score is typically greater than 3 (in absolute value).

$$z = \frac{x - \text{mean}}{\text{standard deviation}}$$

Value	Z-score	Outlier?
52	0.12	No
55	0.35	No
58	0.62	No
120	3.48	Yes
49	−0.21	No

Standardized data

In Excel

1. Calculate the mean (AVERAGE) and standard deviation (STDEV.S).
2. Compute the z-score for each value.
3. Flag as outlier if ABS(z) > 3. 3 std dev

Excel formula (in cell C2)

```
=(A2-AVERAGE($A$2:$A$100)) /
 STDEV.S($A$2:$A$100)
```

3 IQR method

Values outside a typical range

Outliers are values below Q1 − 1.5·IQR or above Q3 + 1.5·IQR.

$$\text{IQR} = \text{Q3} - \text{Q1}$$
$$\text{Lower} = \text{Q1} - 1.5 \cdot \text{IQR}$$
$$\text{Upper} = \text{Q3} + 1.5 \cdot \text{IQR}$$

Value	Outlier?
18	No
21	No
22	No
23	No
85	Yes
90	Yes

In Excel

1. Compute Q1 and Q3 using QUARTILE.
2. Calculate IQR, lower and upper bounds.
3. Flag values outside the bounds.

Excel formulas (example range A2:A100)

```
Q1:    =QUARTILE.INC(A2:A100,1)
Q3:    =QUARTILE.INC(A2:A100,3)
IQR:   =Q3 - Q1
Lower: =Q1 - 1.5*IQR
Upper: =Q3 + 1.5*IQR
Flag (in B2): =IF(OR(A2<Lower,A2>Upper),
                  "Yes","No")
```

4 Modified Z-score

More robust (uses median)

Uses the median and the Median Absolute Deviation (MAD). Consider an outlier if |M| > 3.5.

$$M = 0.6745 \cdot \frac{x - \text{median}}{\text{MAD}}$$

Value	Modified Z	Outlier?
18	−0.12	No
21	0.05	No
22	0.18	No
23	0.31	No
85	4.21	Yes
90	5.03	Yes

In Excel

1. Compute the median (MEDIAN),
2. Calculate the MAD (median of absolute deviations).
3. Compute the modified z-score.
4. Flag as outlier if ABS(M) > 3.5.

Excel formula (in B2)

```
=0.6745*(A2-MEDIAN($A$2:$A$100))
/MEDIAN(ABS($A$2:$A$100)-MEDIAN
($A$2:$A$100)))
```

5 Domain knowledge

Business rules

Sometimes you know what values are unrealistic based on context (e.g. negative age, very high income).

Value	Rule	Outlier?
-5	Age < 0	Yes
220	Age > 120	Yes
35	—	No
105	—	No

In Excel

1. Define business rules (e.g. min/max allowed values).
2. Use IF statements to flag violations.

Excel formula (example for Age in A2)

```
=IF(OR(A2<0,A2>120),"Yes","No")
```

Tip

Combine multiple methods: use statistical techniques to detect unusual values and domain knowledge to validate them.

Next steps

Identify outliers → Investigate (e.g. data entry error?) → Decide what to do (remove, correct, or keep) → Document your decision

Key takeaways

- There is no single best method.
- Visualize your data.
- Use statistical methods and domain knowledge.
- Always investigate before removing outliers.
- Document your choices.

Basic Rule for Identifying Outliers

Rule of thumb for one numeric data column

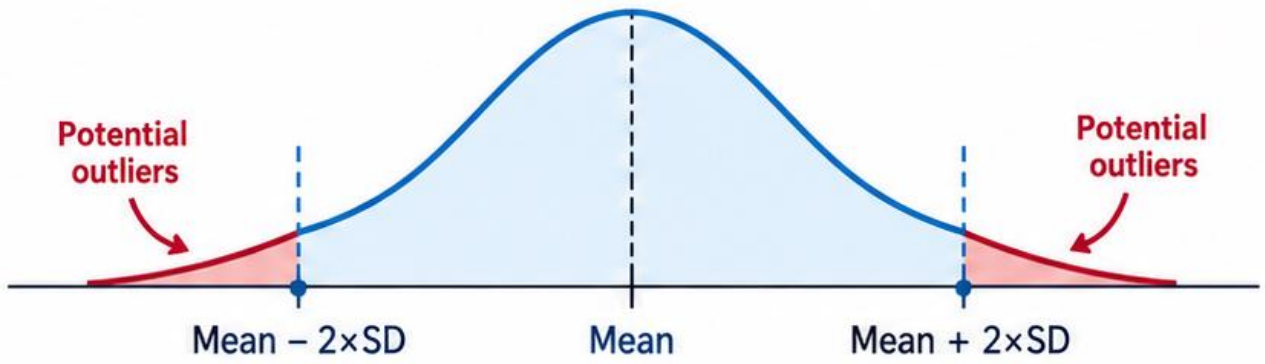
Data column

Value	
3	
7	
8	
9	
10	
10	
11	
12	
13	
14	
-5	← Low outlier
15	
16	
18	
20	
45	← High outlier

The rule

Average	=	Mean
Spread	=	Standard Deviation
Lower threshold	=	Mean − 2 × Standard Deviation
Upper threshold	=	Mean + 2 × Standard Deviation

A value is flagged as a **potential outlier** if it is **below Mean − 2×SD** or **above Mean + 2×SD.**



Steps

1. Compute the *mean.*
2. Compute the standard deviation.
3. $\bar{x} \pm 2\sigma$ Calculate mean − 2×SD and mean + 2×SD.
4. Flag values outside this interval.

This is a simple practical rule, not a universal law.

Synthetic CSV Dataset with Outliers

customers.csv

```
CustomerID,Age,Income_USD,Monthly_Spend_USD,Debt_USD,City
C001,34,42000,1800,6500,Turin
C002,41,51000,2200,7200,Milan
C003,29,39000,1600,4800,Rome
C004,37,47000,2100,5900,Naples
C005,45,56000,2400,8100,Genoa
C006,32,44000,1750,5300,Bologna
C007,38,49500,2050,6100,Florence
C008,27,36000,1500,4200,Palermo
C009,143,52000,2300,7000,Venice
C010,36,450000,2600,8500,Turin
C011,33,46000,12000,6400,Milan
C012,40,54000,2150,-2500,Rome
C013,31,43000,1850,5600,Naples
C014,39,50500,2250,6900,Genoa
```

- Outlier: Unrealistic age (143)
- Outlier: Extremely high income
- Outlier: Very high monthly spend
- Outlier: Negative debt value

`bank_customers_outliers.xlsx` (participants): 1,000 customers × 22 columns (demographics, income, balances, transactions, card, mortgage, score, complaints) with filters and a frozen header, plus a `Data_Dictionary` sheet with the plausible range for each column.

`bank_customers_outliers_TRAINER.xlsx`: same data plus three sheets:

- `Statistics` — mean, median, standard deviation, quartiles, IQR fences and z-score thresholds for the 15 numeric columns, all live formulas on the Customers sheet
- `Checks` — 13 row-level checks (plausible age, tenure vs age, card spend vs limit, mortgage vs payment, dates, duplicates, score out of range…) with a summary of the counts
- `Outlier_Key_Trainer` — the 95 injected anomalies, one per row, with ID, Excel row, columns involved, why it is "to be verified" and how to spot it in Excel

The 8 categories: **A** plausible extreme values (an executive earning €1.85M, a 97-year-old), **B** impossible values (age −3 and 187, a €32M income with extra zeros, an income of "42" in thousands), **C** sentinels and blanks (−999, 9999, empty cells), **D** cross-column inconsistencies (28 years of tenure at age 21, mortgage payment above monthly income, a student earning €240k), **E** future dates and "zero" dates (1900-01-01, 1970-01-01), **F** categorical variants ("lombardia", "Lombardia " with a trailing space, "Male", "Counter"), **G** exact duplicates and near-duplicates with a different ID, **H** a cluster of 12 accounts opened in the same month with 48 transactions of €199 each — normal one by one, anomalous as a group.

A point that lands well in class: on income, the z-score flags **1** outlier only (the €32M inflates the standard deviation and masks everything else), while the IQR rule finds 74. That's the perfect argument for removing sentinels and gross errors first, and only then looking for the "real" outliers.

Dates are formatted as YYYY-MM-DD to stay unambiguous for an international audience; if the participants use US Excel and you'd rather have MM/DD/YYYY, I can switch it.



Watch out for the separator: ; versus , (and comma versus point as the decimal separator) depends on the Windows regional format, not on Office, so if you want the screen to show exactly =IF(A2>1.5, ...) you also need to change the regional format in Settings > Time & Language > Region.

In **Windows 11**, the path is:

Start → Settings → Time & language → Language & region

Here you can find **Regional format**. The selected format, for example **Italian (Italy)** or **English (United States)**, affects Excel, especially:

- `,` or `.` as the decimal separator
- `;` or `,` as the formula argument separator

For example:

Italian (Italy):

```excel
=IF(A2>1,5; "Yes"; "No")
```

English (United States):

```excel
=IF(A2>1.5, "Yes", "No")
```

If you want to change the **separators specifically**, the most direct path in Windows is:

Control Panel → Clock and Region → Region → Additional settings

There you can change the **Decimal symbol** and the **List separator**.

In **Windows 10**,

Start → Settings → Time & Language → Region

Then look for **Regional format**.

If you want to change the separators specifically:

Control Panel → Clock and Region → Region → Additional settings

There you can set:

- **Decimal symbol**
- **List separator**

So the wording "**Settings → Time & Language → Region**" is correct for **Windows 10**.

Rules for deciding whether a point is an outlier on a single dimension

An **outlier** is an observation that lies very far from the bulk of the data. On a single variable, the question is always the same: how far from the "centre" must a point be before we call it anomalous? The rules differ in two choices: how we measure the **centre** (mean or median) and how we measure the **spread**, that is, how scattered the data are.

1. Z-score rule (mean ± k standard deviations)

The **z-score** measures how many standard deviations a point lies from the mean: z = (x − mean) / standard deviation. The **standard deviation** is the typical distance of the values from the mean.

- **Rule:** a point is an outlier if $|z| > 3$. Less strict thresholds, 2.5 or 2, are also used.
- **Why 3:** if the data follow a normal distribution (the classic bell curve), only 0.27% of values exceed this threshold.
- **Limitation:** the mean and standard deviation are influenced by the very outliers we are looking for. An extreme value "inflates" the standard deviation and can hide itself or other outliers. This effect is called **masking**. The rule also assumes roughly normal data.

2. Tukey's rule (IQR, the boxplot method)

This rule is based on **quartiles**:

- **Q1** is the value below which 25% of the data fall;
- **Q3** is the value below which 75% fall;
- the **IQR** (interquartile range) is Q3 − Q1, the width of the middle 50% of the data.
- **Rule:** a point is an outlier if $x < Q1 - 1.5 \cdot IQR$ or $x > Q3 + 1.5 \cdot IQR$. These limits are called **fences**.
- **Extreme outlier:** $3 \cdot IQR$ is used instead of $1.5 \cdot IQR$.
- **Strengths:** it is **robust**, meaning little affected by the outliers themselves, and it does not require normality. It is the rule that produces the isolated points in a **boxplot** (box-and-whisker chart).
- **Limitation:** with strongly **skewed** distributions it flags too many outliers on the long-tail side. Incomes and balances are typical examples: many small values and a few very large ones.

3. Median and MAD rule (modified z-score)

This is the robust version of the z-score:

- instead of the mean we use the **median**, the middle value of the sorted data;
- instead of the standard deviation we use the **MAD** (Median Absolute Deviation), the median of the absolute distances from the median.
- **Modified z-score** (proposed by Iglewicz and Hoaglin): $M = 0.6745 \cdot (x - \text{median}) / \text{MAD}$. A point is an outlier if $|M| > 3.5$.
- **Equivalent variant:** a point is an outlier if x lies outside median $\pm\ 3 \cdot 1.4826 \cdot$ MAD. The factor 1.4826 makes the MAD comparable to the standard deviation when the data are normal. In this form the rule is also known as the **Hampel filter**.
- **Strengths:** it solves the masking problem and is often the best general-purpose choice.

4. Skewness-adjusted boxplot

This is a variant of Tukey's rule, proposed by Hubert and Vandervieren. The fences are widened more on the long-tail side, using the **medcouple**, a robust measure of skewness.

- **When to use it:** for strongly skewed variables, where the classic Tukey rule would raise too many false alarms.
- **Simpler alternative:** apply a **log transformation** to the variable, then use Tukey's rule or the z-score on the transformed values.

5. Chebyshev's inequality

It holds for **any** distribution, including non-normal ones. It states that at most $1/k^2$ of the values can lie more than k standard deviations from the mean.

- **Rule:** we choose k based on the share of data we are willing to flag. For example, $k = 4.5$ corresponds to at most 5% of the data.
- **Strengths and limitations:** it is very conservative and flags few outliers. It has the same masking problem as the z-score, because it uses the mean and standard deviation.

6. Formal statistical tests

These tests provide a **p-value**: the probability of observing a value this extreme if there were in fact no outliers. They all require roughly normal data.

- **Grubbs' test:** checks whether the most extreme value is an outlier. It detects **only one** outlier at a time.
- **Dixon's Q test:** designed for **small** samples, roughly 3 to 30 observations. It compares the gap between the suspect value and its nearest neighbour with the total range of the data.
- **Rosner's generalized ESD:** extends Grubbs to search for **several** outliers, up to a maximum number we set. It avoids the mistake of applying Grubbs repeatedly.
- **Chauvenet's criterion:** a point is an outlier if $n \cdot P(|Z| > |z|) < 0.5$, where n is the number of observations. In practice, we would expect fewer than "half" a value this extreme in the sample. It is historical and rarely recommended today.

7. Percentile-based rules

Values below the 1st percentile or above the 99th (or the 5th and 95th) are treated as outliers. The **p-th percentile** is the value below which p% of the data fall.

- **Caution:** this is not true detection. It always flags the same percentage of data, even when there are no outliers. It is mainly used for **winsorizing**, which replaces extreme values with the percentile value instead of removing them.

8. Domain rules

These are thresholds set from knowledge of the phenomenon, not from statistics. Examples are an age above 120, a negative balance where one is impossible, or an amount above the contractual limit.

- **Strengths:** they are often the most reliable way to catch **data errors**, which are different from extreme but genuine values.

Summary

Rule	Centre / spread	Requires normality?	Robust to masking?	When to use it
Z-score ($\|z\| > 3$)	mean / std. deviation	Yes	No	Normal data, few outliers
Tukey (1.5·IQR)	quartiles / IQR	No	Yes	General rule, boxplots
Modified z-score (MAD)	median / MAD	No	Yes	Recommended general choice
Adjusted boxplot	quartiles + medcouple	No	Yes	Strongly skewed data
Chebyshev	mean / std. deviation	No	No	No assumption about the distribution
Grubbs / Dixon / ESD	mean / std. deviation	Yes	ESD yes, Grubbs no	When a formal test with a p-value is needed
Percentiles	percentiles	No	—	Winsorizing, not true detection
Domain	business thresholds	No	—	Data errors

Final note: a rule only tells us that a point is **suspicious**. Before removing it, we should find out whether it is an error (to be corrected or removed) or a genuine but rare value. A genuine value is often the most interesting information in the data, such as a fraud or a high-risk customer.

Missing Values Identification and Management

Missing values: “unknown” is a data state, not a number

Missingness can distort summaries and models — and may itself contain information.

Missing values

fx `=COUNTBLANK(Table1[Income])`

Customer	Income	Balance	RiskGrade
C1001	65,000	12,500	A
C1002	54,000	8,200	B
C1003		45,500	A
C1004	45,000		C
C1005		165,000	

MCAR

Missing Completely At Random — unrelated to observed or unobserved data.

MAR

Missing At Random conditional on observed information.

MNAR

Missing Not At Random — missingness depends on the missing value or process.

Do not silently replace missing values with zero unless zero is the correct business meaning.

Detecting missing values in Excel

Combine formulas, filters and visual highlighting.

```
=COUNTBLANK(Table1[Income])
```

```
=ISBLANK(C2)
```

```
=COUNTIF(C2:C1000,"")
```

```
=IF(ISBLANK(C2),"Missing","OK")
```

Be aware: a formula that returns "" may look blank but is not always the same as a truly empty cell.

Exception list

fx `=IF(ISBLANK(B4),"Missing","OK")`

Customer	Income	Balance	MissingFlag
C1001	65,000	12,500	OK
C1002	54,000	8,200	OK
C1003		45,500	Missing
C1004	45,000		Missing
C1005		165,000	Missing

Handling missing data: choose a strategy, do not hide the problem

Treatment depends on volume, mechanism, business meaning and downstream use.

Leave missing

Best when “unknown” is meaningful and downstream tools can handle it.

Exclude rows

Reasonable only when missingness is limited and exclusion does not bias the sample.

Impute simple values

Median for skewed numeric data; mode for categories — document the rule.

Create missing indicator

A separate flag can capture information in the fact that a value is missing (eg, NA, MV, etc).

Impute with a predictive model

Useful for complex data, but adds assumptions and validation requirements.

```
Example simple rule: =IF(ISBLANK(C2), MEDIAN($C$2:$C$1000), C2)
```

What are missing values?

Real data is rarely complete. Missing values are common, and it's important to recognise and handle them.

Look at this small dataset:

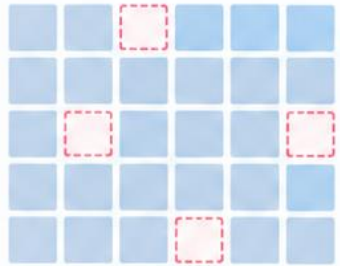
Customer_ID	Age	Region	Balance	Occupation
C001	25	North	1,200	Engineer
C002		South	850	Teacher
C003	37	NA	2,300	
C004	29	East	N/A	Doctor
C005	NULL	West	670	Artist
C006	45		1,100	Manager
C007	52	Central	?	Retired

Do you notice anything unusual?

Missing values are data entries that are absent, unknown, not recorded, or not available. They are often represented as blank cells or markers such as NA, N/A, NULL, or ?.

Missing values are pieces of information that should be there, but are not.

Missing values create gaps in our data...



Gaps mean missing information.

They can also appear in real-world forms, such as incomplete records...



Some information is missing here too.

Why they matter

- can bias summaries
- can distort charts and correlations
- can reduce usable sample size
- must be handled explicitly

Three types of missing values

How they differ, what they mean, and how to handle them

MCAR
Missing Completely At Random

Definition

The probability that a value is missing is unrelated to the variable itself and unrelated to other observed variables.

Intuition / example

Example: a few survey answers are missing because of a random technical glitch.

ID	Age	Income	Score
1	34	—	88
2	—	52	73
3	28	61	—
4	—	—	90
5	41	48	76

Missing values are scattered randomly across the data.

How to handle

Often the least problematic case. You can use listwise deletion when the amount is small, or simple imputation when appropriate.

Lowest risk of bias

MAR
Missing At Random

Definition

The probability that a value is missing depends on other observed variables, but not on the missing value itself once those variables are considered.

Intuition / example

Example: income is more often missing for younger customers; age is observed.

ID	Age	Income	Score
1	22	—	78
2	25	—	85
3	34	56	90
4	47	62	88
5	52	71	76

Missing values in income are more common for younger customers (age is observed).

How to handle

Model-based methods and imputation are usually appropriate: mean within groups, regression imputation, or multiple imputation.

Manageable if related observed variables are available

MNAR
Missing Not At Random

Definition

The probability that a value is missing depends on the unobserved value itself, or on unobserved factors.

Intuition / example

Example: customers with very high debt are less likely to report it.

ID	Debt	Income	Score
1	5	42	80
2	12	58	76
3	28	61	82
4	—	—	74
5	—	—	68

Missing values are concentrated among extreme (typically high) values of the variable itself (e.g., high debt).

How to handle

Most difficult case. Investigate the data-generation process, use domain knowledge, sensitivity analysis, and specialized models when possible.

Highest risk of bias

MCAR: random missingness

MAR: depends on observed data

MNAR: depends on missing/unobserved data

Key takeaway: the right treatment depends on **why** data is missing, not just on how much is missing.

3 Ways to Handle Missing Values (MV) in Excel

Choose the right method based on your data, the amount of missing values and your analysis goals.

1 Delete rows with missing values (Listwise deletion)

Remove every row that contains at least one missing value.

When to use

- Few missing values
- Missing values are completely random (MCAR)
- You want a simple and clean dataset

	A	B	C	D
1	**ID**	**Age**	**Income**	**City**
2	1	25	50000	Rome
3	2		62000	Milan
4	3	34		Turin
5	4	28	58000	Naples

How to do it in Excel

1. Select the data range.
2. Go to Data → Filter.
3. In each column, filter "(Blanks)" and delete the corresponding rows.

Tip

In a dynamic way (newer Excel versions), you can use the **FILTER** function:

```
=FILTER(A2:D100, (A2:A100<>"")*(B2:B100<>"")*
 (C2:C100<>"")*(D2:D100<>""))
```

2 Impute with the column mean or median

Replace missing values with a simple statistic of the column (mean or median).

When to use

- Small to moderate amount of missing values
- Numeric variables
- You need a quick and simple solution

A	B	C
ID	**Age**	**Income**
1	25	50000
2		62000
3	34	
4	28	58000

How to do it in Excel

Use IF to replace blanks with the mean or median of the column.

Using the mean:

```
=IF(B2="",AVERAGE(B:B),B2)
```

If B2 is empty, replace it with the column mean.

Using the median:

```
=IF(B2="",MEDIAN(B:B),B2)
```

If B2 is empty, replace it with the column median.

Tip

Create a new column with the imputed values to keep the original data.

3 Use a predictive model to estimate missing values

Predict missing values using other variables in the dataset (e.g. regression, machine learning).

When to use

- Moderate to large amount of missing values
- Missing values depend on other variables (MAR or MNAR)
- You want more accurate estimates

	A	B	C	D
1	**ID**	**Age**	**Income**	**Experience**
2	1	25	50000	2
3	2		62000	5
4	3	34	54000	8
5	4	28	58000	4

How to do it in Excel (example: linear regression)

1. Go to Data → Data Analysis → Regression.
2. Use the other variables (e.g. Income, Experience) to predict the variable with missing values (e.g. Age).
3. Excel will output the coefficients (intercept and slopes).
4. Use the coefficients in a formula to estimate missing values.

Example formula (after running a regression for Age):

```
=IF(B2="",$E$1+$E$2*C2+$E$3*D2,B2)
```

If B2 is empty, predict Age using Income (C2) and Experience (D2).

Tip

This approach usually gives better results than simple imputation, but it requires more work and domain knowledge.

There is no single best method.

Consider the amount of missing values, the reason why they are missing, and the goal of your analysis.

The two files are ready (same 22 predictors as the outlier dataset, new seed, everything in English):

`bank_customers_missing.xlsx` (participants) — `Customers`: 1,000 customers with 1,598 effectively missing cells (7.3%) under every disguise: empty cells, cells of only spaces, text placeholders (N/A, NA, n.d., ?, -, Unknown), numeric sentinels (-999, 9999, 0, date 1970-01-01), numbers saved as text, 4 ghost records (only the ID), 6 almost-empty records, and structural blanks (no card, no mortgage, dormant account, "NULL = 0 complaints"). Three mechanisms: MCAR, MAR (income missing at 28% for self-employed and business owners vs 3% for the others) and MNAR (the savings balance is missing all the more the higher it is). `Data_Dictionary` documents only the "official" conventions (0 = no card, etc.): everything else is to be discovered.

`bank_customers_missing_TRAINER.xlsx` — same data plus the workflow sheets, all with live formulas:

- `Codes`: the single list of placeholders, read by all the formulas (add a code and the file updates)
- `Missing_Profile`: per column COUNTBLANK, spaces, placeholders, sentinels, text-numbers, % missing, structural share, what to do + chart
- `Missing_Matrix`: 0/1 map with conditional formatting, count per row, ghost / mostly empty flags
- `Clean`: a single formula pattern per column that normalizes everything to a true blank and converts the text-numbers with VALUE
- `Imputed`: the strategies side by side, always next to the clean value and the missing flag (median, mean, group median with VLOOKUP, derivation with DATEDIF, mode, "Unknown" category, spend/limit ratio, constant 0, listwise on 21 vs 8 columns)
- `Strategies`: reference values, missing rate by occupation, and the trainer section that measures each strategy against `Ground_Truth`
- `Missing_Key_Trainer`: key to everything that was injected, cheat sheet of the strategies, classroom outline with the live numbers

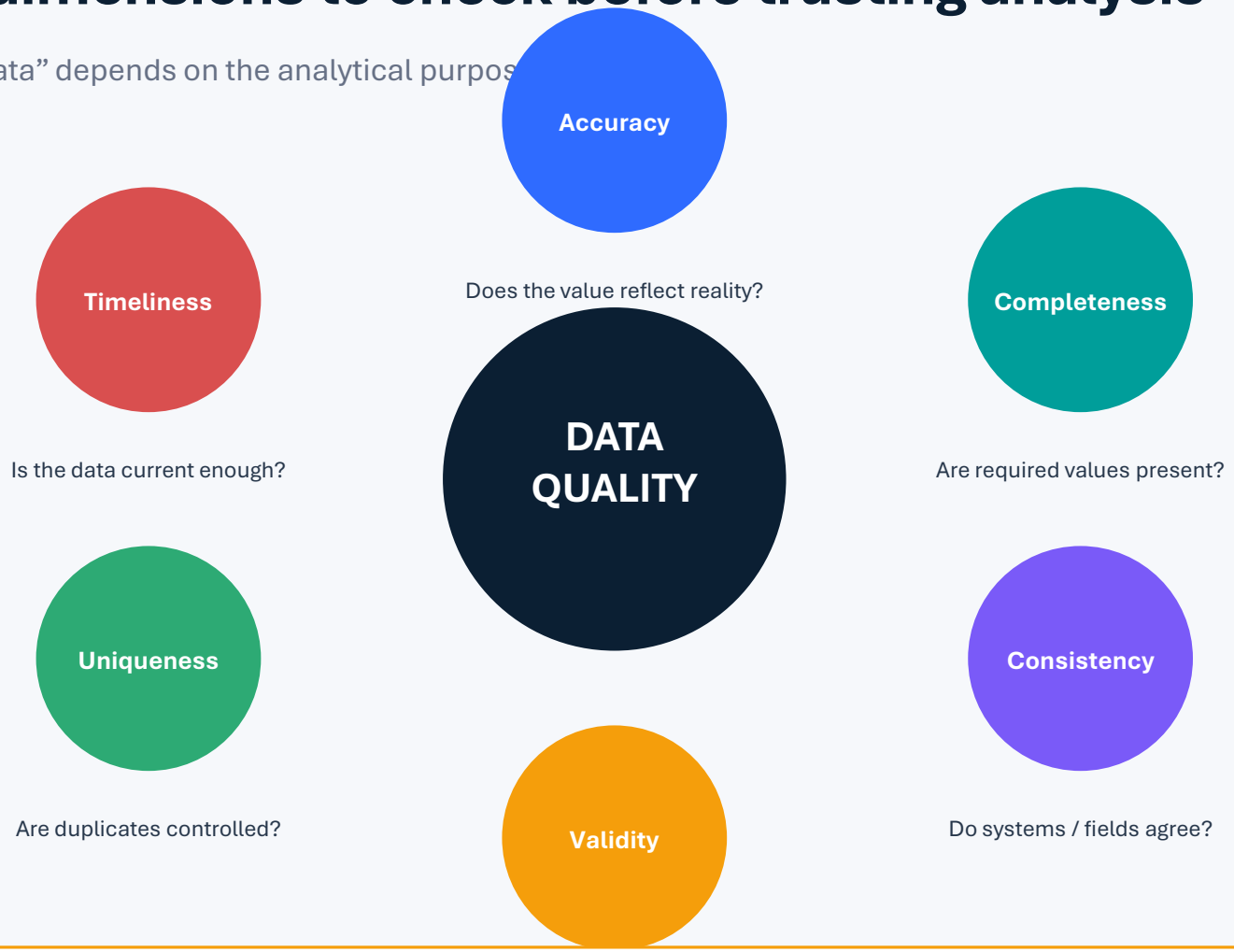
Dates are formatted as YYYY-MM-DD to stay unambiguous for an international audience; if the participants use US Excel and you'd rather have MM/DD/YYYY, I can switch it.

Punti che funzionano in aula (numeri già calcolati nel file): la listwise deletion tiene il 29,5% delle righe (72,3% sulle sole 8 colonne del modello); sul reddito i casi completi danno media 34.752 contro 35.827 vera, la media imputata schiaccia la deviazione standard, la mediana per occupazione recupera 35.514 con metà dell'errore; sul risparmio MNAR le celle mancanti valgono in media 43.669 e noi imputiamo 10.653; la tenure derivata dalla data è esatta 22 volte su 22; la moda del canale indovina il 31%; "NULL = 0 reclami" sbaglia 13 volte su 214.

Allego anche lo script generatore (seed fisso, riproduce entrambi i file). Una nota: nella chat «Dataset outlier per analisi bancaria in Excel» risultava già generata una coppia con questo stesso nome; questa è rifatta da zero, quindi usa i due file insieme — la chiave del trainer vale solo per questo `Customers`.

Data quality: six dimensions to check before trusting analysis

Quality is fitness for use: “good data” depends on the analytical purpos



The same dataset can be “good enough” for one purpose and unacceptable for another.

How data quality changes the analysis

Poor quality can create fake signals, hide real ones and bias decisions.

x Dirty data → distorted evidence

```
fx  =COUNTBLANK(Table1[Income])
```

Issue	Example	Analytical effect
Missing	Income absent for some segments	Biased averages
Wrong type	Amounts imported as text	Rows silently excluded
Duplicate	Customer repeated twice	Inflated counts / balances
Outlier error	Balance ×1000	Mean & correlation distorted
Stale data	Old risk grade	Wrong segmentation

Wrong descriptive statistics

Mean, rate, dispersion and counts become misleading.

Wrong relationships

Spurious or suppressed correlations and segment differences.

Wrong decisions

Mis-prioritised customers, risk, operations or controls.

```
Data quality is part of the model – not a preprocessing footnote.
```

Chapter title



08 – Takeaways

Excel EDA checklist: from raw table to trustworthy insight

A compact workflow to use after the course.

1. Define the question and the unit of analysis.
2. Convert the data to an Excel Table and preserve the raw extract.
3. Check types, units, keys and valid categories.
4. Profile rows, unique values, ranges and missingness.
5. Summarise numeric and categorical variables.
6. Visualise distributions, relationships and time patterns.
7. Investigate outliers and missing values before changing them.
8. Use correlation with scatter plots — not as proof of causality.
9. Assess data quality and document every transformation.
10. Communicate findings, limitations and next questions.

Good Data Science in Excel = clear question + clean structure + statistical thinking + visual evidence + data quality discipline

09 – Predictive Mopdels

Le componenti della regressione

predictor, 'x-variable',
independent variable,
explanatory variable

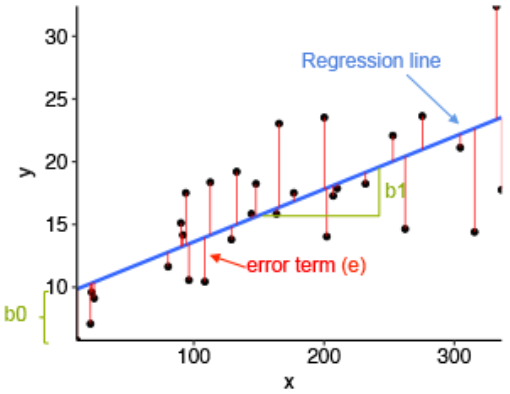
coefficient

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p + \varepsilon$$

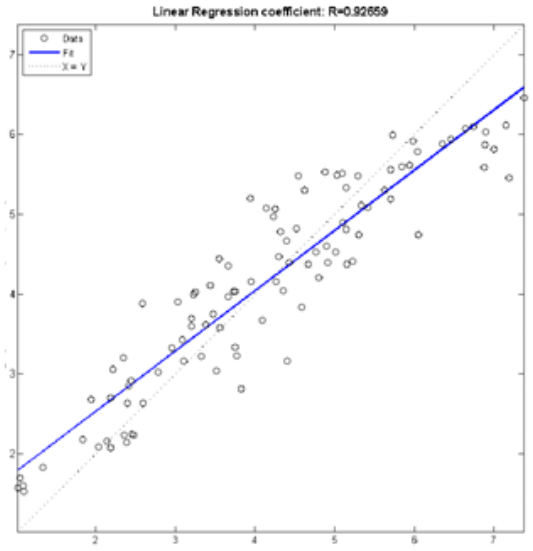
linear predictor

response, dependent variable,
observation, 'y-variable'

random error,
"noise"



nel caso univariato



Retta di regressione
vera vs stimata

link to animation

ID	Income	Balance	Limit	Rating	Cards	Age	Education	Gender	Student	Married	Ethnicity
1	14891	333	3606	283	2	34	11	Male	No	Yes	Caucasian
2	106025	903	6645	483	3	82	15	Female	Yes	Yes	Asian
3	104593	580	7075	514	4	71	11	Male	No	No	Asian
4	148924	964	9504	681	3	36	11	Female	No	No	Asian
5	55882	331	4897	357	2	68	16	Male	No	Yes	Caucasian
6	8018	1151	8047	569	4	77	10	Male	No	No	Caucasian
7	20996	203	3388	259	2	37	12	Female	No	No	African American
8	71408	872	7114	512	2	87	9	Male	No	No	Asian
9	15125	279	3300	266	5	66	13	Female	No	No	Caucasian
10	71061	1350	6819	491	3	41	19	Female	Yes	Yes	African American
11	63095	1407	8117	589	4	30	14	Male	No	Yes	Caucasian
12	15045	0	1311	138	3	64	16	Male	No	No	Caucasian
13	80616	204	5308	394	1	57	7	Female	No	Yes	Asian
14	43682	1081	6922	511	1	49	9	Male	No	Yes	Caucasian
15	19144	148	3291	269	2	75	13	Female	No	No	African American
16	20089	0	2525	200	3	57	15	Female	No	Yes	African American
17	53598	0	3714	286	3	73	17	Female	No	Yes	African American
18	36496	368	4378	339	3	69	15	Female	No	Yes	Asian
19	4957	891	6384	448	1	28	9	Female	No	Yes	Asian
20	42079	1048	6626	479	2	44	9	Male	No	No	Asian
21	177	89	2860	235	4	63	16	Female	No	No	Asian
22	37348	968	6378	458	1	72	17	Female	No	No	Caucasian
23	20103	0	2631	213	3	61	10	Male	No	Yes	African American
24	64027	411	5179	398	5	48	8	Male	No	Yes	African American
25	10742	0	1757	156	3	57	15	Female	No	No	Caucasian
26	1409	671	4323	326	5	25	16	Female	No	Yes	African American
27	42471	654	3625	289	6	44	12	Female	Yes	No	Caucasian
28	32793	467	4534	333	2	44	16	Male	No	No	African American
29	186634	1809	13414	949	2	41	14	Female	No	Yes	African American
30	26813	915	5611	411	4	55	16	Female	No	No	Caucasian
31	34142	863	5666	413	4	47	5	Female	No	Yes	Caucasian
32	28941	0	2733	210	5	43	16	Male	No	Yes	Asian
33	134181	526	7838	563	2	48	13	Female	No	No	Caucasian
34	31367	0	1829	162	4	30	10	Male	No	Yes	Caucasian

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.996879737
R Square	0.99376921
Adjusted F	0.993753555
Standard I	182.4272398
Observati	400

ANOVA

	df	*SS*	*MS*	*F*	*gnificance F*
Regressio	1	2112539666	2112539666	63478.33075	0
Residual	398	13245319.74	33279.69783		
Total	399	2125784986			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-542.92823	22.85025671	-23.7602683	2.46767E-78	-587.851	-498.0059432	-587.8505168	-498.005943
Rating	14.87160712	0.059026245	251.9490638	0	14.75556	14.98764931	14.75556492	14.98764931

